

Calculus of Variations

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1 Generalized extremal problems

Extremal problems, like minimum and maximum problems, have played a major role in the development of calculus. In fact, calculus was more or less invented to solve such problems.

In the language of calculus, the quantity we need to maximize or minimize is a *function* of a real variable, x , and the challenge is to find an x_0 such that

$$f(x) \leq f(x_0) \quad (f(x) \geq f(x_0)) \quad \forall x \neq x_0, \quad (1)$$

If $f(x)$ is a well behaved function, calculus tells us that we only need to look at points x^* such that

$$f'(x^*) = 0. \quad (2)$$

All maximum and minimum points will be found among the set of points that satisfy (2). By well behaved functions, we mean here that $f(x)$ is continuously differentiable and defined on the whole real axis.

There are however many important extremal problems that do not fall into the category described above. In fact, some of these problems are much older than calculus itself.

The so-called *isoperimetric problem* was clearly stated already 200 BCE¹ by the Greek mathematician Zenodorus. The problem consists of finding, among all curves of a fixed length L , the curve that encloses the largest area.

We will start our study of the calculus of variation by introducing several other interesting extremal problems that are beyond the bounds of standard calculus, and then move on to describing mathematical tools which we can use to solve them.

1.1 Curve of shortest length in a plane

Let p and q be two points on the plane. The challenge is to find a curve, C , connecting p and q , which is of shortest possible length.



Figure 1: Curve connecting two points p and q in the plane

In order to state the problem in precise mathematical terms we, introduce

¹BCE=Before Common Era, A religiously neutral substitute for BC

a parametrization $\gamma(t)$ for C ,

$$\begin{aligned}\gamma(t) &= (x(t), y(t)), & 0 \leq t \leq 1, \\ \gamma([0, 1]) &= C, \\ \gamma(0) &= p, & \gamma(1) = q.\end{aligned}$$

Using this parametrization, the length of the curve, $L(C)$, can be written as

$$L(C) = \int_0^1 dt \|\gamma'(t)\| = \int_0^1 dt \sqrt{x'(t)^2 + y'(t)^2}.$$

The challenge is then to find a curve, C_0 , such that

$$L(C) \geq L(C_0),$$

for all C connecting p and q . This looks just like the minimum problem (1) from elementary calculus.

The only difference is that L is *not* a function of a real variable, but is rather a function defined on the set of smooth curved connecting the points p and q .



Figure 2: A functional defined on plane curves

Such a function is called a *functional*. We will in these lecture notes encounter many other functions of this type, with ever larger domains of definition. All such functions will be called functionals.

1.2 Curve of shortest length on a surface

Let S be a surface in $\mathbf{R}^3$ and let p and q be points on S . The challenge is to find a curve, C , on the surface S , connecting p and q , which is of the shortest possible length.

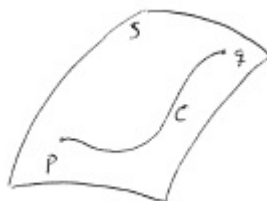


Figure 3: A curve on the surface, S connecting two points p and q on the surface

Let $\gamma(t)$ be a parametrization for C

$$\begin{aligned}\gamma(t) &= (x(t), y(t), z(t)) & 0 \leq t \leq 1, \\ \gamma([0, 1]) &= C \subset S, \\ \gamma(0) &= p, & \gamma(1) = q.\end{aligned}$$

The length of C is

$$L(C) = \int_0^1 dt \sqrt{x'(t)^2 + y'(t)^2 + z'(t)^2},$$

and the challenge is to find a curve, C_0 , such that

$$L(C) \geq L(C_0) \quad \text{for all } C \subset S \text{ connecting } p \text{ and } q.$$

This looks exactly like an extremal problem from regular calculus where we have a *constraint*. In regular calculus such problems are solved using Lagrange multipliers. We will see that a similar approach will work for functionals.

For this particular problem we can remove the constraint using some vector calculus.

Let the surface be parametrized by $\mathbf{x}(u, v)$,

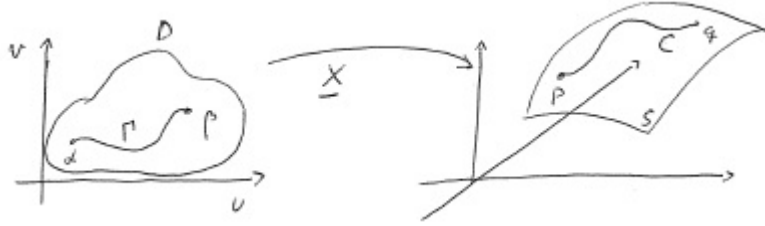


Figure 4: A parametrization for the surface S

where

$$\begin{aligned}\mathbf{x}(u, v) &= (x(u, v), y(u, v), z(u, v)), & (u, v) \in D, \\ \mathbf{x}(D) &= S, \\ \mathbf{x}(\Gamma) &= C, \quad \mathbf{x}(\alpha) = p, \quad \mathbf{x}(\beta) = q.\end{aligned}$$

Let the curve Γ be parametrized by $\gamma(t)$

$$\begin{aligned}\gamma(t) &= (u(t), v(t)), & 0 \leq t \leq 1, \\ \gamma([0, 1]) &= \Gamma, \\ \gamma(0) &= \alpha, & \gamma(1) = \beta.\end{aligned}$$

Then then the curve C is parametrized by

$$\xi(t) = \mathbf{x}(\gamma(t)), \quad 0 \leq t \leq 1.$$

In component form we have

$$\begin{aligned}x(t) &= x(u(t), v(t)), \\y(t) &= y(u(t), v(t)), \\z(t) &= z(u(t), v(t)),\end{aligned}$$

so that

$$\xi(t) = (x(t), y(t), z(t)).$$

Using the chain rule we have

$$\begin{aligned}x'(t) &= \partial_u x(u(t), v(t))u'(t) + \partial_v x(u(t), v(t))v'(t), \\y'(t) &= \partial_u y(u(t), v(t))u'(t) + \partial_v y(u(t), v(t))v'(t), \\z'(t) &= \partial_u z(u(t), v(t))u'(t) + \partial_v z(u(t), v(t))v'(t).\end{aligned}$$

In vector form this can be written as

$$\xi'(t) = \mathbf{T}_u u' + \mathbf{T}_v v',$$

where the vectors $\mathbf{T}_u$ and $\mathbf{T}_v$ are defined by

$$\begin{aligned}\mathbf{T}_u &= \mathbf{T}_u(t) = \mathbf{T}_u(u(t), v(t)) \\&= (\partial_u x(u(t), v(t)), \partial_u y(u(t), v(t)), \partial_u z(u(t), v(t))), \\ \mathbf{T}_v &= \mathbf{T}_v(t) = \mathbf{T}_v(u(t), v(t)) \\&= (\partial_v x(u(t), v(t)), \partial_v y(u(t), v(t)), \partial_v z(u(t), v(t))).\end{aligned}$$

The two vectors $\mathbf{T}_u(u, v)$ and $\mathbf{T}_v(u, v)$ are tangent to the surface S . Using $\mathbf{T}_u$ and $\mathbf{T}_v$ we have

$$\begin{aligned}x'(t)^2 + y'(t)^2 + z'(t)^2 &= \|\xi'(t)\|^2 \\&= \xi'(t) \cdot \xi'(t) = (\mathbf{T}_u u' + \mathbf{T}_v v') \cdot (\mathbf{T}_u u' + \mathbf{T}_v v') \\&= A u'^2 + 2 B u' v' + C v'^2,\end{aligned}\tag{3}$$

where

$$\begin{aligned}A &= A(t) = \mathbf{T}_u \cdot \mathbf{T}_u, & B &= B(t) = \mathbf{T}_u \cdot \mathbf{T}_v, \\C &= \mathbf{T}_v \cdot \mathbf{T}_v.\end{aligned}\tag{4}$$

Observe that the functions A , B , C depends only on the structure of the surface S

$$\begin{aligned}A(u, v) &= \mathbf{T}_u(u, v) \cdot \mathbf{T}_u(u, v), \\B(u, v) &= \mathbf{T}_u(u, v) \cdot \mathbf{T}_v(u, v), \\C(u, v) &= \mathbf{T}_v(u, v) \cdot \mathbf{T}_v(u, v).\end{aligned}$$

We are here abusing the notation in the usual calculus way.

$$A = A(t) = A(u(t), v(t)).$$

Using (3) and (4), our minimum problem is now to find a curve $\Gamma_0 \subset D$, connecting α and β , such that

$$L(\Gamma) \geq L(\Gamma_0) \text{ for all curves in } D \text{ connecting } \alpha \text{ and } \beta,$$

where the functional L is

$$L(\Gamma) = \int_0^1 dt [A(t) u'(t)^2 + 2 B(t) u'(t) v'(t) + C(t) v'(t)^2]^{\frac{1}{2}}.$$

This is now an unconstrained minimum problem.

Observe that the three functions, A , B and C determine the length of *all* curves on the surface S . Thus the three functions determine the *geometry* of the surface S .

The functions A , B and C are called the *metric* coefficients for S and the curve of minimal length, C_0 , connecting p and q is called a *geodesics* for the surface.

These ideas and their generalization play a fundamental role in theoretical physics.

In fact, Einstein's fundamental contribution to gravitational physics was to merge gravitational and inertial forces into a collection of 10 metric coefficients determining the geometry of the four dimensional space-time continuum. From the metric coefficients we can calculate the length of curves connecting points, here called *events*, in space-time. The length of a curve, in this space-time context, is the time it would take an observer, to follow the curve from an event p to another event q . The time in question is the one measured by the observer following the curve. This time is called *proper* time in Einstein's gravitational theory (General Relativity).

The physical postulate is that an observer following the curve C_0 from p to event q , will feel no inertial forces if the curve has maximal length. Since length is proper time, this means that if an observer wants to move from event p to another q , without feeling inertial forces, she should choose a curve that takes as much time as possible as measured by her clock.

These curves are also called geodiscs, but here, in the space-time context, they are curves of maximal, not minimal, length. The metric coefficients are determined by the distribution of mass and energy through the *Einstein field equations*.

1.3 The isoperimetric problem

Let us consider the truly ancient isoperimetric problem. Let C be a curve enclosing a domain D .

We want to express the area of D in terms of C , and for this purpose introduce a vector field $\mathbf{f}$, on the plane given by

$$\mathbf{f}(x, y) = \frac{1}{2}(x, y).$$

Evidently we have

$$\nabla \cdot \mathbf{f} = \partial_x \left(\frac{1}{2} x \right) + \partial_y \left(\frac{1}{2} y \right) = \frac{1}{2} + \frac{1}{2} = 1.$$

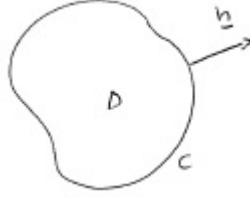


Figure 5: Curve enclosing a domain D

But then, using the divergence form of Green's theorem, we have

$$A = \iint_D dA = \iint_D dA \nabla \cdot \mathbf{f} = \oint_C dl \mathbf{f} \cdot \mathbf{n}.$$

Thus the area of D is a functional of C

$$A = A(C) = \oint_C dl \mathbf{f} \cdot \mathbf{n}. \quad (5)$$

The length of the curve C is determined by the functional

$$L = L(C) = \oint_C dl. \quad (6)$$

The isoperimetric problem consists of, for a fixed length L^* , finding the curve C_0 such that

$$A(C) \leq A(C_0) \quad \forall C, \quad L(C) = L^*.$$

We will now parametrize this problem, and therefore introduce a counter clockwise orientation for the curve C .

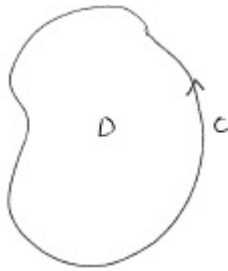


Figure 6: Defining orientation for the curve C

Let γ be a parametrization of the oriented curve C . We thus have

$$\begin{aligned} \gamma(t) &= (x(t), y(t)) & 0 \leq t \leq 1, \\ \gamma([0, 1]) &= C, \\ \gamma(0) &= \gamma(1), \end{aligned} \quad (7)$$

and $\gamma'(t)$ points along the orientation of C . The choice of orientation ensures that

$$\mathbf{n}(t) = \frac{(y'(t), -x'(t))}{\|\gamma'(t)\|}, \quad (8)$$

is a unit normal defined on C that points *out* of the domain D . Using (7) and (8) in (5) and (6) we have

$$\begin{aligned} L(C) &= \oint_C dl = \int_0^1 dt \|\gamma'(t)\| = \int_0^1 dt \sqrt{x'(t)^2 + y'(t)^2}, \\ A(C) &= \oint_C dl \mathbf{f} \cdot \mathbf{n} = \int_0^1 dt \|\gamma'(t)\| \frac{1}{2} (x(t), y(t)) \cdot \frac{(y'(t), -x'(t))}{\|\gamma'(t)\|} \\ &= \frac{1}{2} \int_0^1 dt (x(t) y'(t) - y(t) x'(t)). \end{aligned}$$

Thus, our problem consists in finding functions $x_0(t), y_0(t)$ such that

- i) $x_0(0) = x_0(1), \quad y_0(0) = y_0(1),$
- ii) $\int_0^1 dt \sqrt{x'(t)^2 + y'(t)^2} = L^*,$
- iii) $\frac{1}{2} \int_0^1 dt (x(t) y'(t) - y(t) x'(t))$ is maximal.

1.4 Surface of revolution of minimal area

Let $y(x)$ be a function defined on the interval (x_1, x_2) . Assume $y(x) > 0$ for all $x \in (x_1, x_2)$.

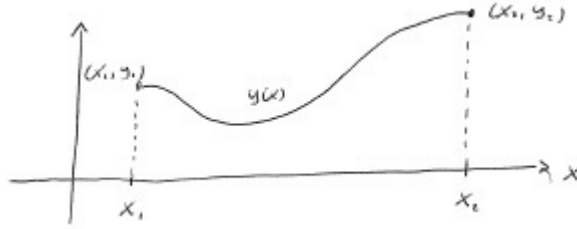


Figure 7: Curve defining surface of revolution

Let $A(y)$ be the area of the surface of revolution that we get by rotating the curve $y(x)$ around the x -axis. The challenge is to find a curve $y(x)$ for which $A(y)$ is minimal for given fixed points (x_1, y_1) and (x_2, y_2) .

Recall from elementary Calculus that the formula for the area $A(y)$ is

$$A(y) = 2\pi \int_{x_1}^{x_2} dx y(x) \sqrt{1 + y'(x)^2}.$$

Thus the challenge is to find a curve $y(x)$, defined on (x_1, x_2) , such that

- i) $y(x_1) = y_1, \quad y(x_2) = y_2,$
- ii) $A(y) = 2 \pi \int_{x_1}^{x_2} dx \, y(x) \sqrt{1 + y'(x)^2}$ is minimal.

1.5 General surface of minimal area

Let a curve C , in $\mathbf{R}^3$ be given. The challenge is to find a surface $S \subset \mathbf{R}^3$ such that

- i) $\partial S = C,$
- ii) $A(S)$ is minimal,

where $A(S)$ is the area of the surface S . Such a surface is called a *minimal surface*. Many important problems in theoretical physics and applied mathematics can be reduced to the problem of finding a minimal surface.

For example, if we dip a piece of string, described by a closed curve in $\mathbf{R}^3$, into a bucket of soap water, the resulting soap film, clinging to the string, will form a minimal surface.

Let us parametrize this problem by introducing a parametrization $\mathbf{x}$ of the surface S . Thus

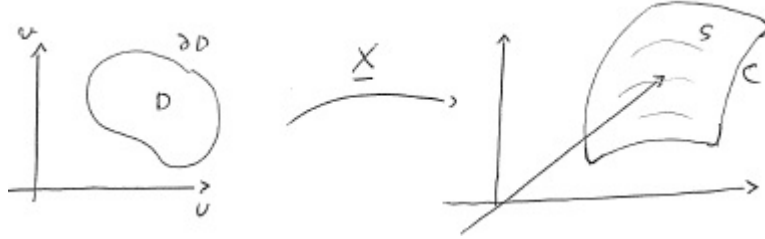


Figure 8: Parametrization of surface S bounded by the curve C

- i) $\mathbf{x} : \mathbf{R}^2 \rightarrow \mathbf{R}^3, \quad \mathbf{x} = \mathbf{x}(u, v),$
- ii) $\mathbf{x}(D) = S,$
- iii) $\mathbf{x}(\partial D) = C.$

From calculus we know that the area of S , $A(S)$, is given by the formula

$$A(S) = \iint_D du \, dv \, \|\mathbf{T}_u \times \mathbf{T}_v\|,$$

where as before

$$\mathbf{T}_u = \frac{\partial \mathbf{x}}{\partial u}, \quad \mathbf{T}_v = \frac{\partial \mathbf{x}}{\partial v}.$$

The challenge is to choose functions $x(u, v)$, $y(u, v)$, $z(u, v)$ with

$$\mathbf{x}(u, v) = (x(u, v), y(u, v), z(u, v)),$$

such that $A(S)$ is minimal under the constraint

$$\mathbf{x}(\partial D) = C.$$

Let us restrict to the case when C is given by the graph of a function $h = h(x, y)$ defined on the boundary, ∂D , of D . A surface of this type is displayed in figure 9. For such surfaces we can use a parametrization of the form

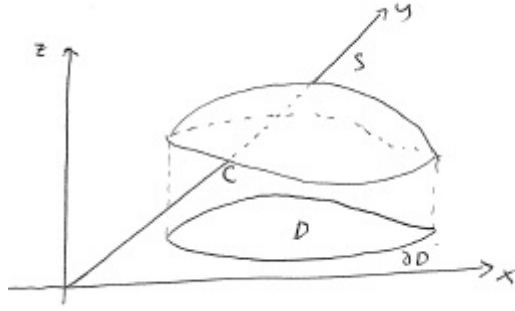


Figure 9: Surface whose boundary is the graph of a function.

$$\mathbf{x}(x, y) = (x, y, f(x, y)).$$

For this parametrization we have

$$\begin{aligned} \mathbf{T}_x &= \frac{\partial \mathbf{x}}{\partial x} = (1, 0, f_x), \\ \mathbf{T}_y &= \frac{\partial \mathbf{x}}{\partial y} = (0, 1, f_y). \end{aligned}$$

Thus

$$\|\mathbf{T}_x \times \mathbf{T}_y\| = \sqrt{1 + f_x^2 + f_y^2}.$$

Therefore the challenge is to find a function $f(x, y)$, defined on a domain D in $\mathbf{R}^2$, such that

$$\begin{aligned} \text{i)} \quad & f(x, y) = h(x, y), \quad (x, y) \in \partial D, \\ \text{ii)} \quad & \iint_D dx \, dy \sqrt{1 + f_x^2 + f_y^2} \quad \text{is minimal.} \end{aligned}$$

Note that in this case the functional is defined on a domain consisting of functions of two variables. This is clearly a very large domain.

1.6 The Fermat Principle

Let c be the speed of light in vacuum. Recall that the speed of light typically depends on the physical properties of the medium it is travelling through. The ratio between the speed of light in vacuum and the speed of light in a medium, v , is a dimensionless number

$$n = \frac{c}{v}, \quad (9)$$

which is called the refractive index of the material. Under normal circumstances $v < c$ so that $n > 1$.

The refractive index typically depends on the frequency of the light, but we will disregard this effect here. Unless the material is homogeneous, the refractive index *will* depend on position

$$n = n(\mathbf{x}).$$

Let us now consider a light ray passing through a medium that has a refractive index $n(\mathbf{x})$.

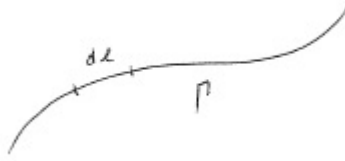


Figure 10: Light ray in a refractive medium.

Using (9) and figure 10, we have

$$\begin{aligned} n &= \frac{c}{\frac{dl}{dt}}, \\ \Updownarrow \\ dt &= \frac{n}{c} dl, \end{aligned}$$

where now dt is the time it takes light to propagate the distance dl along the curve Γ . The total time it takes light to propagate along a curve Γ is then

$$T(\Gamma) = \frac{1}{c} \int_{\Gamma} dl \, n.$$

Fermat's principle states that light follows the path through a medium of refractive index $n(\mathbf{x})$, which takes the *shortest* possible time. Thus, in order to find the path followed by light we must minimize $T(\Gamma)$ over all paths Γ .

Let us parametrize this problem. Let γ be a parametrization for Γ .

$$\begin{aligned} \gamma(t) &= (x(t), y(t), z(t)), \quad 0 \leq t \leq 1, \\ \gamma(0) &= p, \quad \gamma(1) = q. \end{aligned}$$

Thus, in order to find the path followed by light from p to q we must find functions $x(t)$, $y(t)$, $z(t)$ such that

$$T(\Gamma) = \int_0^1 dt \, n(x(t), y(t), z(t)) \sqrt{x'(t)^2 + y'(t)^2 + z'(t)^2},$$

subject to the constraints

$$\begin{aligned} (x(0), y(0), z(0)) &= p, \\ (x(1), y(1), z(1)) &= q, \end{aligned}$$

is as small as possible.

1.7 The brachistochrone problem.

(brachistochrone - shortest time in Greek)

The challenge is to find the arc $(x, y(x))$, which a particle of mass m must follow from (x_1, y_1) to (x_2, y_2) , in order for it to use as little time as possible. The particle is influenced by a constant gravitational field pointing vertically down.

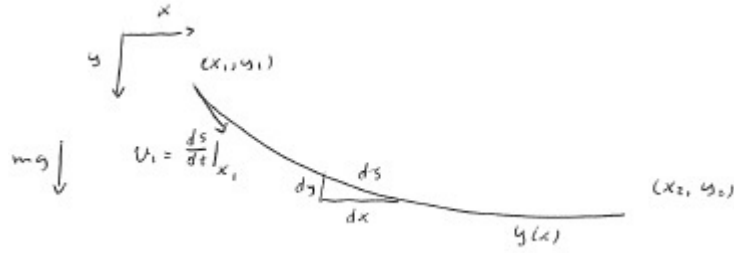


Figure 11: The brachistochrone problem

We have

$$\begin{aligned} v &= \frac{ds}{dt}, \\ \Downarrow \\ dt &= \frac{ds}{v}. \end{aligned}$$

Thus, the time it takes the particle to move from (x_1, y_1) to (x_2, y_2) along the arc $(x, y(x))$ is

$$T(y) = \int_{x_1}^{x_2} dx \, \frac{1}{v} \sqrt{1 + y'^2}.$$

We find a relation between v and y using the conservation of energy. We set the zero for potential energy at $y = y_1$. Then we have

$$\frac{1}{2} m v_1^2 = \frac{1}{2} m v^2 - m g (y - y_1),$$

which leads to

$$v = \sqrt{2g} \sqrt{y - y_0}, \quad y_0 \equiv y_1 - \frac{v_1^2}{2g},$$

and thus

$$T(y) = \frac{1}{\sqrt{2g}} \int_{x_1}^{x_2} dx \sqrt{\frac{1 + y'(x)^2}{y(x) - y_0}}.$$

The challenge is then to minimize $T(y)$ subject to the constraints

$$y(x_1) = y_1, \quad y(x_2) = y_2.$$

1.8 The Action Principle.

Let us consider a system consisting of N mass-points with positions $\{\mathbf{x}_i\}_{i=1}^N$, velocities $\{\mathbf{x}'_i\}_{i=1}^N$ and masses m_i . Let us assume that the mass-points are moving under the influence of a conservative force. Let $V(\mathbf{x}_1, \dots, \mathbf{x}_N)$ be the potential of this conservative force. Then by definition

$$\mathbf{f}_i = \frac{\partial V}{\partial \mathbf{x}_i},$$

is the force acting on particle number i . The kinetic energy of the system of particles is

$$T(\mathbf{x}'_1, \dots, \mathbf{x}'_N) = \sum_{i=1}^N \frac{1}{2} m_i \mathbf{x}'_i{}^2,$$

and the *Lagrangian* of the system is by definition

$$L(\mathbf{x}_1, \dots, \mathbf{x}_N, \mathbf{x}'_1, \dots, \mathbf{x}'_N) = T(\mathbf{x}'_1, \dots, \mathbf{x}'_N) - V(\mathbf{x}_1, \dots, \mathbf{x}_N).$$

A position vector

$$\mathbf{P} = (\mathbf{x}_1, \dots, \mathbf{x}_N) \in \mathbf{R}^{3N},$$

is a *configuration* for the system. Let a curve

$$\mathbf{P}(t) = (\mathbf{x}_1(t), \dots, \mathbf{x}_N(t)), \quad t_1 \leq t \leq t_2,$$

in the *configuration space* $\mathbf{R}^{3N}$ be given. The *action* of the curve is by definition

$$\mathcal{S}(\mathbf{P}) = \int_{t_1}^{t_2} dt L(\mathbf{x}_1(t), \dots, \mathbf{x}_N(t), \mathbf{x}'_1(t), \dots, \mathbf{x}'_N(t)).$$

The action principle says that the path, $\mathbf{P}(t)$, traced out in configuration space by a system of mass-points under the influence of conservative forces, is the one that is *stationary* for the action. By definition, this means that for all curves $\mathbf{q}(t)$ whose size is of order one, $\|\mathbf{q}\| = \mathcal{O}(1)$, we have

$$\mathcal{S}(\mathbf{P} + \epsilon \mathbf{q}) = \mathcal{S}(\mathbf{P}) + \mathcal{O}(\epsilon^2).$$

This approach to dynamics is very different from the usual one where we solve Newton's equations subject to given initial conditions. We will show that they are in fact equivalent. The action principle is also called *Hamilton's principle*.

The action principle is the single most important idea in theoretical physics. *All* fundamental physical models are derived from the action principle, this is true for both classical physics and quantum physics.

1.9 The Maximum Entropy Principle

Let $x_1, \dots, x_n$ be random variables with an associated probability distribution $\rho(x_1, \dots, x_n)$. Let $f_1(x_1, \dots, x_n), \dots, f_p(x_1, \dots, x_n)$ be functions defined on the space of random variables $\mathbf{R}^n$. The functions f_j are our *observables*. Their *expectation* values are as usual defined by

$$\langle f_j \rangle = \int_{\mathbf{R}^n} dV f_j(x_1, \dots, x_n) \rho(x_1, \dots, x_n).$$

The expectation value of a given observable of course depends on which probability distribution ρ , we use. The challenge in statistics is to figure out which probability distribution one should use in any given situation. Let us say that we for some reason, (guesswork, hearsay, ...) believe that a probability distribution ρ_0 accurately represents what we currently know about a given system. The probability distribution ρ_0 is called the *prior distribution*, or just the *prior*.

Let us next assume that we measure the mean values of the observables $f_1, \dots, f_p$ and find the values $c_1, \dots, c_p$. If

$$\langle f_j \rangle_0 = \int_{\mathbf{R}^n} dV f_j(x_1, \dots, x_n) \rho_0(x_1, \dots, x_n) = c_j,$$

for $j = 1, \dots, p$, we are satisfied with our choice of prior. It predicts exactly the mean values that are observed.

But we might not be so lucky. Perhaps

$$\langle f_j \rangle_0 \neq c_j,$$

for at least one j . Our selected ρ_0 is then not the correct one, it predicts expectation values that are not observed. The challenge is to modify ρ_0 into a new distribution ρ that is consistent with *all* the observed mean values.

For this purpose we define a functional $S(\rho)$ by

$$S(\rho) = - \int_{\mathbf{R}^n} dV \rho \ln \left(\frac{\rho}{\rho_0} \right).$$

S is by definition the *relative entropy* of the probability distribution ρ with respect to ρ_0 . We will see later that our use of the word entropy here is consistent with its usage in thermodynamics.

The *maximum entropy* principle states that one should choose the probability distribution that maximizes the functional

$$S(\rho) = - \int_{\mathbf{R}^n} dV \rho \ln \left(\frac{\rho}{\rho_0} \right),$$

subject to the constraints

$$\langle f_j \rangle = \int_{\mathbf{R}^n} dV f_j \rho = c_j.$$

2 The Euler-Lagrange Equations

From examples given in the previous subsections, it should be clear that extremal problems for functionals play an important role in our description of nature. It is now time to find a way to solve such problems.

2.1 One dependent variable

Several of the examples involved a functional of the form

$$T(y) = \int_{t_0}^{t_1} dt L(t, y, y'), \quad (10)$$

with constraints of the form

$$\begin{aligned} y(t_0) &= y_0, \\ y(t_1) &= y_1. \end{aligned} \quad (11)$$

Note that the integrand, defining the functional (10), is in the context of the calculus of variation sometimes called an *integral density*, but more often a *Lagrangian*. The inspiration for the second name came originally from applications of the calculus of variation to mechanical systems. Joseph-Louis Lagrange reshaped the subject of particle mechanics in the 1700's and has had a large influence on how we today think about the subject of mechanics. The name Lagrangian for the integral densities defining functionals has subsequently migrated to field theory, which forms the foundation for fundamental physics in general, and particle physics in particular. In these subjects, the functionals of interest are time integrals over a Lagrangian, which itself is the space integral over a function, which in this context is called a *Lagrangian density*. Theories in fundamental physics are *defined* in terms of Lagrangian densities.

In order to be consistent with the standard usage of the terms Lagrangian and Lagrangian density, we call the integrand in one of our functional for the Lagrangian, with symbol L , if the integral is over time, and Lagrangian density, with symbol $\mathcal{L}$, if the functional is defined by an integral over space and time, or over space alone.

Our challenge is to find a function $y(x)$ that satisfies the constraints (11) and which is extremal for (10).

Let us for a moment return to ordinary calculus. Let $f(x)$ be a function of a real variable, and let $x = x_0$ be some fixed value of x . The function $f(x)$ is by definition differentiable at $x = x_0$ if for $\epsilon \ll 1$ and for all h of order one, we have

$$f(x_0 + \epsilon h) = f(x_0) + \epsilon A(x_0) h + \mathcal{O}(\epsilon^2). \quad (12)$$

$A(x_0)$ is by definition the derivative of f at $x = x_0$. The point x_0 is *stationary* for $f(x)$ if

$$f'(x_0) \equiv A(x_0) = 0. \quad (13)$$

Thus, at a stationary point we have

$$f(x_0 + \epsilon h) = f(x_0) + \mathcal{O}(\epsilon^2). \quad (14)$$

We know that for a differentiable function defined on the real line, all extremal points of a function $f(x)$ are found among the list of stationary points. Recall that extremal points are local or global maximum points or minimum points.

Inspired by (12), (13) and (14) we say that a functional T is differentiable at y if for $\epsilon \ll 1$ and all $\eta(x)$ of order one, we have

$$T(y + \epsilon \eta) = T(y) + \epsilon A(y, \eta) + \mathcal{O}(\epsilon^2), \quad (15)$$

where the map

$$\eta \rightarrow A(y, \eta),$$

is *linear* in η . This linear map is called the *variational derivative* of T at $y(x)$, and in these lecture notes we denote it by $\delta T(y)$, thus

$$\delta T(y)(\eta) = A(y, \eta).$$

The function $y = y(t)$ is stationary for $T(y)$ iff (if and only if)

$$\begin{aligned} \delta T(y) &= 0, \\ \Updownarrow \\ \delta T(y)(\eta) &= 0 \quad \forall \eta(x). \end{aligned}$$

Thus at a stationary point we have

$$T(y + \epsilon \eta) = T(y) + \mathcal{O}(\epsilon^2) \quad \forall \eta(x).$$

Just like in ordinary calculus, extremal points for a functional are found among the list of stationary points. In the calculus of variation, the function $y + \epsilon \eta$ is for any given function η called a *variation* of the function y .

Let us now calculate the variational derivative of the functional (10).

$$\begin{aligned} T(y + \epsilon \eta) &= \int_{t_0}^{t_1} dt L(t, y + \epsilon \eta, y' + \epsilon \eta') \\ &= \int_{t_0}^{t_1} dt \left\{ L(t, y, y') + \frac{\partial L}{\partial y} \epsilon \eta + \frac{\partial L}{\partial y'} \epsilon \eta' + \mathcal{O}(\epsilon^2) \right\} \\ &= T(y) + \epsilon \int_{t_0}^{t_1} dt \left\{ \frac{\partial L}{\partial y} \eta + \frac{\partial L}{\partial y'} \eta' \right\} + \mathcal{O}(\epsilon^2). \end{aligned} \quad (16)$$

The functions $y(t)$ and $y(t) + \epsilon \eta(t)$ must satisfy the constraints (11), only such functions are relevant. Thus we have

$$\begin{aligned} y(t_0) + \epsilon \eta(t_0) &= y_0 = y(t_0) \Rightarrow \eta(t_0) = 0, \\ y(t_1) + \epsilon \eta(t_1) &= y_1 = y(t_1) \Rightarrow \eta(t_1) = 0. \end{aligned} \quad (17)$$

Using integration by parts and (17) we have

$$\begin{aligned} \int_{t_0}^{t_1} dt \frac{\partial L}{\partial y'} \eta' &= \frac{\partial L}{\partial y'} \eta \Big|_{t_0}^{t_1} - \int_{t_0}^{t_1} dt \frac{d}{dt} \left(\frac{\partial L}{\partial y'} \right) \eta \\ &= - \int_{t_0}^{t_1} dt \frac{d}{dt} \left(\frac{\partial L}{\partial y'} \right) \eta. \end{aligned} \quad (18)$$

Inserting (18) into (16) we get

$$T(y + \epsilon \eta) = T(y) + \epsilon \int_{t_0}^{t_1} dt \left\{ \frac{\partial L}{\partial y} - \frac{d}{dt} \left(\frac{\partial L}{\partial y'} \right) \right\} \eta + \mathcal{O}(\epsilon^2) \quad \forall \eta(t).$$

Thus, by the definition of the variational derivative (15), we have

$$\delta T(y)(\eta) = \int_{t_0}^{t_1} dt \left\{ \frac{\partial L}{\partial y} - \frac{d}{dt} \left(\frac{\partial L}{\partial y'} \right) \right\} \eta.$$

We have the following result; the so-called the fundamental lemma of variational calculus.

lemma

Let $a(t)$ be a continuous function and assume that

$$\int_{t_0}^{t_1} dt a(t) \eta(t) = 0, \quad (19)$$

for *all* continuous functions η . Then

$$a(t) = 0 \quad t_0 \leq t \leq t_1.$$

If we apply the lemma (19) we find that $y(t)$ is a stationary point for T iff $y(t)$ satisfies the equation

$$\frac{\partial L}{\partial y} - \frac{d}{dt} \left(\frac{\partial L}{\partial y'} \right) = 0.$$

This equation is called the *Euler-Lagrange* equation.

Example 1. In section 1.4 we discussed the problem of finding a curve which, upon rotation around the x-axis, generates a surface of minimal area.

We observed that such a curve, $y(x)$, minimize the functional

$$A(y) = 2 \pi \int_{x_1}^{x_2} dx y(x) \sqrt{1 + y'(x)^2},$$

subject to the constraints

$$\begin{aligned} y(x_1) &= y_1, \\ y(x_2) &= y_2. \end{aligned}$$

Just as in ordinary calculus the minimum will be found among the stationary points for A .

Let us therefore find the Euler-Lagrange equation for the functional A . For A the Lagrangian density is given by

$$\mathcal{L} = 2 \pi y \sqrt{1 + y'^2}. \quad (20)$$

Thus

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial y} &= 2 \pi \sqrt{1 + y'^2}, \\ \frac{\partial \mathcal{L}}{\partial y'} &= 2 \pi y \frac{y'}{\sqrt{1 + y'^2}}, \end{aligned}$$

and therefore the Euler-Lagrange equation is

$$2 \pi \sqrt{1 + y'^2} - \frac{d}{dt} (2 \pi y \frac{y'}{\sqrt{1 + y'^2}}) = 0. \quad (21)$$

Any stationary point $y(x)$ is a solution to this equation. Equation (21) is a highly nonlinear second order differential equation. In fact, the Euler-Lagrange equation will always be second order for functionals of the form (10).

Note that the Lagrange density $\mathcal{L}$, in formula (20), defining the functional (10), is not totally general since it does not depend explicitly on x .

For such Lagrangians densities we have in general

$$\begin{aligned} \frac{d}{dx} (y' \frac{\partial \mathcal{L}}{\partial y'} - \mathcal{L}) &= y'' \frac{\partial \mathcal{L}}{\partial y'} + y' \frac{d}{dx} (\frac{\partial \mathcal{L}}{\partial y'}) \\ - \frac{\partial \mathcal{L}}{\partial y} y' - \frac{\partial \mathcal{L}}{\partial y'} y'' &= y' (\frac{d}{dx} (\frac{\partial \mathcal{L}}{\partial y'}) - \frac{\partial \mathcal{L}}{\partial y'}) = 0, \end{aligned}$$

where we in the last step used the Euler-Lagrange equation.

This calculation shows that for an $\mathcal{L}$ that does not explicitly depends on x , any extremal of A is a solution to the following *first* order differential equation

$$y' \frac{\partial \mathcal{L}}{\partial y'} - \mathcal{L} = c, \quad (22)$$

where c is a constant. The reduction of order for such special Lagrangian densities is not just a lucky break. Behind this result, there is a hugely important mathematical machine called *Noether's theorem*. It will give similar reductions of order in many other unrelated situations.

Noether's theorem has for almost a century been at the center of the action in theoretical physics, both quantum and classical. In quantum theory Noether's theorem, in the form of *the gauge principle*, is used to derive *all* the fundamental equations in elementary particle physics. These equations, taken together, form what is modestly called *the Standard Model*. This model has predicted the outcome of *all* experiments in fundamental physics since the 1970's. It is the most accurate theory of nature ever created, and it all flows from Noether's theorem.

Using (22) for the Lagrangian density (20), we get

$$\begin{aligned} y' 2 \pi y \frac{y'}{\sqrt{1 + y'^2}} - 2 \pi y \sqrt{1 + y'^2} &= c, \\ \Downarrow \\ - \frac{y}{\sqrt{1 + y'^2}} &= c_1 \equiv \frac{c}{2 \pi}. \end{aligned} \quad (23)$$

Since $y(x)$ from section 1.4 is positive, equation (23) can have solutions only if $c_1 < 0$. We can therefore, without loss of generality, write $c_1 = -\alpha$, where

$\alpha > 0$. Using this we have

$$\begin{aligned}
\frac{y}{\sqrt{1+y'^2}} &= \alpha, \\
&\Downarrow \\
y^2 &= \alpha^2 (1+y'^2), \\
&\Downarrow \\
y'^2 &= \left(\frac{y}{\alpha}\right)^2 - 1, \\
&\Downarrow \\
y' &= \pm \sqrt{\left(\frac{y}{\alpha}\right)^2 - 1}.
\end{aligned} \tag{24}$$

Equation (24) is separable and can be solved. The general solution is

$$y(x) = \alpha \cosh\left(\frac{x}{\alpha} + \beta\right) \quad \alpha > 0, \quad \beta \in \mathbf{R}, \tag{25}$$

where the constants α and β are determined from the conditions

$$y(x_1) = y_1, \quad y(x_2) = y_2.$$

A graph of functions of the type (25) is called a *Catenary*.

By construction, functions of the type (25) are stationary points for the area functional $A(y)$. In ordinary calculus we use the second derivative test in order to decide if a stationary point is a local maximum or a local minimum. Global maximum or minimum will be found among the local maximums or minimums *or* at points where the function is singular (not differentiable). This is true assuming there are no boundary points. If there are boundary points, maximum and minimum can occur there also.

The same rules apply here in the *calculus of variation*, which is the official name for what we are doing. We will however not develop the theory further in this direction and will not discuss the second variational derivative.

The problem of deciding whether a given stationary point gives a global minimum or a global maximum must be investigated in each separate case. Solving this problem can be highly non-trivial.

For the stationary points (25), it is easy to verify that a Catenary passing through the point (x_1, y_1) is given by

$$y(x) = \alpha \cosh\left(\frac{x - x_1}{\alpha} + \cosh^{-1}\left(\frac{y_1}{\alpha}\right)\right).$$

In order to make sure that $y(x)$ also pass through the point (x_2, y_2) we must find an α such that

$$\alpha \cosh\left(\frac{x_2 - x_1}{\alpha} + \cosh^{-1}\left(\frac{y_1}{\alpha}\right)\right) = y_2.$$

This is a transcendental equation for α and depending on coordinates, $(x_1, y_1), (x_2, y_2)$. It might have no positive solutions or several positive solutions.

The following cases are known to occur

- i) There exists no Catenary connecting the points (x_1, y_1) and (x_2, y_2) ,
- ii) There exists exactly one Catenary connecting (x_1, y_1) and (x_2, y_2) ,
- iii) There exists exactly two Catenaries connecting (x_1, y_1) and (x_2, y_2) .

In case **ii)** the unique Catenary is neither a global nor local minimum. In case **iii)** one of the solutions is a local minimum, and this is also a global minimum, if y_1 is large enough in a certain sense. Thus a surface of revolution of minimum area exists only in case **iii)** and only if y_1 is large enough.

Clearly, the Euler-Lagrange equation only gives stationary points and these points are in general not even *local* maximum or minimum. However, recall that in some cases, like the action principle, we are only looking for stationary points.

Example 2. The Brachistochrone, from section 1.7, is defined to be the minimum of the functional

$$T(y) = \frac{1}{\sqrt{2g}} \int_{x_1}^{x_2} dx \left(\frac{1+y'^2}{y-y_0} \right)^{\frac{1}{2}}.$$

Thus, the Lagrangian density for this problem is

$$\mathcal{L} = \frac{1}{\sqrt{2g}} (1+y'^2)^{\frac{1}{2}} (y-y_0)^{-\frac{1}{2}}. \quad (26)$$

Observe that the Lagrangian density(26) does not depend explicitly on the independent variable x . Therefore, using the result from (22), we know that any extremal of the functional T is a solution to the following first order ODE

$$y' \frac{\partial \mathcal{L}}{\partial y'} - \mathcal{L} = c, \quad (27)$$

Differentiating $\mathcal{L}$ we get

$$\frac{\partial \mathcal{L}}{\partial y'} = \frac{1}{\sqrt{2g}} y' (1+y'^2)^{-\frac{1}{2}} (y-y_0)^{-\frac{1}{2}}, \quad (28)$$

and inserting (28) into (27) we get

$$\begin{aligned} y' \frac{\partial \mathcal{L}}{\partial y'} - \mathcal{L} &= c, \\ \Downarrow \\ y'^2 (1+y'^2)^{-\frac{1}{2}} (y-y_0)^{-\frac{1}{2}} - (1+y'^2)^{\frac{1}{2}} (y-y_0)^{-\frac{1}{2}} &= c_1 \equiv c \sqrt{2g}, \\ \Downarrow \\ y'^2 &= 1 + y'^2 + c_1 (1+y'^2)^{\frac{1}{2}} (y-y_0)^{\frac{1}{2}}, \\ \Downarrow \\ c_1 (1+y'^2)^{\frac{1}{2}} (y-y_0)^{\frac{1}{2}} &= -1. \end{aligned} \quad (29)$$

For there to be a solution we must have $c_1 < 0$. Let $c_1 = -\beta$, $\beta > 0$. From (29) we then get

$$\begin{aligned}
(1 + y'^2)^{\frac{1}{2}} (y - y_0)^{\frac{1}{2}} &= \frac{1}{\beta}, \\
&\Downarrow \\
1 + y'^2 &= \frac{1}{\beta^2 (y - y_0)}, \\
&\Downarrow \\
y'^2 &= \frac{1 - \beta^2 (y - y_0)}{\beta^2 (y - y_0)}, \\
&\Downarrow \\
y' &= \left(\frac{\alpha^2 - (y - y_0)}{y - y_0} \right)^{\frac{1}{2}} \quad \alpha^2 = \frac{1}{\beta^2} > 0, \quad (30)
\end{aligned}$$

where we have used the positive root since we know that for the current example $y' > 0$.

The equation (30) is separable, and we get the following implicit solution

$$\int dy \left(\frac{y - y_0}{\alpha^2 - (y - y_0)} \right)^{\frac{1}{2}} = x + c, \quad (31)$$

where c is an arbitrary integration constant.

The integral in (31) can be solved using a trigonometric substitution

$$y - y_0 = \alpha^2 \sin^2 \frac{\theta}{2}. \quad (32)$$

Inserting (32) into (31) give us

$$\begin{aligned}
\alpha^2 \int d\theta \sin^2 \frac{\theta}{2} &= x + c, \\
&\Downarrow \\
\frac{1}{2} \alpha^2 (\theta - \sin \theta) &= x + c.
\end{aligned}$$

Thus our solution is

$$\begin{aligned}
x &= \frac{1}{2} \alpha^2 (\theta - \sin \theta) - c, \\
y &= y_0 + \frac{1}{2} \alpha^2 (1 - \cos \theta).
\end{aligned}$$

This is a parametric representation of a type of curve called a *cycloid*. One can show that there is a unique cycloid passing through any pair of points (x_1, y_1) , (x_2, y_2) with $y_1 < y_2$ and that the unique cycloid is a global minimum for the propagation time functional $T(y)$.

2.2 Several dependent variables

Several of our examples involved functionals of the general form

$$T(y_1, \dots, y_n) = \int_{t_0}^{t_1} dt L(t, y_1, \dots, y_n, y'_1, \dots, y'_n).$$

By analogy with multi-variable calculus, we consider independent variations $y_i + \epsilon \eta_i$ for all the functions $\{y_i(x)\}_{i=1}^n$. Proceeding like for the case of one function, we get

$$\begin{aligned}
& T(y_1 + \epsilon \eta_1, \dots, y_n + \epsilon \eta_n) \\
&= \int_{t_0}^{t_1} dt L(t_1, y_1 + \epsilon \eta_1, \dots, y_n + \epsilon \eta_n, y'_1 + \epsilon \eta'_1, \dots, y'_n + \epsilon \eta'_n) \\
&= \int_{t_0}^{t_1} dt \left\{ L(t_1, y_1, \dots, y_n, y'_1, \dots, y'_n) + \frac{\partial L}{\partial y_1} \epsilon \eta_1 \right. \\
&\quad + \dots + \frac{\partial L}{\partial y_n} \epsilon \eta_n + \frac{\partial L}{\partial y'_1} \epsilon \eta'_1 + \dots + \frac{\partial L}{\partial y'_n} \epsilon \eta'_n \left. \right\} \\
&\quad + \mathcal{O}(\epsilon^2) \\
&= T(y_1, \dots, y_n) \\
&\quad + \epsilon \int_{t_0}^{t_1} dt \left\{ \left(\frac{\partial L}{\partial y_1} - \frac{d}{dt} \left(\frac{\partial L}{\partial y'_1} \right) \right) \eta_1 + \dots + \left(\frac{\partial L}{\partial y_n} - \frac{d}{dt} \left(\frac{\partial L}{\partial y'_n} \right) \right) \eta_n \right\} + \mathcal{O}(\epsilon^2),
\end{aligned}$$

where we in the last step has used integration by parts and the fact that

$$\eta_j(t_0) = \eta_j(t_1) = 0 \quad j = 1, 2, \dots, n, \quad (33)$$

The relations (33) follows from the fact we have constraints

$$y_j(t_0) = y_j^0, \quad y_j(t_1) = y_j^1,$$

by an argument that is identical to the one in equation (17). A stationary point for the functional $T(y_1, \dots, y_n)$ is determined by the condition that

$$\begin{aligned}
& \int_{t_0}^{t_1} dt \left\{ \left(\frac{\partial L}{\partial y_1} - \frac{d}{dt} \left(\frac{\partial L}{\partial y'_1} \right) \right) \eta_1 + \dots + \left(\frac{\partial L}{\partial y_n} - \frac{d}{dt} \left(\frac{\partial L}{\partial y'_n} \right) \right) \eta_n \right\} \\
&= 0,
\end{aligned}$$

for *all* functions $\{\eta_i\}_{i=1}^n$. Since there by assumption are no dependencies among the functions $\eta_i(t)$, we conclude, using the fundamental lemma (19), that $(y_1, \dots, y_n)$ is a stationary point for $T(y_1, \dots, y_n)$ only if

$$\frac{\partial L}{\partial y_i} - \frac{d}{dt} \left(\frac{\partial L}{\partial y'_i} \right) = 0 \quad i = 1, 2, \dots, n. \quad (34)$$

These are the Euler-Lagrange equations for the functional $T(y_1, \dots, y_n)$. Observe that, in general, (34) are n coupled non-linear second order differential equations.

Example 3. In 1.1. we found that the problem of finding the shortest curve in the plane, connecting two fixed points $p = (x_0, y_0)$ and $q = (x_1, y_1)$, amounted to minimizing the functional

$$T(x, y) = \int_0^1 dt \sqrt{x'^2 + y'^2},$$

subject to the constraints

$$(x(0), y(0)) = p, \quad (x(1), y(1)) = q.$$

The Lagrangian density is

$$\mathcal{L}(x, y) = \sqrt{x'^2 + y'^2}.$$

Observe that

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial x} &= 0, & \frac{\partial \mathcal{L}}{\partial y} &= 0, \\ \frac{\partial \mathcal{L}}{\partial x'} &= \frac{x'}{\sqrt{x'^2 + y'^2}}, & \frac{\partial \mathcal{L}}{\partial y'} &= \frac{y'}{\sqrt{x'^2 + y'^2}}. \end{aligned}$$

Thus the Euler-Lagrange equations are

$$\begin{aligned} \frac{d}{dt} \left(\frac{x'}{\sqrt{x'^2 + y'^2}} \right) &= 0, \\ &\Downarrow \\ \frac{x'}{\sqrt{x'^2 + y'^2}} &= c_1, \\ \frac{d}{dt} \left(\frac{y'}{\sqrt{x'^2 + y'^2}} \right) &= 0, \\ &\Downarrow \\ \frac{y'}{\sqrt{x'^2 + y'^2}} &= c_2. \end{aligned} \tag{35}$$

From (35) we find that

$$\begin{aligned} \frac{dy}{dx} &= \frac{\frac{dy}{dt}}{\frac{dx}{dt}} = \frac{c_2}{c_1}, \\ &\Downarrow \\ y &= \frac{c_2}{c_1} x + c_3, \end{aligned}$$

and the curve passing through p and q is

$$y(x) = \frac{y_1 - y_0}{x_1 - x_0} (x - x_0) + y_0.$$

This is a straight line and is evidently the curve of minimal length. We of course already knew that the straight line is the shortest curve connecting two points in the plane.

In 1.2 we asked the same question for two points on a surface in $\mathbf{R}^3$. Here the answer is not obvious, a straight line in $\mathbf{R}^3$ will not work unless S is a plane. We are in some sense seeking a curve on a general curved surface that is the analogy of straight lines in $\mathbf{R}^3$.

When the family of such curves has been found we can use them to construct analogues of triangles, squares etc., on the curved surface and ask geometrical questions like: What is the sum of the internal angles of a triangle on a given surface S ? We can in fact develop a whole analogue to Euclidean geometry for plane figures, for any surface, not only plane surfaces.

This has been done for many kind of surfaces and is of obvious practical importance for the case of a sphere. (Fuel efficient long distance transport)

Example 4. In section 1.6 we introduced the Fermat principle determining the light-rays in a material of variable refractive index.

The functional was

$$T(x, y, z) = \int_0^1 dt \, n(x(t), y(t), z(t)) \sqrt{x'^2(t) + y'^2(t) + z'^2(t)}.$$

The Lagrangian is

$$L = n(x, y, z) \sqrt{x'^2 + y'^2 + z'^2} = n \|\mathbf{x}'\|$$

Observe that

$$\begin{aligned} \frac{\partial L}{\partial x} &= \partial_x n \|\mathbf{x}'\|, & \frac{\partial L}{\partial x'} &= \frac{n x'}{\|\mathbf{x}'\|}, \\ \frac{\partial L}{\partial y} &= \partial_y n \|\mathbf{x}'\|, & \frac{\partial L}{\partial y'} &= \frac{n y'}{\|\mathbf{x}'\|}, \\ \frac{\partial L}{\partial z} &= \partial_z n \|\mathbf{x}'\|, & \frac{\partial L}{\partial z'} &= \frac{n z'}{\|\mathbf{x}'\|}. \end{aligned}$$

The Euler-Lagrange equations are

$$\begin{aligned} \partial_x n \|\mathbf{x}'\| - \frac{d}{dt} \left(\frac{n x'}{\|\mathbf{x}'\|} \right) &= 0, \\ \partial_y n \|\mathbf{x}'\| - \frac{d}{dt} \left(\frac{n y'}{\|\mathbf{x}'\|} \right) &= 0, \\ \partial_z n \|\mathbf{x}'\| - \frac{d}{dt} \left(\frac{n z'}{\|\mathbf{x}'\|} \right) &= 0. \end{aligned} \tag{36}$$

Consider the case of a homogeneous medium where $n(x, y, z) = n_0$. For this case (36) simplifies into

$$\begin{aligned} \frac{d}{dt} \left(\frac{x'}{\|\mathbf{x}'\|} \right) &= 0, \\ &\Downarrow \\ \frac{dx}{dt} &= c_1 \|\mathbf{x}'\|, \\ \frac{d}{dt} \left(\frac{y'}{\|\mathbf{x}'\|} \right) &= 0, \\ &\Downarrow \\ \frac{dy}{dt} &= c_2 \|\mathbf{x}'\|, \\ \frac{d}{dt} \left(\frac{z'}{\|\mathbf{x}'\|} \right) &= 0, \\ &\Downarrow \\ \frac{dz}{dt} &= c_3 \|\mathbf{x}'\|. \end{aligned}$$

Using x as a new independent variables assuming $c_1 \neq 0$ we get in the usual way

$$\begin{aligned}\frac{dy}{dx} &= \frac{\frac{dy}{dt}}{\frac{dx}{dt}} = \frac{c_2}{c_1}, \\ &\Downarrow \\ y &= \frac{c_2}{c_1} x + c_4, \\ \frac{dz}{dx} &= \frac{\frac{dz}{dt}}{\frac{dx}{dt}} = \frac{c_3}{c_1}, \\ &\Downarrow \\ z &= \frac{c_3}{c_1} x + c_5.\end{aligned}$$

Thus the light-rays in a homogeneous medium are straight lines. We of course know this from elementary physics.

For the general case, we introduce path-length as a new parameter

$$s(t) = \int_{t_0}^t dt' \|\mathbf{x}'(t')\|.$$

Using this as our curve parameter we have, using the chain rule, that

$$\frac{d}{dt} = \|\mathbf{x}'\| \frac{d}{ds}.$$

Thus we have

$$\begin{aligned}\partial_x n \|\mathbf{x}'\| - \frac{d}{dt} \left(\frac{n x'}{\|\mathbf{x}'\|} \right) &= 0, \\ &\Downarrow \\ \partial_x n \|\mathbf{x}'\| - \frac{d}{dt} \left(\frac{n}{\|\mathbf{x}'\|} \frac{dx}{dt} \right) &= 0, \\ &\Downarrow \\ \partial_x n \|\mathbf{x}'\| - \|\mathbf{x}'\| \frac{d}{ds} \left(\frac{n}{\|\mathbf{x}'\|} \|\mathbf{x}'\| \frac{d}{ds} \right) &= 0, \\ &\Downarrow \\ \partial_x n \|\mathbf{x}'\| - \|\mathbf{x}'\| \frac{d}{ds} \left(n \frac{dx}{ds} \right) &= 0, \\ &\Downarrow \\ \partial_x n - \frac{d}{ds} \left(n \frac{dx}{ds} \right) &= 0.\end{aligned}$$

We rewrite the other two Euler-Lagrange equations in the same way. Collecting the three scalar equations into one vector equation we have

$$\frac{d}{ds} \left(n \frac{d\mathbf{x}}{ds} \right) = \nabla n. \quad (37)$$

This is the fundamental equation for *ray optics*.

We are not going to solve this equation but will make a general observation.

Since we are using path-length parametrization, the tangent vector

$$\mathbf{T} = \frac{d\mathbf{x}}{ds},$$

is a *unit* vector. This means that

$$\begin{aligned} \mathbf{T} \cdot \mathbf{T} &= 1, \\ \Downarrow \\ \frac{d\mathbf{T}}{ds} \cdot \mathbf{T} &= 0, \\ \Downarrow \\ \alpha \cdot \mathbf{T} &= 0, \end{aligned}$$

where $\alpha = \frac{d\mathbf{T}}{ds}$ is the *curvature vector* for the curve $\mathbf{x}(s)$. It is normal to the curve and points in the direction in which the curve bends

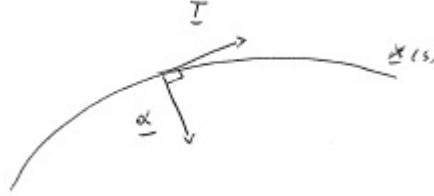


Figure 12: Unit tangent $\mathbf{T}$ and curvature vector α , for a light ray.

From the fundamental equation for ray optics (37) we have

$$\begin{aligned} \frac{dn}{ds} \mathbf{T} + n \frac{d\mathbf{T}}{ds} &= \nabla n, \\ \Downarrow \\ \alpha &= \frac{\nabla n}{n} - \frac{d}{ds}(\ln n) \mathbf{T}, \\ \Downarrow \\ 0 < \alpha \cdot \alpha &= \frac{1}{n} \alpha \cdot \nabla n, \\ \Downarrow \\ \alpha \cdot \nabla n &> 0. \end{aligned}$$

Thus the curvature vector for the light-ray points in the direction of increasing refractive index. This means that light-rays in an inhomogeneous medium always curve towards regions of higher refractive index. This fact has many important physical consequences. Hot air close to the ground is less dense and has lower refractive index than the higher colder air. The light coming from the sky will therefore bend upwards, The sky will appear as "water" on the ground to an observer. This is illustrated in figure 13.

Another consequence of practical importance is that light rays will be confined to move within a region of higher refractive index. Optical fibers, which form the physical backbone of the internet, work because of this fact.

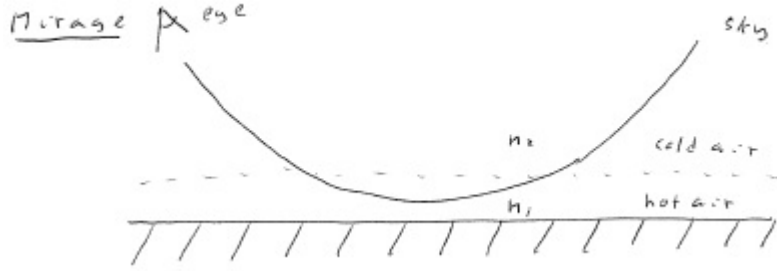


Figure 13: Explaining the occurrence of a Mirage using ray optics.

Example 5. In 1.8. we introduced the action principle for a system of mass-points moving under the influence of conservative forces.

The action functional was

$$\mathcal{S}(\mathbf{x}_1, \dots, \mathbf{x}_N) = \int_{t_0}^{t_1} dt \{T(\mathbf{x}'_1(t), \dots, \mathbf{x}'_N(t)) - V(\mathbf{x}_1(t), \dots, \mathbf{x}_N(t))\}, \quad (38)$$

where T is the kinetic energy of the system of mass-points and V is the potential determining the conservative forces.

The Lagrangian is thus

$$L = \frac{1}{2} \sum_{i=1}^n m_i \mathbf{x}'_i{}^2 - V(\mathbf{x}_1, \dots, \mathbf{x}_n). \quad (39)$$

Observe that

$$\frac{\partial L}{\partial \mathbf{x}_i} = -\frac{\partial V}{\partial \mathbf{x}_i}, \quad \frac{\partial L}{\partial \mathbf{x}'_i} = m_i \mathbf{x}'_i,$$

and thus the Euler-Lagrange equations are

$$\begin{aligned} \frac{\partial L}{\partial \mathbf{x}_i} - \frac{d}{dt} \left(\frac{\partial L}{\partial \mathbf{x}'_i} \right) &= 0 & i = 1, \dots, n, \\ \Downarrow \\ -\frac{\partial V}{\partial \mathbf{x}_i} - m_i \mathbf{x}''_i &= 0, \\ \Downarrow \\ m_i \mathbf{x}''_i &= -\frac{\partial V}{\partial \mathbf{x}_i} \equiv \mathbf{f}_i. \end{aligned}$$

This is Newton's law for n mass-points, m_i , moving under the influence of conservative forces

$$\mathbf{f}_i = -\frac{\partial V}{\partial \mathbf{x}_i}.$$

Note that the Lagrangian (39) does not depend explicitly on time. Let

$$E = \sum_{i=1}^n \mathbf{x}'_i \cdot \frac{\partial L}{\partial \mathbf{x}'_i} - \mathcal{L}.$$

Then, using the Euler-Lagrange equation we have

$$\begin{aligned}
\frac{dE}{dt} &= \sum_{i=1}^n \left\{ \mathbf{x}_i'' \cdot \frac{\partial L}{\partial \mathbf{x}_i'} + \mathbf{x}_i' \cdot \frac{d}{dt} \left(\frac{\partial L}{\partial \mathbf{x}_i'} \right) \right\} \\
&\quad - \sum_{i=1}^n \left\{ \frac{\partial L}{\partial \mathbf{x}_i} \cdot \mathbf{x}_i' - \frac{\partial L}{\partial \mathbf{x}_i'} \cdot \mathbf{x}_i'' \right\} \\
&= \sum_{i=1}^n \mathbf{x}_i' \cdot \left\{ \frac{d}{dt} \left(\frac{\partial L}{\partial \mathbf{x}_i'} \right) - \frac{\partial L}{\partial \mathbf{x}_i} \right\} = 0,
\end{aligned}$$

and thus E is a constant of the motion for any solution of the Euler-Lagrange equation, and therefore for any stationary point for the action functional (38). Note that

$$\begin{aligned}
E &= \sum_{i=1}^n \mathbf{x}_i' \cdot (m_i \mathbf{x}_i') - \sum_{i=1}^n \frac{1}{2} m_i \mathbf{x}_i'^2 + V \\
&= \frac{1}{2} \sum_{i=1}^n m_i \mathbf{x}_i'^2 + V = T + V.
\end{aligned} \tag{40}$$

Thus, the conserved quantity is as expected the total energy of the system of mass-points. The function E in (40) is called the *Hamiltonian* for the system and is traditionally written using the letter H . Thus

$$\begin{aligned}
L &= T - V && \text{-Lagrangian,} \\
H &= T + V && \text{-Hamiltonian.}
\end{aligned}$$

As we have just seen, the action principle leads to Euler-Lagrange equations that are equivalent to Newton's law. Thus from this point of view, nothing appears to have been gained from using the action principle.

However, the action principle, when it is taken together with the theory of canonical transformations, is a much more efficient tool for solving mechanical problems than Newton's law.

Also, insights gained from the use of the action principle in mechanics played a major role in the development of quantum mechanics. Because of this, the action principle plays an important and ever increasing role in theoretical (and practical!) physics today.

2.3 Constraints

Let us consider the general problem of finding the extremum of a functional

$$T(y) = \int_{t_0}^{t_1} dt L(t, y, y'), \tag{41}$$

subject to the constraint

$$J(y) = \int_{t_0}^{t_1} dt j(t, y, y') = c. \tag{42}$$

We solve this problem using the Lagrange multiplier approach. Let us recall how this approach works for the case of function on $\mathbf{R}^n$.

For this case, $f : \mathbf{R}^n \rightarrow \mathbf{R}$, $g : \mathbf{R}^n \rightarrow \mathbf{R}$ are functions on $\mathbf{R}^n$. The challenge is to find a stationary point of f subject to the constraint

$$g(x_1, \dots, x_n) = c. \quad (43)$$

In calculus one proves that stationary points for the function f under the constraint (43), defined by the function g , can be found by introducing a new function

$$h(x_1, \dots, x_n) = f(x_1, \dots, x_n) + \lambda (g(x_1, \dots, x_n) - c),$$

and then seek an unconstrained stationary point for h . The parameter λ is called a *Lagrange multiplier*.

We thus solve the equation

$$\begin{aligned} \nabla h &= 0, \\ \Downarrow \\ \nabla(f + \lambda(g - c)) &= 0, \\ \Downarrow \\ \nabla f &= -\lambda \nabla g. \end{aligned}$$

The value for λ is chosen so that the constraint

$$g(x_1, \dots, x_n) = c.$$

is satisfied.

The exact same approach works for constrained variational problems. We will not prove this fact.

Thus in order to find a stationary point for the functional (41) subject to the constraint (42), we introduce the new functional

$$K(y) = T(y) + \lambda (J(y) - c),$$

and find the unconstrained stationary points for K . Using the notation δR for the variational derivatives of a functional R , we have

$$\begin{aligned} \delta K &= 0, \\ \Downarrow \\ \delta T + \lambda \delta J &= 0. \end{aligned} \quad (44)$$

The value of the Lagrange multiplier is chosen so that

$$J(y^*) = c,$$

where y^* is a solution of (44). Observe that we have

$$\begin{aligned} K(y) &= \int_{t_0}^{t_1} dt L(t, y, y') + \lambda \left(\int_{t_0}^{t_1} dt j(t, y, y') - c \right) \\ &= \int_{t_0}^{t_1} dt \{L(t, y, y') + \lambda (j(t, y, y') - \frac{c}{T})\}, \end{aligned}$$

where $T = t_1 - t_0$. Thus the Lagrangian for K is

$$L^* = L + \lambda \left(j - \frac{c}{T} \right).$$

Stationary points for the functional K are as usual solutions to the corresponding Euler-Lagrange equations

$$\frac{\partial L^*}{\partial y} - \frac{d}{dt} \left(\frac{\partial L^*}{\partial y'} \right) = 0.$$

The Euler-Lagrange equation is here a scalar ODE of second order, and therefore the general solution contains two integration constants. These two integration constants, together with λ , are determined by the following three constraints

$$\begin{aligned} y(t_0) &= y_0, \\ y(t_1) &= y_1, \\ J(y) &= c. \end{aligned}$$

This procedure can obviously be extended to any number of constraints and any number of dependent variables: If we want to find stationary points for the functional

$$T(y_1, \dots, y_n) = \int_{t_0}^{t_1} dt \, L(t, y_1, \dots, y_n, y'_1, \dots, y'_n),$$

subject to the constraints

$$J_p(y_1, \dots, y_n) = c_p \quad p = 1, \dots, m \quad ,$$

where

$$J_p(y_1, \dots, y_n) = \int_{t_0}^{t_1} dt \, j_p(t, y_1, \dots, y_n, y'_1, \dots, y'_n),$$

we solve the Euler-Lagrange equation for the functional

$$K(y_1, \dots, y_n) = T(y_1, \dots, y_n) + \sum_p \lambda_p (J_p(y_1, \dots, y_n) - c_p).$$

The Lagrangian corresponding to K is

$$L^* = L + \sum_p \lambda_p \left(j_p - \frac{c_p}{T} \right),$$

where $T = t_1 - t_0$. The $2n$ integration constants, together with the m Lagrange multipliers $\{\lambda_p\}_{p=1}^m$, are determined by the conditions

$$\left. \begin{aligned} y_i(t_0) &= y_i^0 \\ y_i(t_1) &= y_i^1 \end{aligned} \right\} i = 1, \dots, n \quad ,$$

$$J_p(y_1, \dots, y_n) = c_p \quad p = 1, \dots, m \quad .$$

Example 6. Let us consider the isoperimetric problem from section 1.3. The solution to this problem was there reduced to maximizing the functional

$$2 T = \int_0^1 dt (x(t) y'(t) - y(t) x'(t)),$$

subject to the constraint

$$J = \int_0^1 dt \sqrt{x'(t)^2 + y'(t)^2} = L^*.$$

We introduce the Lagrangian

$$L = x y' - y x' + \lambda (\sqrt{x'^2 + y'^2} - L^*),$$

and observe that

$$\begin{aligned} \frac{\partial L}{\partial x} &= y', & \frac{\partial L}{\partial y} &= -x', \\ \frac{\partial L}{\partial x'} &= -y + \frac{\lambda x'}{\sqrt{x'^2 + y'^2}}, \\ \frac{\partial L}{\partial y'} &= x + \frac{\lambda y'}{\sqrt{x'^2 + y'^2}}. \end{aligned}$$

The Euler-Lagrange equations are then

$$\begin{aligned} \frac{\partial L}{\partial x} - \frac{d}{dt} \left(\frac{\partial L}{\partial x'} \right) &= 0, \\ \Downarrow \\ y' - \frac{d}{dt} \left(-y + \frac{\lambda x'}{\sqrt{x'^2 + y'^2}} \right) &= 0, \end{aligned} \tag{45}$$

and

$$\begin{aligned} \frac{\partial L}{\partial y} - \frac{d}{dt} \left(\frac{\partial L}{\partial y'} \right) &= 0, \\ \Downarrow \\ -x' - \frac{d}{dt} \left(x + \frac{\lambda y'}{\sqrt{x'^2 + y'^2}} \right) &= 0. \end{aligned} \tag{46}$$

Equations (45) and (46) can be integrated once to yield

$$\begin{aligned} 2 y - \frac{\lambda x'}{\sqrt{x'^2 + y'^2}} &= c_1, \\ 2 x + \frac{\lambda y'}{\sqrt{x'^2 + y'^2}} &= c_2, \\ \Downarrow \\ y - \frac{c_1}{2} &= \frac{1}{2} \frac{\lambda x'}{\sqrt{x'^2 + y'^2}}, \\ x - \frac{c_2}{2} &= -\frac{1}{2} \frac{\lambda y'}{\sqrt{x'^2 + y'^2}}. \end{aligned} \tag{47}$$

Squaring and adding the two equations (47) we get

$$(y - \frac{c_1}{2})^2 + (x - \frac{c_2}{2})^2 = \frac{1}{4} \lambda^2, \quad (48)$$

which we recognize to be the equation for a circle. Thus all extremals are circles. The radius of the circle (48) is $R = \frac{1}{2} \lambda$. Thus the constraint is satisfied if

$$\begin{aligned} 2 \pi R &= L^*, \\ \Downarrow \\ \lambda &= \frac{L^*}{\pi}. \end{aligned}$$

2.4 Several independent variables

Consider a functional of the form

$$T(u) = \int_D dx dy \mathcal{L}(x, y, u, u_x, u_y).$$

The challenge is to find the stationary points for T subject to the constraint

$$u|_{\partial D} = f.$$

We proceed like before by introducing a variation

$$v = u + \epsilon \eta, \quad \eta = \eta(x, y),$$

where, as always, the η is assumed to be of order one, and ϵ is a small quantity.

Observe that since the boundary condition has to be fixed we have

$$v|_{\partial D} = u|_{\partial D} + \epsilon \eta|_{\partial D} = f \Rightarrow \eta|_{\partial D} = 0. \quad (49)$$

For the functional T we now have

$$\begin{aligned} T(u + \epsilon \eta) &= \int_D dx dy \mathcal{L}(x, y, u + \epsilon \eta, u_x + \epsilon \eta_x, u_y + \epsilon \eta_y) \\ &= \int_D dx dy \left\{ \mathcal{L}(x, y, u, u_x, u_y) + \frac{\partial \mathcal{L}}{\partial u} \epsilon \eta + \frac{\partial \mathcal{L}}{\partial u_x} \epsilon \eta_x \right. \\ &\quad \left. + \frac{\partial \mathcal{L}}{\partial u_y} \epsilon \eta_y \right\} + \mathcal{O}(\epsilon^2) \\ &= T(u) + \epsilon \int_D dx dy \left\{ \frac{\partial \mathcal{L}}{\partial u} \eta + \frac{\partial \mathcal{L}}{\partial u_x} \eta_x + \frac{\partial \mathcal{L}}{\partial u_y} \eta_y \right\} \\ &\quad + \mathcal{O}(\epsilon^2) \\ &= \mathbf{T}(u) + \epsilon \int_D dx dy \left\{ \frac{\partial \mathcal{L}}{\partial u} - \partial_x \left(\frac{\partial \mathcal{L}}{\partial u_x} \right) - \partial_y \left(\frac{\partial \mathcal{L}}{\partial u_y} \right) \right\} \eta \\ &\quad + \mathcal{O}(\epsilon^2), \end{aligned}$$

where we in the last step have used Green's theorem in divergence form and the boundary condition (49) on η .

Using the fundamental lemma we conclude that u is a stationary point for T iff it satisfies the following Euler-Lagrange equation.

$$\frac{\partial \mathcal{L}}{\partial u} - \partial_x \left(\frac{\partial \mathcal{L}}{\partial u_x} \right) - \partial_y \left(\frac{\partial \mathcal{L}}{\partial u_y} \right) = 0.$$

Example 7. In section 1.5 we introduced the notion of a minimal surface. We consider here the simplified situation where the surface is the graph of a function $u(x, y)$ over a domain D in the plane. The challenge is to minimize the functional

$$T(u) = \int_D dx dy \sqrt{1 + u_x^2 + u_y^2},$$

subject to the constraint

$$u|_{\partial D} = h(x, y).$$

The Lagrangian density is

$$\mathcal{L} = \sqrt{1 + u_x^2 + u_y^2},$$

and we have

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial u} &= 0, \\ \frac{\partial \mathcal{L}}{\partial u_x} &= \frac{u_x}{\sqrt{1 + u_x^2 + u_y^2}}, \\ \frac{\partial \mathcal{L}}{\partial u_y} &= \frac{u_y}{\sqrt{1 + u_x^2 + u_y^2}}. \end{aligned}$$

The Euler-Lagrange equation is thus

$$-\partial_x \frac{u_x}{\sqrt{1 + u_x^2 + u_y^2}} - \partial_y \frac{u_y}{\sqrt{1 + u_x^2 + u_y^2}} = 0,$$

which can be rewritten as

$$u_{xx} + u_{yy} = 2u_x u_y u_{xy} - u_x^2 u_{yy} - u_y^2 u_{xx} \quad (50)$$

The boundary condition for u is

$$u|_{\partial D} = h.$$

The equation (50) is a non-linear second order partial differential equation and is not by any means easy to solve.

However, if the boundary curve is horizontal

$$h(x, y) = h_0,$$

the boundary value problem clearly has the unique solution

$$u(x, y) = h_0, \quad (x, y) \in D.$$

This is flat and is obviously of minimum area among all surfaces with flat boundary curve $h = h_0$. If the boundary curve is not constant but varies little on the scale of h_0

$$h = h_0 + \epsilon k \quad \epsilon \ll 1,$$

we seek a solution that is a small modification of $u(x, y) = h_0$

$$u(x, y) = h_0 + \epsilon v \quad \epsilon \ll 1.$$

The function v must then satisfy the equation

$$v_{xx} + v_{yy} = \epsilon^2 (2v_x v_y v_{xy} - v_x^2 v_{yy} - v_y^2 v_{xx}).$$

This equation we can solve approximately using a perturbation expansion. Expansions of this type will be discussed later in these lecture notes.

Example 8. In section 1.9 we introduced the maximum entropy principle. The aim of this section is to derive the maximum entropy distribution by solving the corresponding Euler-Lagrange equation. It turns out that special cases of the resulting probability distribution form the foundation for statistical mechanics and thermodynamics, information theory and probably also elementary particle physics through its mathematical grounding in quantum field theory. The current section is an excerpt of a more detailed treatment of the maximum entropy principle given in an Appendix.

As you recall, the maximum entropy principle states that, if what we know about a system S , prior to a measurement, is described by a probability distribution ρ_0 , and we measure the mean c_j , of p observables f_j , for S , then the probability distribution that best incorporates this new information about the system, is the one that maximizes the functional

$$S(\rho) = - \int_{\mathbf{R}^n} dV \rho \ln \left(\frac{\rho}{\rho_0} \right), \quad (51)$$

under the constraints

$$\langle f_j \rangle = \int_{\mathbf{R}^n} dV f(x_1, \dots, x_n) \rho(x_1, \dots, x_n) = c_j.$$

Since ρ must be a probability distribution we have one more constraint

$$\langle 1 \rangle = \int_{\mathbf{R}^n} dV \rho(x_1, \dots, x_n) = 1,$$

and we thus have $p+1$ constraints and therefore introduce an extended functional

$$T(\rho) = S(\rho) + \lambda_0 \langle 1 \rangle + \sum_{j=1}^p \lambda_j \langle f_j \rangle$$

Note that we could have written

$$T(\rho) = S(\rho) + \lambda_0 (\langle 1 \rangle - 1) + \sum_{j=1}^p \lambda_j (\langle f_j \rangle - c_j),$$

in order to make the values of the constraints explicit, like we have done on previous occasions. However, all constant terms vanish when we take variational derivative, so we might as well drop them.

The Lagrangian density corresponding to the extended functional $T(\rho)$ is then

$$\mathcal{L} = -\rho \ln\left(\frac{\rho}{\rho_0}\right) + \lambda_0 \rho + \sum_{j=1}^p \lambda_j f_j \rho$$

Observe that $\mathcal{L}$ does not depend on any derivatives of ρ . The Euler-Lagrange equation for T is therefore simply

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial \rho} &= 0, \\ \Downarrow \\ -\ln\left(\frac{\rho}{\rho_0}\right) - 1 + \lambda_0 + \sum_{j=1}^p \lambda_j f_j &= 0, \\ \Downarrow \\ \rho &= \frac{\rho_0}{Z} \exp\left(\sum_j \lambda_j f_j\right), \end{aligned}$$

where we have defined $Z = \exp(1 - \lambda_0)$. In order for the constraint $\langle 1 \rangle = 1$ to be satisfied, we must have

$$\begin{aligned} \langle 1 \rangle &= 1, \\ \Downarrow \\ \int_{\mathbf{R}^n} dV \frac{\rho_0}{Z} \exp\left(\sum_j \lambda_j f_j\right) &= 1, \\ \Downarrow \\ Z &= Z(\lambda_1, \dots, \lambda_p) = \int_{\mathbf{R}^n} dV \rho_0 \exp\left\{\sum_j \lambda_j f_j\right\}, \end{aligned}$$

and the stationary distribution is

$$\rho(x_1, \dots, x_n) = \frac{\rho_0(x_1, \dots, x_n)}{Z(\lambda_1, \dots, \lambda_p)} \exp\left(\sum_{j=1}^p \lambda_j f_j(x_1, \dots, x_n)\right). \quad (52)$$

ρ is called the *maximum entropy distribution* and Z is the *partition function*. Note that we have not proved that it in fact is the distribution that gives a maximum for S , but this can be done.

The Lagrange multipliers $\lambda_1, \dots, \lambda_p$ are chosen so that all the constraints are satisfied

$$\langle f_j \rangle = \int_{\mathbf{R}^n} dV f_j(x_1, \dots, x_n) \rho(x_1, \dots, x_n) = c_j \quad j = 1, \dots, p. \quad (53)$$

The system of equations (53) consists of p equations for the p quantities λ_j . As it turns out, we almost never need to know the distribution ρ from (52), it is enough to know the partition function. Observe that

$$\begin{aligned}
\langle f_j \rangle &= \int_{\mathbf{R}^n} dV f_j \rho \\
&= \frac{1}{Z} \int_{\mathbf{R}^n} dV f_j \rho_0 \exp \left(\sum_{i=1}^p \lambda_i f_i \right) \\
&= \frac{1}{Z} \int_{\mathbf{R}^n} dV \partial_{\lambda_j} \{ \rho_0 \exp \left(\sum_{i=1}^p \lambda_i f_i \right) \} \\
&= \frac{1}{Z} \partial_{\lambda_j} \int_{\mathbf{R}^n} dV \rho_0 \exp \left(\sum_{i=1}^p \lambda_i f_i \right) \\
&= \frac{1}{Z} \partial_{\lambda_j} Z = \partial_{\lambda_j} \ln Z,
\end{aligned} \tag{54}$$

and thus we can find the mean of all the quantities f_j by taking partial derivatives of the partition function with respect to the Lagrangian multipliers. Moreover, we also have

$$\begin{aligned}
\partial_{\lambda_j \lambda_k} \ln Z &= \partial_{\lambda_j} \left(\frac{1}{Z} \partial_{\lambda_k} Z \right) \\
&= -\frac{1}{Z^2} \partial_{\lambda_j} Z \partial_{\lambda_k} Z + \frac{1}{Z} \partial_{\lambda_j \lambda_k} Z \\
&= -\partial_{\lambda_j} \ln Z \partial_{\lambda_k} \ln Z + \frac{1}{Z} \int_{\mathbf{R}^n} dV f_j f_k \rho_0 \exp \left(\sum_i \lambda_i f_i \right) \\
&= -\partial_{\lambda_j} \ln Z \partial_{\lambda_k} \ln Z + \langle f_j f_k \rangle.
\end{aligned}$$

Thus

$$\langle f_j f_k \rangle = \partial_{\lambda_j} \ln Z \partial_{\lambda_k} \ln Z + \partial_{\lambda_j \lambda_k} \ln Z$$

In a similar way *all* correlation coefficients $\langle f_1^{n_1} \dots f_p^{n_p} \rangle$ can be expressed through derivatives of the partition function.

Let us consider the special case when our underlying space is the classical state space for a mechanical system with n degrees of freedom. This could for example consist of n mass points. We thus have a state space $\mathbf{R}^{6n}$ since we need 3 position coordinates $\mathbf{x} = (x_1, x_2, x_3)$, and 3 velocity coordinates $\mathbf{v} = (v_1, v_2, v_3)$, or equivalently three momentum coordinates $\mathbf{p} = m\mathbf{v}$, for each particle in order to uniquely specify the state of the system.

Let $H = H(\mathbf{x}_1, \dots, \mathbf{x}_n, \mathbf{p}_1, \dots, \mathbf{p}_n)$ be the Hamiltonian for the system of mass points. Recall that the value of the Hamiltonian on any given state $(\mathbf{x}_1, \dots, \mathbf{x}_n, \mathbf{p}_1, \dots, \mathbf{p}_n)$, is the total energy of that state.

When n is large it is very hard, and also mostly useless, to try to track the exact state $(\mathbf{x}_1(t), \dots, \mathbf{x}_n(t), \mathbf{p}_1(t), \dots, \mathbf{p}_n(t))$ of a system of mass points.

For such a large system it is more useful to consider a probability distribution $\rho(\mathbf{x}_1, \dots, \mathbf{x}_n, \mathbf{p}_1, \dots, \mathbf{p}_n)$ on the state-space. This point of view is in particular

useful when we want to find the fundamental equations describing the motion of fluids.

Here we will assume that we have some prior distribution ρ_0 and the observation of the mean value of the total energy, H , of the system

$$\langle H \rangle = E.$$

Using the maximal entropy principle we are lead to select the following probability distribution

$$\rho(\mathbf{x}_1, \dots, \mathbf{x}_n, \mathbf{p}_1, \dots, \mathbf{p}_n) = \frac{\rho_0}{Z} \exp(\lambda H).$$

In this situation one usually redefines λ by writing

$$\lambda = -\frac{1}{k T},$$

where k is the Boltzmann constant and T is a new parameter.

What we then get is the well known *Gibb's ensemble* from statistical physics

$$\rho = \frac{\rho_0}{Z(T)} \exp\left\{-\frac{1}{kT} H\right\}. \quad (55)$$

The parameter T is determined by

$$\begin{aligned} \langle H \rangle &= E, \\ \Downarrow \\ k T^2 \partial_T \ln Z &= E, \end{aligned} \quad (56)$$

where we have used the chain rule

$$\partial_\lambda = k T^2 \partial_T,$$

in the general formula (54).

The Gibb's ensemble is the foundation of statistical physics. All results in statistical physics flows from formula (55). Statistical physics is also the foundation of thermodynamics, so all conclusions from that subject also flow from formula (55). In this context, T is the temperature of the system of mass-points and (56) is nothing but the *equation of state*.

An interesting insight here is that the temperature of a thermodynamic system is in fact a Lagrange multiplier!! This is a profound insight that to this day has not been fully understood or explored.

An extended discussion of the maximum entropy principle, and how it relates to foundational problems in statistical physics, is included in an Appendix.

From this example, it appears as if it might be useful to think of any application of the maximal entropy principle as an extension of the methods of statistical mechanics to systems that has absolutely nothing to do with the motion of mass points.

This wide general applicability of the methods of statistical physics has also lead to deep questions and insights into the nature and significance of the assumption of equilibrium that appears to underline the application of the Gibb's ensemble in statistical physics.

There is also the intriguing fact that the very same functional (51) used in the maximum entropy principle, is also the foundation of information theory which was discovered by Shannon in 1948. This connection between information theory and statistical mechanics (and thermodynamics) has lead to deep insights into the role of information in our fundamental physical theories.

The general nature and wide applicability of the maximum entropy principle has been described well by E.T. Jaynes in many papers and the monumental book "Probability theory: The Logic of Science".

As if all this is not impressive enough for one single principle, it is also a very intriguing fact that when one looks deep into the heart of fundamental physics, in the form of quantum field theory, one again finds an appropriately generalized form of the Gibb's ensemble! The whole computational engine in the theory of quantum fields revolve around this generalized Gibb's ensemble.

What on earth is going on...

3 Equations of Variational Type

We have seen that stationary points for functionals are solutions to the Euler-Lagrange equations corresponding to the functional. The exact structure of the Euler-Lagrange equations and their number depends on the functional. We have seen several examples of differential equations of the Euler-Lagrange type in the previous sections of these lecture notes. For example have we found that the differential equations (21) and (50) are of Euler-Lagrange type.

In this section we will ask which (systems of) differential equations are Euler-Lagrange equations for some functional. This is an important question to ask, because many structural properties of differential equations can be decided if we know that they are Euler-Lagrange equations for some functional.

Equations that are Euler-Lagrange equations for a functional are said to be *variational*.

3.1 Real valued functions

Example 9. Let $T(u)$ be the functional

$$T(u) = \iint_D dx dy \left(\frac{1}{2} u_x^2 + \frac{1}{2} u_y^2 \right),$$

with a constraint

$$u|_{\partial D} = f.$$

The Lagrangian density is

$$\mathcal{L} = \frac{1}{2} u_x^2 + \frac{1}{2} u_y^2,$$

and we have

$$\frac{\partial \mathcal{L}}{\partial u} = 0, \quad \frac{\partial \mathcal{L}}{\partial u_x} = u_x, \quad \frac{\partial \mathcal{L}}{\partial u_y} = u_y,$$

so the Euler-Lagrange equation is

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial u} - \partial_x \left(\frac{\partial \mathcal{L}}{\partial u_x} \right) - \partial_y \left(\frac{\partial \mathcal{L}}{\partial u_y} \right) &= 0, \\ \Downarrow \\ u_{xx} + u_{yy} &= 0. \end{aligned}$$

Thus the 2D Laplace equation is variational. The same is true in 3D or in any dimension for that matter. The same is also true for Poisson's equation in any dimension.

Example 10. Consider the functional

$$T(u) = \int_D dx dy \int_{t_0}^{t_1} dt \left\{ \frac{1}{2} u_t^2 - \frac{1}{2} c^2 u_x^2 - \frac{1}{2} c^2 u_y^2 \right\}$$

We are looking for functions $u(x, y, t)$ that are stationary points with respect to variations that vanish on the boundaries to the domain of integration

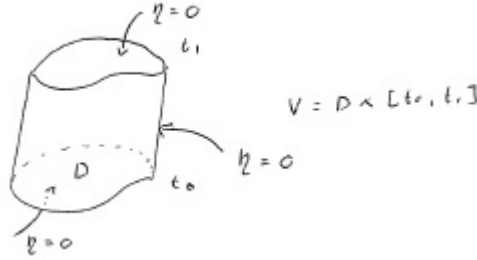


Figure 14: The integration domain of the functional whose Euler-Lagrange equation is the wave equation.

$$\begin{aligned} T(u + \epsilon \eta) &= \int_D dx dy \int_{t_0}^{t_1} dt \left\{ \frac{1}{2} (u_t + \epsilon \eta_t)^2 - \frac{1}{2} c^2 (u_x + \epsilon \eta_x)^2 \right. \\ &\quad \left. - \frac{1}{2} c^2 (u_y + \epsilon \eta_y)^2 \right\} \\ &= \int_D dx dy \int_{t_0}^{t_1} dt \left\{ \frac{1}{2} u_t^2 - \frac{1}{2} c^2 u_x^2 - \frac{1}{2} c^2 u_y^2 \right. \\ &\quad \left. + \epsilon u_t \eta_t - c^2 \epsilon u_x \eta_x - c^2 \epsilon u_y \eta_y \right\} + \mathcal{O}(\epsilon^2) \\ &= T(u) + \epsilon \int_D dx dy \int_{t_0}^{t_1} dt \left\{ u_t \eta_t - c^2 \nabla u \cdot \nabla \eta \right\} \\ &\quad + \mathcal{O}(\epsilon^2) \\ &= T(u) + \epsilon \left\{ \int_D dx dy \int_{t_0}^{t_1} dt ((u_t \eta)_t - u_{tt} \eta) \right. \\ &\quad \left. - c^2 \int_{t_0}^{t_1} dt \int_D dx dy (\nabla \cdot (\nabla u \eta) - \nabla^2 u \eta) \right\} + \mathcal{O}(\epsilon^2) \end{aligned}$$

↓

$$\begin{aligned}
T(u + \epsilon \eta) &= T(u) + \epsilon \left\{ \int_D dx dy (u_t \eta) \Big|_{t_0}^{t_1} - \iint_D dx dy \int_{t_0}^{t_1} dt u_{tt} \eta \right. \\
&\quad \left. - c^2 \int_{t_0}^{t_1} dt \int_{\partial D} dl \eta \nabla u \cdot \mathbf{n} + c^2 \int_{t_0}^{t_1} dt \int_D dx dy \nabla^2 u \eta \right\} \\
&\quad + \mathcal{O}(\epsilon^2) \\
&= T(u) + \epsilon \int_D dx dy \int_{t_0}^{t_1} dt \{-u_{tt} + c^2 \nabla^2 u\} \eta + \mathcal{O}(\epsilon^2),
\end{aligned}$$

where we in the last step have used the boundary conditions for η . Since $\eta(x, y, t)$ is arbitrary within the domain $D \times [t_0, t_1]$ we can use the fundamental lemma to conclude that u is a stationary point for T only if

$$u_{tt} - c^2 \nabla^2 u = 0.$$

This proves that the 2D wave equation is variational. The same is true for wave equations in any number of spatial dimensions.

Example 11. Consider the functional

$$T(u) = \int_D dx dy \int_{t_0}^{t_1} dt \left\{ \frac{1}{2} u_t^2 - \frac{1}{2} c^2 u_x^2 - \frac{1}{2} c^2 u_y^2 - \frac{1}{2} m u^2 \right\} \quad m > 0$$

Using the exact same approach as in example 10, it is easy to show that u is extremal for variation η vanishing on the boundary of $D \times [t_0, t_1]$ only if

$$u_{tt} - c^2 \nabla^2 u + m u = 0$$

This is the (real) Klein-Gordon equation. Thus the Klein-Gordon equation is variational.

This is all fine, you might say, but where did the functionals in examples 9-11 come from? Is there a useful general method for deciding if a given (system of) differential equations is variational, and to find an appropriate functional if they are variational?

Sadly, no such general method exists. The functionals must be constructed by trial and error.

3.2 Complex valued functions

Many important equations in theoretical physics and applied mathematics involve complex valued functions. The prime example is of course the quantum mechanical Schrödinger equation.

In order to decide whether such equations are variational, or not, we need to extend the calculus of variation to functions whose values are complex.

Example 12. Let us consider the complex second order differential equation

$$A_{tt} = |A|^2 A \quad A(t) \in \mathbf{C}. \quad (57)$$

Introduce the real and imaginary parts of A

$$A(t) = u(t) + i v(t). \quad (58)$$

Inserting (58) into (57), we get

$$u_{tt} + i v_{tt} = (u^2 + v^2) (u + i v).$$

Separating real and imaginary parts we get two coupled real equations

$$\begin{aligned} u_{tt} &= (u^2 + v^2) u, \\ v_{tt} &= (u^2 + v^2) v. \end{aligned} \tag{59}$$

Consider the functional

$$T(u, v) = \int_{t_0}^{t_1} dt \left(\frac{1}{2} u_t^2 + \frac{1}{2} v_t^2 + \frac{1}{4} u^4 + \frac{1}{2} u^2 v^2 + \frac{1}{4} v^4 \right). \tag{60}$$

The Lagrangian is

$$L = \frac{1}{2} u_t^2 + \frac{1}{2} v_t^2 + \frac{1}{4} u^4 + \frac{1}{2} u^2 v^2 + \frac{1}{4} v^4, \tag{61}$$

and the corresponding Euler-Lagrange equations are

$$\begin{aligned} \frac{\partial L}{\partial u} - \frac{d}{dt} \left(\frac{\partial L}{\partial u_t} \right) &= 0, \\ \Downarrow \\ u^3 + v^2 u - u_{tt} &= 0, \\ \Downarrow \\ u_{tt} &= (u^2 + v^2) u, \\ \frac{\partial L}{\partial v} - \frac{d}{dt} \left(\frac{\partial L}{\partial v_t} \right) &= 0, \\ \Downarrow \\ v^3 + u^2 v - v_{tt} &= 0, \\ \Downarrow \\ v_{tt} &= (u^2 + v^2) v. \end{aligned}$$

This shows that (59) is variational with the corresponding functional defined in (60). We could now *define* a complex equation like (57) to be variational if the system of real equations we get when separating into real and imaginary parts is variational.

It would however be better if we could define what it means for a complex equation to be variational *without* first separating it into real and imaginary parts.

Observe that the Lagrangian (61) can be written more compactly as

$$L = \frac{1}{2} u_t^2 + \frac{1}{2} v_t^2 + \frac{1}{4} (u^2 + v^2)^2. \tag{62}$$

We have

$$\begin{aligned} u + i v &= A, & u - i v &= A^*, \\ u &= \frac{1}{2} (A + A^*), & v &= \frac{1}{2i} (A - A^*). \end{aligned} \tag{63}$$

Inserting (63) into the Lagrangian (62), we get

$$\begin{aligned}
L &= \frac{1}{8} (A_t + A_t^*)^2 - \frac{1}{8} (A_t - A_t^*)^2 \\
&\quad + \frac{1}{4} (A A^*)^2 \\
&= \frac{1}{8} A_t^2 + \frac{1}{4} A_t A_t^* + \frac{1}{8} A_t^{*2} - \frac{1}{8} A_t^2 + \frac{1}{4} A_t A_t^* \\
&\quad - \frac{1}{8} A_t^{*2} + \frac{1}{4} A^2 A^{*2} \\
&= \frac{1}{2} A_t A_t^* + \frac{1}{4} A^2 A^{*2}.
\end{aligned}$$

Since a factor of 2 makes no difference for the Euler-Lagrange equation, we might as well use the Lagrangian

$$L = A_t A_t^* + \frac{1}{2} A^2 A^{*2}. \quad (64)$$

In this Lagrangian density, A is the only dependent variable. A^* is of course calculated by taking the complex conjugate of A . However let us disregard this fact and assume that A and A^* can be varied *independently*. Then, any functional of the form

$$T(A, A^*) = \int_{t_0}^{t_1} dt L(A, A^*, A_t, A_t^*),$$

for some Lagrangian L , will in the usual way lead to Euler-Lagrange equations

$$\begin{aligned}
\frac{\partial L}{\partial A^*} - \frac{d}{dt} \left(\frac{\partial L}{\partial A_t^*} \right) &= 0, \\
\frac{\partial L}{\partial A} - \frac{d}{dt} \left(\frac{\partial L}{\partial A_t} \right) &= 0.
\end{aligned}$$

For our particular Lagrangian (64) we have

$$\begin{aligned}
\frac{\partial L}{\partial A} &= A A^{*2}, & \frac{\partial L}{\partial A_t} &= A_t^*, \\
\frac{\partial L}{\partial A^*} &= A^2 A^*, & \frac{\partial L}{\partial A_t^*} &= A_t.
\end{aligned}$$

The Euler-lagrange equations are thus

$$\begin{aligned}
A A^{*2} - A_{tt}^* &= 0, \\
&\Downarrow \\
A_{tt}^* &= |A|^2 A^*, \\
A^2 A^* - A_{tt} &= 0, \\
&\Downarrow \\
A_{tt} &= |A|^2 A,
\end{aligned}$$

which is our original equation (57) and its complex conjugate.

Thus, following this formal procedure, where we assume that A and A^* can be

varied independently, we have again proved that (57) is variational, and we have done this without separating the problem into real and imaginary parts.

This procedure will always work and its conclusions are equivalent to what we get by separating into real and imaginary part.

However, an important caveat is that we must use this formal procedure *only* for Lagrangians that are *real valued*. We must thus always make sure that

$$L^* = L$$

Example 13. Consider the functional

$$T(A, A^*) = \int_D dx dy \int_{t_0}^{t_1} dt \mathcal{L}(A, A^*, A_t, A_t^*, A_x, A_x^*, A_y, A_y^*),$$

where $\mathcal{L} = \mathcal{L}^*$. Following the procedure from example 12 with dependent variables A and A^* we get the Euler-Lagrange equations

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial A} - \partial_t \left(\frac{\partial \mathcal{L}}{\partial A_t} \right) - \partial_x \left(\frac{\partial \mathcal{L}}{\partial A_x} \right) - \partial_y \left(\frac{\partial \mathcal{L}}{\partial A_y} \right) &= 0, \\ \frac{\partial \mathcal{L}}{\partial A^*} - \partial_t \left(\frac{\partial \mathcal{L}}{\partial A_t^*} \right) - \partial_x \left(\frac{\partial \mathcal{L}}{\partial A_x^*} \right) - \partial_y \left(\frac{\partial \mathcal{L}}{\partial A_y^*} \right) &= 0. \end{aligned} \quad (65)$$

As a matter of fact, we only need one of the equations (65) since the first one is just the complex conjugate of the second one.

Her we choose to use the second equation. Consider the special Lagrangian density

$$\mathcal{L} = A_t A_t^* - c^2 A_x A_x^* - c^2 A_y A_y^* - m A A^*. \quad (66)$$

We have

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial A^*} &= -m A, & \frac{\partial \mathcal{L}}{\partial A_t^*} &= A_t, \\ \frac{\partial \mathcal{L}}{\partial A_x^*} &= -c^2 A_x, & \frac{\partial \mathcal{L}}{\partial A_y^*} &= -c^2 A_y, \end{aligned}$$

and the Euler -Lagrange equation is

$$\begin{aligned} -m A - A_{tt} + c^2 A_{xx} + c^2 A_{yy} &= 0, \\ \Downarrow \\ A_{tt} - c^2 \nabla^2 A + m A &= 0. \end{aligned}$$

This shows that the 2D complex Klein-Gordon equation is variational with Lagrangian density defined in (66). The same is true for the 1D and 3D cases.

Example 14. Consider a Lagrangian density

$$\mathcal{L} = i \frac{\hbar}{2} (\psi^* \psi_t - \psi \psi_t^*) - \frac{\hbar^2}{2m} (\psi_x \psi_x^* + \psi_y \psi_y^*) - V(x, y) \psi \psi^*. \quad (67)$$

We evidently have $\mathcal{L} = \mathcal{L}^*$. Observe that

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial \psi^*} &= i \frac{\hbar}{2} \psi_t - V \psi, & \frac{\partial \mathcal{L}}{\partial \psi_t^*} &= -i \frac{\hbar}{2} \psi, \\ \frac{\partial \mathcal{L}}{\partial \psi_x^*} &= -\frac{\hbar^2}{2m} \psi_x, & \frac{\partial \mathcal{L}}{\partial \psi_y^*} &= -\frac{\hbar^2}{2m} \psi_y, \end{aligned}$$

and the Euler-Lagrange equation is

$$\begin{aligned}
i \frac{\hbar}{2} \partial_t \psi - V \psi - \partial_t \left(-\frac{\hbar}{2} i \psi \right) \\
- \partial_x \left(-\frac{\hbar^2}{2m} \psi_x \right) - \partial_y \left(-\frac{\hbar^2}{2m} \psi_y \right) &= 0, \\
\Updownarrow \\
i \hbar \partial_t \psi &= -\frac{\hbar^2}{2m} \nabla^2 \psi + V \psi.
\end{aligned}$$

This shows that the Schrödinger equation is variational with Lagrangian density defined in (67). It is easy to show that the stationary Schrödinger equation is also variational.

4 Noether's Theorem

Noether's theorem creates a one-to-one correspondence between *conserved quantities* of variational equations and *symmetries* of the corresponding functionals.

The theorem was proved by the mathematician Emmy Noether in 1915. It has been described as:

"One of the most important mathematical theorems ever proved in guiding the development of modern physics"

4.1 One dependent variable

Let us introduce the theorem in the simplest possible context. We consider a functional of the form

$$T(y) = \int_{t_0}^{t_1} dt L(t, y, y'). \quad (68)$$

Let us consider some variation

$$y \rightarrow y + \epsilon \eta, \quad (69)$$

where $\eta = \eta(t)$ now is some *specific* function. Inserting (69) into (68) we get

$$\begin{aligned}
T(y + \epsilon \eta) &= \int_{t_0}^{t_1} dt L(t, y + \epsilon \eta, y' + \epsilon \eta') \\
&= \int_{t_0}^{t_1} dt \left\{ L(t, y, y') + \frac{\partial L}{\partial y} \epsilon \eta + \frac{\partial L}{\partial y'} \epsilon \eta' \right\} + \mathcal{O}(\epsilon^2) \\
&= T(y) + \epsilon \int_{t_0}^{t_1} dt \left\{ \frac{\partial L}{\partial y} \eta + \frac{\partial L}{\partial y'} \eta' \right\} + \mathcal{O}(\epsilon^2).
\end{aligned}$$

We now introduce the key idea of invariance.

The functional $T(y)$ is *invariant* under the variation (69) if there exists a function $F(t)$ such that

$$\frac{\partial L}{\partial y} \eta + \frac{\partial L}{\partial y'} \eta' = \frac{dF}{dt}. \quad (70)$$

If $F = 0$ we say that the Lagrangian L is invariant. Let us next consider a more general variation

$$y(t) \rightarrow y(t) + \epsilon(t) \eta(t), \quad (71)$$

where T is invariant with respect to the variation (69) and $\epsilon(t)$ is a function that is numerically small, $|\epsilon(t)| \ll 1$, vanishes at the boundary points t_0 and t_1 , but which is otherwise arbitrary. Inserting the variation (71) into the functional (68) we get

$$\begin{aligned} T(y + \epsilon \eta) &= \int_{t_0}^{t_1} dt L(t, y + \epsilon \eta, y' + \epsilon \eta' + \epsilon' \eta) \\ &= \int_{t_0}^{t_1} dt \left\{ L + \epsilon \left(\frac{\partial L}{\partial y} \eta + \frac{\partial L}{\partial y'} \eta' \right) + \eta \frac{\partial L}{\partial y'} \epsilon' \right\} + \mathcal{O}(\epsilon^2) \\ &= T(y) + \int_{t_0}^{t_1} dt \left\{ \epsilon \frac{dF}{dt} - \frac{d}{dt} \left(\eta \frac{\partial L}{\partial y} \right) \epsilon \right\} + \mathcal{O}(\epsilon^2) \\ &= T(y) + \int_{t_0}^{t_1} dt \epsilon \left\{ \frac{dF}{dt} - \frac{d}{dt} \left(\eta \frac{\partial L}{\partial y'} \right) \right\} + \mathcal{O}(\epsilon^2) \\ &= T(y) + \int_{t_0}^{t_1} dt \epsilon \frac{dj}{dt} + \mathcal{O}(\epsilon^2), \end{aligned} \quad (72)$$

where we have used (70) in line three, and where the *Noether current* $j(t)$ is defined to be

$$j = F - \eta \frac{\partial L}{\partial y'}. \quad (73)$$

Equation (72) is true for *any* $y(t)$. In particular it is true for a $y(t)$ that is a stationary point for the functional T . But if y is stationary we must have

$$T(y + \tilde{\epsilon} \tilde{\eta}) = T(y) + \mathcal{O}(\tilde{\epsilon}^2), \quad (74)$$

for any $\tilde{\eta}$ of order one and small number $\tilde{\epsilon}$.

If we let $\tilde{\epsilon}$ measure the size of $\epsilon(t)$ and define

$$\tilde{\eta}(t) = \frac{\epsilon(t)}{\tilde{\epsilon}} \eta(t),$$

then the variation (71) is exactly of the form

$$y(t) \rightarrow y(t) + \tilde{\epsilon} \tilde{\eta}(t). \quad (75)$$

Therefore, when y is a stationary point for the functional $T(y)$, we must from (72) have

$$\int_{t_0}^{t_1} dt \epsilon(t) \frac{dj}{dt} = 0, \quad (76)$$

and this holds for all $\epsilon(t)$ that vanishes on the boundaries. The fundamental lemma then implies that

$$\frac{dj}{dt} = 0,$$

or in other words, the Noether current corresponding to an invariant for a functional is conserved for any stationary point of the functional. Such stationary $y(t)$'s satisfy, as we recall, the Euler- Lagrange equations corresponding to the functional. This is one instance of Noether's theorem

Example 15. Let us consider a functional

$$T(y) = \int_{t_0}^{t_1} dt L(y, y'),$$

thus L does not depend explicitly on t .

We now consider an infinitesimal translation of the variable t

$$t \rightarrow t + \epsilon. \quad (77)$$

The translation (77) induces a corresponding variation of $y(t)$ that we find using Taylor's formula

$$y(t) \rightarrow y(t + \epsilon) = y(t) + \epsilon y'(t) + \dots \quad (78)$$

Thus we have a variation of the form (69) with $\eta(t) = y'(t)$.

Observe that using this particular variation we have

$$\begin{aligned} \frac{\partial L}{\partial y} \eta + \frac{\partial L}{\partial y'} \eta' &= \frac{\partial L}{\partial y} y' + \frac{\partial L}{\partial y'} y'' \\ &= \frac{\partial L}{\partial t} + \frac{\partial L}{\partial y} y' + \frac{\partial L}{\partial y'} y'' = \frac{dL}{dt}. \end{aligned}$$

Thus the functional is invariant under the variation (78) *because* L does not depend explicitly on t .

The conserved Noether current corresponding to the variation (78) is then, from (73),

$$j = L - y' \frac{\partial L}{\partial y'}.$$

Let us verify directly that j is indeed conserved. Using the Euler-Lagrange equations we have

$$\begin{aligned} \frac{dj}{dt} &= \frac{\partial L}{\partial y} y' + \frac{\partial L}{\partial y'} y'' - y'' \frac{\partial L}{\partial y'} - y' \frac{d}{dt} \left(\frac{\partial L}{\partial y'} \right) \\ &= y' \left(\frac{\partial L}{\partial y} - \frac{d}{dt} \left(\frac{\partial L}{\partial y'} \right) \right) = 0. \end{aligned}$$

Example 16. The motion of a mass-point m under the influence of a conservative, time invariant force

$$F = -\frac{\partial V}{\partial x}, \quad V = V(x),$$

is determined by Newton's law

$$m x'' = -\frac{\partial V}{\partial x}.$$

We have seen that this equation is variational with Lagrangian

$$L = \frac{1}{2} m x'^2 - V(x), \quad (79)$$

and we observe that L is invariant under time translation because V does not depend on time.

By applying the general result from example 15, we have the following conserved Noether current

$$\begin{aligned} j &= L - x' \frac{\partial L}{\partial x'} \\ &= \frac{1}{2} m x'^2 - V(x) - m x'^2 \\ &= -\left(\frac{1}{2} m x'^2 + V(x)\right) = -E(t), \end{aligned}$$

where $E(t)$ is the total *energy* of the mass-point.

Thus the energy is conserved because the Lagrangian (79) of the action functional does not depend on time and is thus invariant under time translation.

This link between energy conservation and invariance under time translation for the Lagrangians holds in general.

The reason why energy conservation plays such a prominent role in our description of nature is because we insist that our natural laws should look the same for all observers, even if they live at different times. This is part of the essence of what we mean by the term *natural law*.

Thus, energy conservation is not actually a part of nature, but is rather a consequence of how we choose to describe nature.

4.2 Several dependent variables

Let us next consider the case when we have several dependent variables

$$T(y_1, y_2, \dots, y_n) = \int_{t_0}^{t_1} dt L(t, y_1, \dots, y_n, y'_1, \dots, y'_n). \quad (80)$$

We introduce a variation

$$y_i \rightarrow y_i + \epsilon \eta_i \quad i = 1, \dots, n, \quad (81)$$

where $\eta_i = \eta_i(t)$ is a specific set of n functions and $\epsilon \ll 1$.

Inserting the variation (81) into the functional (80) we get

$$\begin{aligned}
& T(y_1 + \epsilon \eta_1, \dots, y_n + \epsilon \eta_n) \\
&= \int_{t_0}^{t_1} dt L(t, y_1 + \epsilon \eta_1, \dots, y_n + \epsilon \eta_n, y'_1 + \epsilon \eta'_1, \dots, y'_n + \epsilon \eta'_n) \\
&= \int_{t_0}^{t_1} dt \left\{ L(t, y_1, \dots, y_n, y'_1, \dots, y'_n) + \frac{\partial L}{\partial y_1} \epsilon \eta_1 + \dots + \frac{\partial L}{\partial y_n} \epsilon \eta_n \right. \\
&\quad \left. + \frac{\partial L}{\partial y'_1} \epsilon \eta'_1 + \dots + \frac{\partial L}{\partial y'_n} \epsilon \eta'_n \right\} + \mathcal{O}(\epsilon^2) \\
&= T(y_1, \dots, y_n) + \epsilon \int_{t_0}^{t_1} dt \left\{ \frac{\partial L}{\partial y_1} \eta_1 + \dots + \frac{\partial L}{\partial y_n} \eta_n + \frac{\partial L}{\partial y'_1} \eta'_1 \right. \\
&\quad \left. + \dots + \frac{\partial L}{\partial y'_n} \eta'_n \right\} + \mathcal{O}(\epsilon^2).
\end{aligned}$$

We now define the functional T to be invariant under the variation (81) if there exists a function $F(t)$ such that

$$\frac{\partial L}{\partial y_1} \eta_1 + \dots + \frac{\partial L}{\partial y_n} \eta_n + \frac{\partial L}{\partial y'_1} \eta'_1 + \dots + \frac{\partial L}{\partial y'_n} \eta'_n = \frac{dF}{dt}. \quad (82)$$

If $F = 0$ we say that the Lagrangian L is invariant.

Let us next introduce the more general variation

$$y_i(t) \rightarrow y_i(t) + \epsilon(t) \eta_i(t), \quad (83)$$

where the functions $\epsilon(t)$ satisfies the properties preceding (72).

Inserting the variation (83) into the functional (80) we get

$$\begin{aligned}
& T(y_1 + \epsilon \eta_1, \dots, y_n + \epsilon \eta_n) \\
&= \int_{t_0}^{t_1} dt L(t, y_1 + \epsilon \eta_1, \dots, y_n + \epsilon \eta_n, y'_1 \\
&\quad + \epsilon \eta'_1 + \epsilon' \eta_1, \dots, y'_n + \epsilon \eta'_n + \epsilon' \eta_n) \\
&= \int_{t_0}^{t_1} dt \left(L(t, y_1, \dots, y_n, y'_1, \dots, y'_n) + \epsilon(t) \left\{ \frac{\partial L}{\partial y_1} \eta_1 + \frac{\partial L}{\partial y'_1} \eta'_1 \right. \right. \\
&\quad \left. \left. + \dots + \frac{\partial L}{\partial y_n} \eta_n + \frac{\partial L}{\partial y'_n} \eta'_n \right\} + \epsilon'(t) \left\{ \eta_1 \frac{\partial L}{\partial y'_1} + \right. \right. \\
&\quad \left. \left. \dots + \eta_n \frac{\partial L}{\partial y'_n} \right\} + \mathcal{O}(\epsilon^2) \right) \\
&= T(y_1, \dots, y_n) + \int_{t_0}^{t_1} dt \epsilon(t) \left\{ \frac{dF}{dt} - \frac{d}{dt} \left(\eta_1 \frac{\partial L}{\partial y'_1} + \right. \right. \\
&\quad \left. \left. \dots + \eta_n \frac{\partial L}{\partial y'_n} \right) \right\} + \mathcal{O}(\epsilon^2) \\
&= T(y_1, \dots, y_n) + \int_{t_0}^{t_1} dt \epsilon(t) \frac{dj}{dt} + \mathcal{O}(\epsilon^2),
\end{aligned}$$

where the Noether current is

$$j = F - \eta_1 \frac{\partial L}{\partial y'_1} - \dots - \eta_n \frac{\partial L}{\partial y'_n} \quad (84)$$

By an argument identical to (74)-(76) we conclude that the Noether current is conserved

$$\frac{dj}{dt} = 0,$$

for any solutions to the Euler - Lagrange equations

$$\frac{\partial L}{\partial y_i} - \frac{d}{dt} \left(\frac{\partial L}{\partial y'_i} \right) = 0 \quad i = 1, \dots, n .$$

This is Noether's theorem for the functional (80).

Example 17. Let us consider the Lagrangian

$$L = my'_1 y'_2 - (y_1 - 2y_2)^2 \quad (85)$$

The Lagrangian L , does not depend explicitly on time.

In general, let us consider a Lagrangian of the form $L = L(y_1, \dots, y_n, y'_1, \dots, y'_n)$, which does not depend explicitly on time.

Observe than an infinitesimal time translation

$$t \rightarrow t + \epsilon,$$

induces, like in (78), variations of the form

$$y_i(t) \rightarrow y_i(t) + \epsilon y'_i(t), \quad (86)$$

and we observe that the functional (80) is invariant under the variation (86)

$$\begin{aligned} & \frac{\partial L}{\partial y_1} \eta_1 + \dots + \frac{\partial L}{\partial y_n} \eta_n + \frac{\partial L}{\partial y'_1} \eta'_1 + \dots + \frac{\partial L}{\partial y'_n} \eta'_n \\ &= \frac{\partial L}{\partial y_1} y'_1 + \dots + \frac{\partial L}{\partial y_n} y'_n + \frac{\partial L}{\partial y'_1} y''_1 + \dots + \frac{\partial L}{\partial y'_n} y''_n \\ &= \frac{dL}{dt}. \end{aligned}$$

Thus we have the following conserved Noether current.

$$j = L - y'_1 \frac{\partial L}{\partial y'_1} - \dots - y'_n \frac{\partial L}{\partial y'_n}. \quad (87)$$

Applying this result to our particular Lagrangian (85) we have the following conserved Noether current

$$\begin{aligned} j &= L - y'_1 \frac{\partial L}{\partial y'_1} - y'_2 \frac{\partial L}{\partial y'_2} \\ &= my'_1 y'_2 - (y_1 - 2y_2)^2 - y'_1 my'_2 - y'_2 my'_1 \\ &= -(my'_1 y'_2 + (y_1 - 2y_2)^2). \end{aligned}$$

Let us verify by direct calculations that this is indeed a conserved quantity for the Euler-Lagrange equations corresponding to the Lagrangian (85).

The Euler-Lagrange equations are

$$\begin{aligned} my_1'' &= 4(y_1 - 2y_2), \\ my_2'' &= -2(y_1 - 2y_2). \end{aligned}$$

Using these equations we have

$$\begin{aligned} \frac{dj}{dt} &= -(my_1''y_2' + my_1'y_2'' + 2(y_1 - 2y_2)(y_1' - 2y_2')) \\ &= -(4(y_1 - 2y_2)y_2' - 2y_1'(y_1 - 2y_2) + 2(y_1 - 2y_2)(y_1' - 2y_2')) = 0. \end{aligned}$$

Let us next consider the variation

$$y_1 \rightarrow y_1 + \epsilon\eta_1, \quad (88)$$

$$y_2 \rightarrow y_2 + \epsilon\eta_2, \quad (89)$$

where $\eta_1 = a$, $\eta_2 = \frac{1}{2}a$ and where a is a real number of order one. For this particular variation we have

$$\begin{aligned} &\frac{\partial L}{\partial y_1}\eta_1 + \frac{\partial L}{\partial y_2}\eta_2 + \frac{\partial L}{\partial y_1'}\eta_1' + \frac{\partial L}{\partial y_2'}\eta_2' \\ &= \frac{\partial L}{\partial y_1}\eta_1 + \frac{\partial L}{\partial y_2}\eta_2 \\ &= -2(y_1 - 2y_2)a + 4(y_1 - 2y_2)\frac{1}{2}a = 0. \end{aligned}$$

Thus, not only the functional, but also the Lagrangian itself is invariant under the variation (89). Then, according to (87), we have the following conserved Noether current

$$\begin{aligned} j &= -\frac{\partial L}{\partial y_1'}\eta_1 - \frac{\partial L}{\partial y_2'}\eta_2 \\ &= -my_2'a - my_1'\frac{1}{2}a \\ &= -ma(y_2' + \frac{1}{2}y_1'). \end{aligned}$$

Example 18. Consider the Lagrangian

$$L = y_1'^2 + y_2'^2 + y_1^2 + 2y_1^2y_2^2 + y_2^4. \quad (90)$$

The Euler-Lagrange equations for this system are

$$y_1'' = 2y_1^3 + 2y_1y_2^2, \quad (91)$$

$$y_2'' = 2y_2^3 + 2y_1^2y_2. \quad (92)$$

We can consider y_1 and y_2 to be coordinates of points (y_1, y_2) in the plane, $\mathcal{R}^2$. A counterclockwise rotation of an angle θ in the plane is, with respect to these coordinates, given by the transformation

$$\begin{aligned} y_1(\theta) &= y_1 \cos(\theta) - y_2 \sin(\theta), \\ y_2(\theta) &= y_1 \sin(\theta) + y_2 \cos(\theta). \end{aligned}$$

For an infinitesimal angle $\theta = \epsilon$ we get a variation of the form

$$\begin{aligned} y_1 &\rightarrow y_1 + \epsilon\eta_1, \\ y_2 &\rightarrow y_2 + \epsilon\eta_2, \end{aligned} \quad (93)$$

where $\eta_1 = -y_2$, and $\eta_2 = y_1$.

It is straight forward to verify that

$$\frac{\partial L}{\partial y_1}\eta_1 + \frac{\partial L}{\partial y_2}\eta_2 + \frac{\partial L}{\partial y_1'}\eta_1' + \frac{\partial L}{\partial y_2'}\eta_2' = 0,$$

so that the Lagrangian (90) is indeed invariant with respect to the variation (93). Then according to (87) we have the following conserved Noether current

$$\begin{aligned} j &= -\frac{\partial L}{\partial y_1'}\eta_1 - \frac{\partial L}{\partial y_2'}\eta_2 \\ &= 2y_2y_1' - 2y_1y_2'. \end{aligned}$$

Using the Euler-Lagrange equations we can verify by direct calculation that j is indeed a conserved quantity

$$\begin{aligned} \frac{dj}{dt} &= 2y_2y_1'' - 2y_1y_2'' \\ &= 2y_2(2y_1^3 + 2y_1y_2^2) - 2y_1(2y_2^3 + 2y_1^2y_2) \\ &= 4y_2y_1^3 + 4y_1y_2^3 - 4y_1y_2^3 - 4y_1^3y_2 = 0. \end{aligned}$$

In the following three examples we will consider systems consisting of a collection of mass points with positions $\mathbf{x}_i \in \mathcal{R}^3$, velocities $\mathbf{x}_i' \in \mathcal{R}^3$ and quantities of mass m_i . Such systems can already be treated using the general theory developed on pages 48 and 49, however it might be clearer to rather redo the derivation of the Euler-Lagrange equation, the condition of invariance and the Noether current, starting with a functional of the form

$$T(\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n) = \int_{t_0}^{t_1} dt L(t, \mathbf{x}_1, \dots, \mathbf{x}_n, \mathbf{x}_1', \dots, \mathbf{x}_n'). \quad (94)$$

We will not do this here, but leave it to the reader. The resulting Euler-Lagrange equations are

$$\frac{\partial L}{\partial \mathbf{x}_i} - \frac{d}{dt} \frac{\partial L}{\partial \mathbf{x}_i'} = 0, \quad i = 1, \dots, n. \quad (95)$$

The condition for the functional (94) to be invariant under the variation

$$\mathbf{x}_i \rightarrow \mathbf{x}_i + \epsilon\eta_i, \quad i = 1, \dots, n, \quad (96)$$

is now

$$\frac{\partial L}{\partial \mathbf{x}_1} \cdot \eta_1 + \dots + \frac{\partial L}{\partial \mathbf{x}_n} \cdot \eta_n + \frac{\partial L}{\partial \mathbf{x}_1'} \cdot \eta_1' + \dots + \frac{\partial L}{\partial \mathbf{x}_n'} \cdot \eta_n' = \frac{dF}{dt}, \quad (97)$$

and the conserved Noether current is

$$j = F - \eta_1 \cdot \frac{\partial L}{\partial \mathbf{x}_1'} - \dots - \eta_n \cdot \frac{\partial L}{\partial \mathbf{x}_n'}. \quad (98)$$

Example 19. In example 5 we discussed the action principle for systems of mass points moving under the influence of conservative forces. The functional for this situation was of the form

$$T(\mathbf{x}_1, \dots, \mathbf{x}_n) = \int_{t_0}^{t_1} dt L(t, \mathbf{x}_1, \dots, \mathbf{x}_n, \mathbf{x}'_1, \dots, \mathbf{x}'_n), \quad (99)$$

where the Lagrangian is

$$L = \frac{1}{2} \sum_{i=1}^n m_i \mathbf{x}'_i{}^2 - V(t, \mathbf{x}_1, \dots, \mathbf{x}_n). \quad (100)$$

Let us assume that the potential does not depend explicitly on time. Then the functional (99) is invariant with respect to variations of the form (96), where $\eta_i = \mathbf{x}'_i$. The corresponding Noether current is then from (98)

$$\begin{aligned} j &= L - \sum_{i=1}^n \mathbf{x}'_i \cdot \frac{\partial L}{\partial \mathbf{x}'_i} \\ &= \frac{1}{2} \sum_{i=1}^n m_i \mathbf{x}'_i{}^2 - V - \sum_{i=1}^n m_i \mathbf{x}'_i{}^2 \\ &= -\left(\frac{1}{2} \sum_{i=1}^n m_i \mathbf{x}'_i{}^2 + V\right) = -E, \end{aligned}$$

where E is the total energy of the system of mass-points.

Thus, we see again that energy conservation exists because we insist on natural laws that appear the same for all observers, independently of when they live.

Example 20. Let us next assume that the the Lagrangian (100) is invariant with respect to a variation of the form (96), where $\eta_i = \mathbf{a}_i$. Here, $\{\mathbf{a}_i\}_{i=1}^n$ are vector in $\mathcal{R}^3$. According to (97), this invariance holds only if

$$\begin{aligned} \frac{\partial L}{\partial \mathbf{x}_1} \cdot \eta_1 + \dots + \frac{\partial L}{\partial \mathbf{x}_n} \cdot \eta_n + \frac{\partial L}{\partial \mathbf{x}'_1} \cdot \eta'_1 + \dots + \frac{\partial L}{\partial \mathbf{x}'_n} \cdot \eta'_n &= 0 \\ \Downarrow \\ \frac{\partial L}{\partial \mathbf{x}_1} \cdot \eta_1 + \dots + \frac{\partial L}{\partial \mathbf{x}_n} \cdot \eta_n &= 0 \\ -\frac{\partial V}{\partial \mathbf{x}_1} \cdot \mathbf{a}_1 - \dots - \frac{\partial V}{\partial \mathbf{x}_n} \cdot \mathbf{a}_n &= 0 \end{aligned} \quad (101)$$

Given this property for the potential, we have according to (98) the following conserved Noether current

$$j = -\sum_i \mathbf{a}_i \cdot \frac{\partial L}{\partial \mathbf{x}'_i} = -\sum_i m_i \mathbf{a}_i \cdot \mathbf{x}'_i.$$

The most common situation is when $\mathbf{a}_i = \mathbf{a} \quad \forall i$, for some fixed vector $\mathbf{a}$. For this case the Noether current is

$$j = -\mathbf{a} \cdot \sum_i m_i \mathbf{x}'_i.$$

Thus the component of the total momentum in the direction of $\mathbf{a}$ is conserved, if the potential satisfy the condition

$$\mathbf{a} \cdot \sum_i \frac{\partial V}{\partial \mathbf{x}_i} = 0.$$

This would be for example be the case if the potential is invariant with respect to translation of the origin along the direction $\mathbf{a}$. If the invariance (102) holds for three vectors $\mathbf{a}, \mathbf{b}, \mathbf{c}$, that span $\mathbf{R}^3$, we can conclude that the total momentum of the system of mass points is conserved

$$\mathbf{P} = \sum_i m_i \mathbf{x}_i'. \quad (102)$$

This conservation law holds for example if the potential only depends on differences of the vectors $\mathbf{x}_i$.

For this case, the laws of motion for mass-points looks the same for all observers, independently of where in space they are located. This invariance with respect to location in space is something we *choose* to impose on *all* our fundamental natural laws. The consequence of this *choice* is that we will have conservation of momentum

Example 21. Recall that counterclockwise rotations of angle θ around some axis $\mathbf{k}$ in $\mathcal{R}^3$ can be written in the following way

$$\mathbf{x} \rightarrow \mathbf{x} \cos \theta + (\mathbf{k} \times \mathbf{x}) \sin \theta + \mathbf{k} (\mathbf{k} \cdot \mathbf{x}) (1 - \cos \theta).$$

This is *Rodrigue's formula*. For an infinitesimal rotation angle we get

$$\mathbf{x} \rightarrow \mathbf{x} + \epsilon (\mathbf{k} \times \mathbf{x}). \quad (103)$$

The rotation of coordinates (103), induces a corresponding variation

$$\mathbf{x}_i(t) \rightarrow \mathbf{x}_i(t) + \epsilon \eta_i, \quad (104)$$

where

$$\eta_i = \mathbf{k} \times \mathbf{x}_i.$$

Let us assume that the Lagrangian for the system of mass-points is invariant with respect the variation (104). For this to be true we must, according to (97), have

$$\begin{aligned} \sum_i \frac{\partial L}{\partial \mathbf{x}_i} \cdot \eta_i + \frac{\partial L}{\partial \mathbf{x}_i'} \cdot \eta_i' &= 0 \\ \Downarrow \\ - \sum_i \frac{\partial V}{\partial \mathbf{x}_i} \cdot (\mathbf{k} \times \mathbf{x}_i) + \sum_i m_i \mathbf{x}_i' \cdot (\mathbf{k} \times \mathbf{x}_i') &= 0 \\ \Downarrow \\ - \sum_i \frac{\partial V}{\partial \mathbf{x}_i} \cdot (\mathbf{k} \times \mathbf{x}_i) &= 0. \end{aligned} \quad (105)$$

It is easy to verify that this last identity holds if the potential is invariant with respect to rotation around the axis $\mathbf{k}$. The conserved Noether current corresponding to the variation (104) is according to (98)

$$\begin{aligned}
j &= - \sum_{i=1}^n (\mathbf{k} \times \mathbf{x}_i) \cdot \frac{\partial L}{\partial \mathbf{x}'_i} \\
&= - \sum_{i=1}^n (\mathbf{k} \times \mathbf{x}_i) \cdot (m_i \mathbf{x}'_i) \\
&= - \sum_{i=1}^n \mathbf{k} \cdot (\mathbf{x}_i \times m_i \mathbf{x}'_i) \\
&= -\mathbf{k} \cdot \sum_{i=1}^n \mathbf{x}_i \times m_i \mathbf{x}'_i = -\mathbf{k} \cdot \mathbf{L},
\end{aligned}$$

where $\mathbf{L}$ is the total angular momentum of the system of mass-points.

Thus, if the Lagrangian is invariant with respect to rotations around some axis $\mathbf{k}$, then the component of the total angular momentum along $\mathbf{k}$ is conserved.

If the system is invariant with respect to rotations around three separate axes, that together span $\mathbf{R}^3$, the total angular momentum is conserved. This implies that the potential is invariant with respect to all possible rotations in $\mathbf{R}^3$.

Like for the location in time and space, this invariance with respect to direction, which is called *isotropy*, is something we *choose* to impose on all our fundamental natural laws.

The consequence of this choice is that we will have conservation of angular momentum.

Our natural laws in general involve fields and partial differential equations, Maxwell's equations for the electromagnetic field is an example that comes to mind.

In all these cases, invariance with respect to location in space and time and with respect to direction in space will lead to conserved Noether currents that we, by convention, call energy, momentum and angular momentum of the fields.

4.3 Several independent variables

Let us first consider a functional of the form

$$T(u) = \int_D dx dy \mathcal{L}(x, y, u, u_x, u_y), \quad (106)$$

and let us consider a variation

$$u(x, y) \rightarrow u(x, y) + \epsilon \eta(x, y), \quad (107)$$

where η is some *specific* function. Inserting (107) into (106) we get

$$\begin{aligned}
T(u + \epsilon \eta) &= \int_D dx dy \mathcal{L}(x, y, u + \epsilon \eta, u_x + \epsilon \eta_x, u_y + \epsilon \eta_y) \\
&= \int_D dx dy \left\{ \mathcal{L}(x, y, u, u_x, u_y) + \frac{\partial \mathcal{L}}{\partial u} \epsilon \eta + \frac{\partial \mathcal{L}}{\partial u_x} \epsilon \eta_x \right. \\
&\quad \left. + \frac{\partial \mathcal{L}}{\partial u_y} \epsilon \eta_y \right\} + \mathcal{O}(\epsilon^2) \\
&= T(u) + \epsilon \int_D dx dy \left\{ \frac{\partial \mathcal{L}}{\partial u} \eta + \frac{\partial \mathcal{L}}{\partial u_x} \eta_x + \frac{\partial \mathcal{L}}{\partial u_y} \eta_y \right\} + \mathcal{O}(\epsilon).
\end{aligned}$$

The functional (106) is said to be invariant under the variation (107), if there exists functions $F_1(x, y)$, $F_2(x, y)$ such that

$$\frac{\partial \mathcal{L}}{\partial u} \eta + \frac{\partial \mathcal{L}}{\partial u_x} \eta_x + \frac{\partial \mathcal{L}}{\partial u_y} \eta_y = \partial_x F_1 + \partial_y F_2 \quad (108)$$

If $F_1 = F_2 = 0$ we say that the Lagrangian density $\mathcal{L}$, is invariant under the variation (107).

Let us now assume that the functional T is invariant under a variation of the form (107) for some specific choice of $\eta(x, y)$. Using this $\eta(x, y)$, we consider the more general variation

$$u(x, y) \rightarrow u(x, y) + \epsilon(x, y) \eta(x, y), \quad (109)$$

where $\epsilon(x, y)$ is a numerically small function that vanished on the boundary of D but is otherwise arbitrary. Inserting (109) into the functional (106) we get

$$\begin{aligned}
T(u + \epsilon \eta) &= \int_D dx dy \mathcal{L}(x, y, u + \epsilon \eta, u_x + \epsilon \eta_x + \epsilon_x \eta, u_y + \epsilon \eta_y + \epsilon_y \eta) \\
&= \int_D dx dy \left\{ \mathcal{L}(x, y, u, u_x, u_y) + \frac{\partial \mathcal{L}}{\partial u} \epsilon \eta + \frac{\partial \mathcal{L}}{\partial u_x} (\epsilon \eta_x + \epsilon_x \eta) \right. \\
&\quad \left. + \frac{\partial \mathcal{L}}{\partial u_y} (\epsilon \eta_y + \epsilon_y \eta) \right\} + \mathcal{O}(\epsilon^2) \\
&= T(u) + \int_D dx dy \left\{ \epsilon \left(\frac{\partial \mathcal{L}}{\partial u} \eta + \frac{\partial \mathcal{L}}{\partial u_x} \eta_x + \frac{\partial \mathcal{L}}{\partial u_y} \eta_y \right) \right. \\
&\quad \left. + \eta \frac{\partial \mathcal{L}}{\partial u_x} \epsilon_x + \eta \frac{\partial \mathcal{L}}{\partial u_y} \epsilon_y \right\} + \mathcal{O}(\epsilon^2) \\
&= T(u) + \int_D dx dy \left\{ \epsilon (\partial_x F_1 + \partial_x F_2) - \partial_x \left(\eta \frac{\partial \mathcal{L}}{\partial u_x} \right) \epsilon \right. \\
&\quad \left. - \partial_y \left(\eta \frac{\partial \mathcal{L}}{\partial u_y} \right) \epsilon \right\} + \mathcal{O}(\epsilon^2), \quad (110)
\end{aligned}$$

where we have used the divergence theorem and the boundary conditions on ϵ in addition to the definition (108) of invariance of T with respect to the variation (107).

Thus from (110) we have

$$T(u + \epsilon \eta) = T(u) + \int_D dx dy \epsilon \{ \partial_x j_1 + \partial_y j_2 \} + \mathcal{O}(\epsilon^2), \quad (111)$$

where the Noether current $\mathbf{j} = (j_1, j_2)$ is

$$\begin{aligned} j_1 &= F_1 - \eta \frac{\partial \mathcal{L}}{\partial u_x}, \\ j_2 &= F_2 - \eta \frac{\partial \mathcal{L}}{\partial u_y}. \end{aligned} \quad (112)$$

Equation (111) holds for all u . If u is extremal, all variations of $T(u)$ vanish, in particular they vanish for variations of the form (109). Using a 2D version of the fundamental lemma we conclude that the Noether current satisfies

$$\partial_x j_1 + \partial_y j_2 = 0. \quad (113)$$

This kind of identity is in general called a *local conservation law*.

The 2D Laplace equation In example 9 we discussed the functional

$$T(u) = \iint_D dx \, dy \left\{ \frac{1}{2} u_x^2 + \frac{1}{2} u_y^2 \right\}.$$

The Lagrangian density $\mathcal{L}$, is here

$$\mathcal{L} = \frac{1}{2} u_x^2 + \frac{1}{2} u_y^2,$$

and we found that the corresponding Euler-Lagrange equation is the Laplace equation

$$u_{xx} + u_{yy} = 0.$$

Observe that $\mathcal{L}$ invariant under the variation

$$u(x, y) \rightarrow u(x, y) + \epsilon.$$

Thus $\eta = 1$ and the components of the Noether current are

$$\begin{aligned} j_1 &= -\frac{\partial \mathcal{L}}{\partial u_x} = -u_x, \\ j_2 &= -\frac{\partial \mathcal{L}}{\partial u_y} = -u_y. \end{aligned}$$

The local conservation law (113) is thus

$$\begin{aligned} \partial_x j_1 + \partial_y j_2 &= 0, \\ \Downarrow \\ u_{xx} + u_{yy} &= 0. \end{aligned}$$

This invariance clearly does not tell us anything new since it's local conservation law is just the Laplace equation itself.

We also observe that $\mathcal{L}$ is independent of x and y .

Let us consider an arbitrary infinitesimal translation in the plane

$$\begin{aligned} x &\rightarrow x + \epsilon a, \\ y &\rightarrow y + \epsilon b, \end{aligned} \quad (114)$$

$$a^2 + b^2 = 1.$$

Thus $\mathbf{n} = (a, b)$ is a unit vector determining the direction of the translation. The infinitesimal translation (114) induces the following variation in u

$$u(x, y) \rightarrow u(x, y) + \epsilon \eta(x, y),$$

where

$$\eta(x, y) = (a u_x + b u_y).$$

Observe that for this η we have

$$\begin{aligned} & \frac{\partial \mathcal{L}}{\partial u} \eta + \frac{\partial \mathcal{L}}{\partial u_x} \eta_x + \frac{\partial \mathcal{L}}{\partial u_y} \eta_y \\ & \frac{\partial \mathcal{L}}{\partial u} (a u_x + b u_y) + \frac{\partial \mathcal{L}}{\partial u_x} (a u_{xx} + b u_{xy}) + \frac{\partial \mathcal{L}}{\partial u_y} (a u_{xy} + b u_{yy}) \\ & = a \left(\frac{\partial \mathcal{L}}{\partial u} u_x + \frac{\partial \mathcal{L}}{\partial u_x} u_{xx} + \frac{\partial \mathcal{L}}{\partial u_y} u_{xy} \right) \\ & = b \left(\frac{\partial \mathcal{L}}{\partial u} u_y + \frac{\partial \mathcal{L}}{\partial u_x} u_{xy} + \frac{\partial \mathcal{L}}{\partial u_y} u_{yy} \right) \\ & = \partial_x (a \mathcal{L}) + \partial_y (b \mathcal{L}), \end{aligned}$$

since $\mathcal{L}$ does not depend explicitly on x and y .

Thus from (108) we conclude that

$$F_1 = a \mathcal{L}, \quad F_2 = b \mathcal{L},$$

and the components of the Noether current are

$$\begin{aligned} j_1 &= a \mathcal{L} - (a u_x + b u_y) \frac{\partial \mathcal{L}}{\partial u_x} \\ &= a \mathcal{L} - (a u_x + b u_y) u_x, \\ j_2 &= b \mathcal{L} - (a u_x + b u_y) \frac{\partial \mathcal{L}}{\partial u_y} \\ &= b \mathcal{L} - (a u_x + b u_y) u_y, \end{aligned}$$

and we can conclude that the Laplace equation has the local conservation law

$$\partial_x j_1 + \partial_y j_2 = 0 \tag{115}$$

Let us show by direct calculations that this is indeed a local conservation law that holds for all solutions to the Laplace equation.

We have

$$\begin{aligned}
& \partial_x j_1 + \partial_y j_2 = 0 \\
& \quad \Updownarrow \\
& a \partial_x \left(\frac{1}{2} u_x^2 + \frac{1}{2} u_y^2 \right) - \partial_x (a u_x^2 + b u_x u_y) \\
& + b \partial_y \left(\frac{1}{2} u_x^2 + \frac{1}{2} u_y^2 \right) - \partial_y (a u_x u_y + b u_y^2) = 0 \\
& \quad \Updownarrow \\
& a u_x u_{xx} + a u_y u_{xy} - 2 a u_x u_{xx} - b u_{xx} u_y \\
& \quad - b u_x u_{xy} + b u_x u_{xy} + b u_y u_{yy} \\
& - a u_{xy} u_y - a u_x u_{yy} - 2 b u_y u_{yy} = 0 \\
& \quad \Updownarrow \\
& -(a u_x + b u_y) (u_{xx} + u_{yy}) = 0.
\end{aligned}$$

Thus, the local conservation law (115) clearly holds for any solution to the Laplace equation.

The 1D wave equation Let us consider the well known 1D wave equation

$$u_{tt} - c^2 u_{xx} = 0.$$

This equation describe oscillations, $y = u(x, t)$, transverse to the x -axis, confined to a 2D plane, which we without loss of generality assume is the xy -coordinate plane in a xyz -Cartesian coordinate system in 3D physical space.

From example 10 we have seen that the 1D wave equation is the Euler-Lagrange equation for the functional

$$T(u) = \int_{x_0}^{x_1} dx \mathcal{L}.$$

where the Lagrangian density is

$$\mathcal{L} = \frac{1}{2} u_t^2 - \frac{1}{2} c^2 u_x^2.$$

Let us start by introducing an infinitesimal time translation

$$t \rightarrow t + \epsilon.$$

This time translation induces a variation of the form

$$u(t, x) \rightarrow u(t, x) + \epsilon u_t(t, x).$$

For this variation $\eta = u_t$ and we have

$$\begin{aligned}
& \frac{\partial \mathcal{L}}{\partial u} \eta + \frac{\partial \mathcal{L}}{\partial u_t} \eta_t + \frac{\partial \mathcal{L}}{\partial u_x} \eta_x \\
& = u_t (u_{tt}) + (-c^2 u_x) u_{xt} \\
& = u_t u_{tt} - c^2 u_x u_{xt} \\
& = \partial_t \left(\frac{1}{2} u_t^2 - \frac{1}{2} c^2 u_x^2 \right),
\end{aligned}$$

and thus, according to (108), the functional T is invariant with

$$\begin{aligned} F_1 &= \frac{1}{2} u_t^2 - \frac{1}{2} c^2 u_x^2, \\ F_2 &= 0. \end{aligned}$$

The components of the conserved Noether current are according to (112)

$$\begin{aligned} j_1 &= \frac{1}{2} u_t^2 - \frac{1}{2} c^2 u_x^2 - u_t u_t \\ &= -\left(\frac{1}{2} u_t^2 + \frac{1}{2} c^2 u_x^2\right), \\ j_2 &= -u_t (-c^2 u_x) = c^2 u_t u_x, \end{aligned}$$

and the local conservation law is

$$\partial_t j_1 + \partial_x j_2 = 0. \quad (116)$$

Introducing the quantities

$$\begin{aligned} e &= \frac{1}{2} u_t^2 + \frac{1}{2} c^2 u_x^2, \\ f &= -c^2 u_t u_x, \end{aligned} \quad (117)$$

the local conservation law (116) can be rewritten in the form

$$\partial_t e + \partial_x f = 0.$$

Next, define a quantity E by

$$\begin{aligned} E(t) &= \int_{x_0}^{x_1} dx e(t, x) \\ &= \int_{x_0}^{x_1} dx \left(\frac{1}{2} u_t^2 + \frac{1}{2} c^2 u_x^2 \right). \end{aligned}$$

For this quantity we evidently have

$$\frac{dE}{dt} = \int_{x_0}^{x_1} dx \partial_t e = - \int_{x_0}^{x_1} dx \partial_x f = c^2 (u_t u_x) \Big|_{x_0}^{x_1}.$$

Let us assume that u satisfies one of the following boundary conditions

$$u(t, x_0) = u(t, x_1) = 0, \quad \text{Dirichlet}$$

$$u_x(t, x_0) = u_x(t, x_1) = 0, \quad \text{Neumann}$$

$$x_0 = -\infty, \quad x_1 = +\infty, \quad \text{and } u(x, t) \text{ vanish at } \pm\infty$$

If this the case, we can conclude that

$$\begin{aligned} \frac{dE}{dt} &= 0, \\ &\Updownarrow \\ E &= \text{const}, \end{aligned}$$

and thus

$$E = \int_{x_0}^{x_1} dx \left(\frac{1}{2} u_t^2 + \frac{1}{2} c^2 u_x^2 \right),$$

is a *conserved quantity* for any solution to the wave equation.

It is in general true that, whenever we have a functional of time dependent fields, and the Lagrangian of the functional does not depend explicitly on time, the space integral of the time component of the Noether current will be a conserved quantity for appropriate boundary conditions at the spatial boundary. In general, for any application of Noether's theorem, the derived local conservation law will lead to a conserved quantity only for some select boundary conditions.

Inspired by the situation for systems of mass points, we *define* the space integral of the time component of the Noether current to be the total energy of the system of time dependent fields. Note that frequently one needs to scale the Noether current in such a way that the time component is positive, in that way ensuring that the total energy is a positive quantity.

The (scaled) time component of the Noether current is defined to be the *energy density* and the (scaled) space component(s) are called the *energy flux density*. For this particular example, $e(t, x)$ is the energy density, and $f(t, x)$ is the energy flux density.

In some cases, the formula for the energy density of a field system is known from physical modeling. In all such cases the energy density derived from the modeling is equal or proportional to the energy density defined using time translation invariance and Noether's theorem.

For the current example we know that the 1D wave equation is a model for small vibrations of a string. With respect to this example we recognize the first term in (117) to be proportional to the kinetic energy, K , and the second term to be proportional to the potential energy, V , of a small piece of the string.

Thus for this example we clearly we have

$$e \propto K + V.$$

Let us next look at translation along a direction $\mathbf{a} = (a, b)$, $a^2 + b^2 = 1$, in the xy -coordinate plane

$$\mathbf{x} = (x, y) \rightarrow \mathbf{x} + \epsilon \mathbf{a}.$$

This translation in the xy -coordinate planes induces the following variation on the function $u(t, x)$

$$\begin{aligned} u(t, x) &\rightarrow u(t, x + \epsilon a) + \epsilon b \\ &= u(t, x) + \epsilon a u_x(t, x) + \epsilon b \\ &= u + \epsilon (a u_x + b). \end{aligned}$$

Thus, for this variation $\eta = a u_x + b$ and we have

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial u} \eta + \frac{\partial \mathcal{L}}{\partial u_t} \eta_t + \frac{\partial \mathcal{L}}{\partial u_x} \eta_x \\ &= u_t (a u_{xt}) - c^2 u_x (a u_{xx}) \\ &= \left(\frac{1}{2} a u_t^2 - \frac{1}{2} c^2 a u_x^2 \right)_x. \end{aligned}$$

According to (108), the functional T is therefore invariant with

$$\begin{aligned} F_1 &= 0, \\ F_2 &= \frac{1}{2}au_t^2 - \frac{1}{2}c^2au_x^2. \end{aligned}$$

The components of the conserved Noether current are according to (112)

$$\begin{aligned} j_1 &= F_1 - \eta \frac{\partial \mathcal{L}}{\partial u_t} \\ &= -(au_x + b)u_t \\ &= -au_tu_x - bu_t, \\ j_2 &= F_2 - \eta \frac{\partial \mathcal{L}}{\partial u_x} \\ &= \frac{1}{2}au_t^2 - \frac{1}{2}c^2au_x^2 - (au_x + b)(-c^2u_x) \\ &= \frac{1}{2}au_t^2 + \frac{1}{2}c^2au_x^2 + bc^2u_x. \end{aligned}$$

and the local conservation law is

$$\partial_t j_1 + \partial_x j_2 = 0,$$

which can be rewritten in the form

$$\mathbf{a} \cdot (\partial_t \mathbf{p} + \partial_x \mathbf{f}) = 0, \quad (118)$$

where we have defined

$$\begin{aligned} \mathbf{p} &= (u_tu_x, u_t), \\ \mathbf{f} &= -\left(\frac{1}{2}u_t^2 + \frac{1}{2}c^2u_x^2, c^2u_x\right). \end{aligned} \quad (119)$$

Since (118) holds for all unit vectors $\mathbf{a}$, we have

$$\partial_t \mathbf{p} + \partial_x \mathbf{f} = 0$$

Again, inspired by the situation for systems of mass points, we define $\mathbf{p}(t, x)$ to be the *momentum density* and $\mathbf{f}(t, x)$ to be the *momentum flux density*.

Defining

$$\mathbf{P} = \int_{x_0}^{x_1} dx \, \mathbf{p}(t, x)$$

We have

$$\begin{aligned} \frac{d\mathbf{P}}{dt} &= \int_{x_0}^{x_1} dx \, \partial_t \mathbf{p} \\ &= - \int_{x_0}^{x_1} dx \, \partial_x \mathbf{f} \\ &= -\mathbf{f}(t, x)|_{x_0}^{x_1}. \end{aligned}$$

It is easy to verify that neither Dirichlet conditions, nor Neuman conditions, at the boundary points x_0, x_1 ensure that $\mathbf{f}(t, x)|_{x_0}^{x_1}$ equal zero. Thus, for these common boundary conditions for the 1D wave equation, the quantity $\mathbf{P}$ is not conserved. However, it is conserved if $x_0 \rightarrow -\infty, x_1 \rightarrow \infty$, and we assume that $u(x, t)$ is localized in the sense that u, u_t, u_x approach zero fast enough when $x \rightarrow \pm\infty$. It is also conserved if the interval is finite, and we can extend the function $u(x, t)$ to the whole real line as a periodic continuously differentiable function of period $x_1 - x_0$.

For these kind of boundary conditions, the conserved quantity $\mathbf{P}$ is by definition the *total momentum* for the system

As a conclusion of this example, let us consider an infinitesimal rotation of the xy -coordinate plane around the z -axis.

$$(x, y) \rightarrow (x, y) + \epsilon (-y, x).$$

This translation in the xy -coordinate planes induces the following variation on the function $u(t, x)$

$$u(t, x) \rightarrow u(t, x) + \epsilon x.$$

Thus, for this variation $\eta = x$ and we have

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial u} \eta + \frac{\partial \mathcal{L}}{\partial u_t} \eta_t + \frac{\partial \mathcal{L}}{\partial u_x} \eta_x \\ = -c^2 u_x \\ = (-c^2 u)_x. \end{aligned}$$

Therefore, according to (108), the functional T is invariant with

$$\begin{aligned} F_1 &= 0, \\ F_2 &= -c^2 u. \end{aligned}$$

The components of the conserved Noether current are according to (112)

$$\begin{aligned} j_1 &= F_1 - \eta \frac{\partial \mathcal{L}}{\partial u_t} \\ &= -xu_t, \\ j_2 &= F_2 - \eta \frac{\partial \mathcal{L}}{\partial u_x} \\ &= -c^2 u - x(-c^2 u_x) \\ &= c^2 xu_x - c^2 u. \end{aligned}$$

The local conservation law can then be written in the form

$$\partial_t m + \partial_x f_m = 0,$$

where

$$\begin{aligned} m(t, x) &= xu_t, \\ f_m(t, x) &= c^2(u - xu_x), \end{aligned}$$

are by definition the *angular momentum density* and the *angular momentum flux density*. The total angular momentum for the system is

$$M(t) = \int_{x_0}^{x_1} dx \, m(t, x), \quad (120)$$

and using the local conservation law we have

$$\begin{aligned} \frac{dM}{dt} &= \int_{x_0}^{x_1} dx \, \partial_t m(t, x) \\ &= - \int_{x_0}^{x_1} dx \, \partial_x f_m(t, x) = -f_m(t, x)|_{x_0}^{x_1}. \end{aligned}$$

In general, the total angular momentum is not conserved. It is however conserved if $x_0 \rightarrow -\infty$, $x_1 \rightarrow \infty$, and we assume that $u(x, t)$ is localized. Note that for this case, assuming the the interval is finite, extending u to the whole real line as a continuously differentiable function of period $x_1 - x_0$, does *not* ensure that the total angular momentum is conserved.

As a cross check on our definitions of momentum density and angular momentum density for this system, recall that for systems of mass point of total momentum $\mathbf{p}$, the angular momentum is given in terms of the momentum by the formula $\mathbf{m} = \mathbf{x} \times \mathbf{p}$. For the current system, the 1D wave equation, we have found the momentum density in (119).

Using the notation $\hat{\mathbf{x}}$, $\hat{\mathbf{y}}$ and $\hat{\mathbf{z}}$ for unit vector of the xyz -Cartesian coordinate system, we have

$$\begin{aligned} \mathbf{x} \times \mathbf{p} &= x \hat{\mathbf{x}} \times (u_t u_x \hat{\mathbf{x}} + u_t \hat{\mathbf{y}}) \\ &= x u_t \hat{\mathbf{z}} = m \hat{\mathbf{z}}. \end{aligned}$$

So, $m(t, x)$ is indeed the angular momentum density for the system. Let us next consider functionals of the form

$$T(u) = \int_D dx \, dy \int_{t_0}^{t_1} dt \, \mathcal{L}(t, x, y, u, u_t, u_x, u_y). \quad (121)$$

Following the by now familiar procedure, we define (121) to be invariant under a variation

$$u(t, x, y) \rightarrow u(t, x, y) + \epsilon \eta(t, x, y),$$

if there exists functions F_1, F_2 and F_3 such that

$$\frac{\partial \mathcal{L}}{\partial u} \eta + \frac{\partial \mathcal{L}}{\partial u_t} \eta_t + \frac{\partial \mathcal{L}}{\partial u_x} \eta_x + \frac{\partial \mathcal{L}}{\partial u_y} \eta_y = \partial_t F_1 + \partial_x F_2 + \partial_y F_3.$$

The conserved Noether current components are

$$\begin{aligned} j_1 &= F_1 - \eta \frac{\partial \mathcal{L}}{\partial u_t}, \\ j_2 &= F_2 - \eta \frac{\partial \mathcal{L}}{\partial u_x}, \\ j_3 &= F_3 - \eta \frac{\partial \mathcal{L}}{\partial u_y}, \end{aligned} \quad (122)$$

and the local conservation law is

$$\partial_t j_1 + \partial_x j_2 + \partial_y j_3 = 0.$$

This is Noether's theorem for the functional (121).

2D wave equation The 2D wave equation has been seen, in example 10, to be variational with Lagrangian

$$\mathcal{L} = \frac{1}{2} u_t^2 - \frac{1}{2} c^2 u_x^2 - \frac{1}{2} c^2 u_y^2.$$

This Lagrangian density is translation invariant with respect to time and space and this will lead to conserved Noether currents whose spatial integral of the time component of the current will be the total energy and the total momentum. Let us start with time translation

$$t \rightarrow t + \epsilon,$$

which leads to the variation

$$u(t, x, y) \rightarrow u(t, x, y) + \epsilon u_t(t, x, y).$$

In a calculation entirely similar to the one in example ?? we find that the functional is invariant with $F_1 = \mathcal{L}$, $F_2 = F_3 = 0$.

The components of the Noether current are then from (122)

$$\begin{aligned} j_1 &= \frac{1}{2} u_t^2 - \frac{1}{2} c^2 u_x^2 - \frac{1}{2} c^2 u_y^2 - u_t (u_t) \\ &= -\left(\frac{1}{2} u_t^2 + \frac{1}{2} c^2 u_x^2 + \frac{1}{2} c^2 u_y^2\right), \\ j_2 &= -u_t (-c^2 u_x) = c^2 u_t u_x, \\ j_3 &= -u_t (-c^2 u_y) = c^2 u_t u_y, \end{aligned}$$

and the conservation law is

$$\partial_t j_1 + \partial_x j_2 + \partial_y j_3 = 0.$$

Defining the energy density to be

$$e = \frac{1}{2} u_t^2 + \frac{1}{2} c^2 u_x^2 + \frac{1}{2} c^2 u_y^2,$$

and the energy flux density to be

$$\mathbf{f} = -c^2 u_t \nabla u,$$

where ∇ is the 2D gradient operator, we have

$$\partial_t e + \nabla \cdot \mathbf{f} = 0. \tag{123}$$

For the total energy inside a domain $D \subset \mathbf{R}^2$ we have

$$E = \int_D dx dy e,$$

and using (123) we get

$$\begin{aligned}\frac{dE}{dt} &= \int_D dx dy \partial_t e = - \int_D dx dy \nabla \cdot \mathbf{f} \\ &= - \int_{\partial D} dl \mathbf{f} \cdot \mathbf{n}\end{aligned}\tag{124}$$

The sign convention chosen for $\mathbf{f}$ ensures that $\mathbf{f} \cdot \mathbf{n} > 0$ means that energy is *leaving* the domain D . This convention is standard.

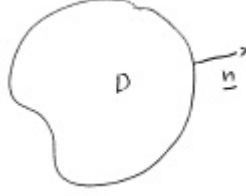


Figure 15: The integration domain for the functional and convention for unit normal on the boundary

Equation (124) clearly expresses conservation of energy; the only way energy can change in D is if energy is escaping across the boundary. We get conservation of energy if the energy flux is zero on ∂D , which would happen if for example $u|_D$ is independent of time, or if $\partial_{\mathbf{n}u}|_D = 0$, or if $D = \mathbb{R}^2$ and solutions vanish fast enough at infinity.

$$\frac{d}{dt} \int_D dx dy \left\{ \frac{1}{2} u_t^2 + \frac{1}{2} c^2 u_x^2 + \frac{1}{2} c^2 u_y^2 \right\} = 0.$$

Let us next look at translation along a direction $\mathbf{a} = (a, b)$, $a^2 + b^2 = 1$

$$\mathbf{x} = (x, y) \rightarrow \mathbf{x} + \epsilon \mathbf{a}.$$

The corresponding variation is clearly

$$u(t, x, y) \rightarrow u(t, x, y) + \epsilon(a u_x + b u_y),$$

and for this variation we have

$$\begin{aligned}& \frac{\partial \mathcal{L}}{\partial u} \eta + \frac{\partial \mathcal{L}}{\partial u_t} \eta_t + \frac{\partial \mathcal{L}}{\partial u_x} \eta_x + \frac{\partial \mathcal{L}}{\partial u_y} \eta_y \\ &= \frac{\partial \mathcal{L}}{\partial u_t} (a u_{xt} + b u_{yt}) + \frac{\partial \mathcal{L}}{\partial u_x} (a u_{xx} + b u_{xy}) + \frac{\partial \mathcal{L}}{\partial u_y} (a u_{xy} + b u_{yy}) \\ &= a \left(\frac{\partial \mathcal{L}}{\partial u_t} u_{xt} + \frac{\partial \mathcal{L}}{\partial u_x} u_{xx} + \frac{\partial \mathcal{L}}{\partial u_y} u_{xy} \right) \\ &+ b \left(\frac{\partial \mathcal{L}}{\partial u_t} u_{yt} + \frac{\partial \mathcal{L}}{\partial u_x} u_{yx} + \frac{\partial \mathcal{L}}{\partial u_y} u_{yy} \right) \\ &= a \partial_x \mathcal{L} + b \partial_y \mathcal{L}.\end{aligned}$$

So the functional is invariant with $F_1 = 0$, $F_2 = a \mathcal{L}$, $F_3 = b \mathcal{L}$. The components of the Noether current are from (122)

$$\begin{aligned}
j_1 &= -(a u_x + b u_y) u_t = -u_t (a u_x + b u_y), \\
j_2 &= a \left(\frac{1}{2} u_t^2 - \frac{1}{2} c^2 u_x^2 - \frac{1}{2} c^2 u_y^2 \right) - (a u_x + b u_y) (-c^2 u_x) \\
&= a \frac{1}{2} u_t^2 - a \frac{1}{2} c^2 u_x^2 - a \frac{1}{2} c^2 u_y^2 + a c^2 u_x^2 + b c^2 u_x u_y \\
&= \frac{1}{2} a u_t^2 + \frac{1}{2} a c^2 u_x^2 - \frac{1}{2} a c^2 u_y^2 + b c^2 u_x u_y, \\
j_3 &= b \left(\frac{1}{2} u_t^2 - \frac{1}{2} c^2 u_x^2 - \frac{1}{2} c^2 u_y^2 \right) - (a u_x + b u_y) (-c^2 u_y) \\
&= \frac{1}{2} b u_t^2 - \frac{1}{2} b c^2 u_x^2 + \frac{1}{2} b c^2 u_y^2 + a c^2 u_x u_y,
\end{aligned}$$

and the local conservation law is

$$\partial_t j_1 + \partial_x j_2 + \partial_y j_3 = 0.$$

Inserting the expressions for the components of the Noether current into this local conservation law produces a very messy expression. However, if we introduce a vector $\mathbf{P}$ and a Cartesian tensor f of rank 2 by

$$\begin{aligned}
\mathbf{P} &= -u_t \nabla u, \\
f &= c^2 \nabla u \nabla u - \frac{1}{2} c^2 \text{Tr}(\nabla u \nabla u) I + \frac{1}{2} u_t^2 I,
\end{aligned}$$

where I is the identity matrix, we can write the local conservation law as

$$\partial_t \mathbf{a} \cdot \mathbf{P} + \nabla \cdot (\mathbf{a} \cdot f) = 0. \quad (125)$$

The argument leading up to (125) is true for all vectors $\mathbf{a}$. Therefore we have the local conservation law

$$\partial_t \mathbf{P} + \nabla \cdot f = 0. \quad (126)$$

By definition, $\mathbf{P}$ is the momentum density and f is the momentum flux density. The total momentum inside some domain D is

$$\mathcal{P} = \int_D dx dy \mathbf{P},$$

and from (126) we get

$$\frac{d\mathcal{P}}{dt} = \int_D dx dy \partial_t \mathbf{P} = - \int_{\partial D} dl f \cdot \mathbf{n}. \quad (127)$$

All of this can be generalized to the 3D wave equation. Then (127) will involve a surface integral of the momentum flux over a 2D surface ∂D bounding a 3D domain $D \subset \mathbf{R}^3$.

Formula (127), and formulas like it, for other field systems, have important practical applications.

Let us for example assume that an object, filling a domain $D \subset \mathbf{R}^2$, is embedded in a wave field satisfying the 2D-wave equation. Let us apply formula



Figure 16:

(127) to $\mathbf{R}^2 - D$ and assume that the wave field is localized so that we get no contribution from the boundary at infinity. The only boundary to $\mathbf{R}^2 - D$ is then ∂D . The outward normal to $\mathbf{R}^2 - D$ points into D .

Since total momentum is conserved, we know that momentum lost from $\mathbf{R}^3 - D$ to D through the boundary ∂D , must be gained by D . Similarly momentum gained by $\mathbf{R} - D$ must be lost by D .

If $\mathbf{Q}$ is the total momentum inside D we thus have, using (127), that

$$\frac{d\mathbf{Q}}{dt} = -\frac{d\mathbf{P}}{dt} = \int_{\partial D} dl \mathbf{f} \cdot \mathbf{n} = - \int_{\partial D} dl \mathbf{f} \cdot \mathbf{s},$$

where now $\mathbf{s} = -\mathbf{n}$ is the normal to ∂D pointing *out* of D .

Recall that change of momentum per. unit time is in fact *force*. Thus the force, $\mathbf{F} = \frac{d\mathbf{Q}}{dt}$, acting on the embedded object D in the wave field is

$$\mathbf{F} = - \int_{\partial D} dl \mathbf{f} \cdot \mathbf{s}.$$

This is an important result since it allows us, for example, to calculate the surface-wave induced mechanical forces on ship hulls and stationary structures like oil platforms or structures that generate electric power from surface waves.

We end our exposition of the calculus of variation and Noether's theorem here. This is not because this is the end of the story for Noether's theorem. For Noether's theorem, and also for the calculus of variations, we have barely scratched the surface. Beyond these notes there is a whole universe to explore.

5 Exercises

1. Find all extremals for the functional

$$I(y) = \int_{t_0}^{t_1} dt e^t \sqrt{1 + (y')^2}$$

2. Find the unique extremal for

$$I(y) = \int_0^1 dt (y'^2 - y^2 + 2ty)$$

satisfying the constraints $y(0) = 0, y(1) = 1$.

3. Show that there is no extremal for the functional

$$I(y) = \int_{-1}^1 dt t^4 (y')^2$$

that satisfy the constraints $y(-1) = -1, y(1) = 1$.

4. Find a unique smooth extremal for

$$I(y) = \int_2^3 dt y^2 (1 - y')^2$$

that satisfy the constraints $y(2) = 1, y(3) = \sqrt{3}$.

5. Derive the Euler-Lagrange equation for a functional of the general form

$$I(y) = \int_{t_0}^{t_1} dt \mathcal{L}(t, y, y', y'')$$

Find the unique extremal to the functional

$$I(y) = \int_0^1 dt (y'')^2$$

that satisfy the constraints

$$\begin{aligned} y(0) &= y'(0) = 0 \\ y(1) &= y'(1) = 0 \end{aligned}$$

6. Find the extremals for the functional

$$I(y) = \int_0^1 dt (y'')^2$$

subject to the constraints

$$J(y) = \int_0^1 dt y = L$$

and $y(0) = 0, y(1) = 2, y'(0) = 1, y'(1) = -1$.

7. Find the extremals of the functional

$$I(y) = \int_0^1 dt (y')^2$$

subject to the constraints

$$\begin{aligned} J_1(y) &= \int_0^1 dt y = 2 \\ J_2(y) &= \int_0^1 dt y t = \frac{1}{2} \end{aligned}$$

and $y(0) = y(1) = 0$.

8. On page 40 in these lecture notes, we discussed extremals of real valued functionals whose domain of definition consisted of a complex valued function A . We argued that the Euler-Lagrange equations could be found by varying the field A and its complex conjugate A^* as if they were independent variables. Let us consider such functionals of the form

$$I(A, A^*) = \int_{t_0}^{t_1} dt \mathcal{L}(t, A, A^*, A_t, A_t^*) \quad (128)$$

Show that the Euler-Lagrange equations for this type of functional are

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial A} - \frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial A_t} \right) &= 0 \\ \frac{\partial \mathcal{L}}{\partial A^*} - \frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial A_t^*} \right) &= 0 \end{aligned}$$

Since these two equations are complex conjugate of each other we only need one of them. Find the Euler-Lagrange equations corresponding to the functionals

a)

$$I(A, A^*) = \int_{t_0}^{t_1} dt (A_t A_t^* + A A^* A A^*) \quad (129)$$

b)

$$I(A, A^*) = \int_{t_0}^{t_1} dt \left(\frac{1}{2} i (A A_t^* - A^* A_t) + \frac{1}{2} A A^* A A^* \right) \quad (130)$$

9. A functional of the type (128) is said to be invariant under variations of the form

$$\begin{aligned} A(t) &\mapsto A(t) + \varepsilon \eta(t) \\ A^*(t) &\mapsto A^*(t) + \varepsilon \eta^*(t) \end{aligned} \quad (131)$$

if there exists a function $F(t)$ such that

$$\frac{\partial \mathcal{L}}{\partial A} \eta + \frac{\partial \mathcal{L}}{\partial A^*} \eta^* + \frac{\partial \mathcal{L}}{\partial A_t} \eta_t + \frac{\partial \mathcal{L}}{\partial A_t^*} \eta_t^* = \frac{dF}{dt}$$

If $F = 0$ we say that the Lagrangian is invariant. Show that the conserved Noether current corresponding to the variation (131) is

$$j = F - \eta \frac{\partial \mathcal{L}}{\partial A_t} - \eta^* \frac{\partial \mathcal{L}}{\partial A_t^*}$$

Show that the Lagrangian in both functionals (129) and (130) are invariant under an infinitesimal rotation of the complex phase of A .

$$\begin{aligned} A(t) &\mapsto A(t) + i\varepsilon A(t) \\ A^*(t) &\mapsto A^*(t) - i\varepsilon A^*(t) \end{aligned} \quad (132)$$

Find the conserved Noether current corresponding to the variation (132) for the two functionals (129) and (130) and show directly using the Euler-Lagrange equations that the Noether currents are indeed conserved.

10. Consider real valued functionals of the form

$$I(A, A^*) = \int_D dx dy \int_{t_0}^{t_0} dt \mathcal{L}(t, A, A^*, A_t, A_t^*, A_x, A_x^*, A_y, A_y^*) \quad (133)$$

Show that the Euler-Lagrange equations for this functional are

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial A} - \partial_t \left(\frac{\partial \mathcal{L}}{\partial A_t} \right) - \partial_x \left(\frac{\partial \mathcal{L}}{\partial A_x} \right) - \partial_y \left(\frac{\partial \mathcal{L}}{\partial A_y} \right) &= 0 \\ \frac{\partial \mathcal{L}}{\partial A^*} - \partial_t \left(\frac{\partial \mathcal{L}}{\partial A_t^*} \right) - \partial_x \left(\frac{\partial \mathcal{L}}{\partial A_x^*} \right) - \partial_y \left(\frac{\partial \mathcal{L}}{\partial A_y^*} \right) &= 0 \end{aligned}$$

Find the Euler-Lagrange equation corresponding to the functional

$$I(A, A^*) = \int_D dx dy \int_{t_0}^{t_0} dt (A_t A_t^* - c^2 A_x A_x^* - c^2 A_y A_y^* - m A A^*) \quad (134)$$

This equation is called the complex Klein-Gordon equation and describe, among other things, charged spin-less elementary particles. In this context m is the mass of the elementary particle.

11. A functional of the type (133) is said to be invariant under variations of the form

$$\begin{aligned} A(t, x, y) &\mapsto A(t, x, y) + \varepsilon \eta(t, x, y) \\ A^*(t, x, y) &\mapsto A^*(t, x, y) + \varepsilon \eta^*(t, x, y) \end{aligned} \quad (135)$$

if there exists functions $F_j = F_j(t, x, y)$ for $j = 1, 2, 3$ such that

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial A} \eta + \frac{\partial \mathcal{L}}{\partial A^*} \eta^* + \frac{\partial \mathcal{L}}{\partial A_t} \eta_t + \frac{\partial \mathcal{L}}{\partial A_t^*} \eta_t^* + \frac{\partial \mathcal{L}}{\partial A_x} \eta_x \\ + \frac{\partial \mathcal{L}}{\partial A_x^*} \eta_x^* + \frac{\partial \mathcal{L}}{\partial A_y} \eta_y + \frac{\partial \mathcal{L}}{\partial A_y^*} \eta_y^* = \partial_t F_1 + \partial_x F_2 + \partial_y F_3 \end{aligned}$$

If $F_1 = F_2 = F_3 = 0$, we say that the Lagrangian is invariant. Show that the components of the Noether current corresponding to the variation (135) are

$$\begin{aligned} j_1 &= F_1 - \eta \frac{\partial \mathcal{L}}{\partial A_t} - \eta^* \frac{\partial \mathcal{L}}{\partial A_t^*} \\ j_2 &= F_2 - \eta \frac{\partial \mathcal{L}}{\partial A_x} - \eta^* \frac{\partial \mathcal{L}}{\partial A_x^*} \\ j_3 &= F_3 - \eta \frac{\partial \mathcal{L}}{\partial A_y} - \eta^* \frac{\partial \mathcal{L}}{\partial A_y^*} \end{aligned}$$

The conservation law for the Noether current is

$$\partial_t j_1 + \partial_x j_2 + \partial_y j_3 = 0$$

12. Consider the following two functionals

$$I(\psi, \psi^*) = \int_D dx dy \int_{t_0}^{t_0} dt \left(\frac{\hbar}{2} i (\psi^* \psi_t - \psi \psi_t^*) \right) \quad (136)$$

$$- \frac{\hbar^2}{2m} (\psi_x \psi_x^* + \psi_y \psi_y^*) - V(x, y) \psi \psi^* \quad (137)$$

$$I(\psi, \psi^*) = \int_D dx dy \int_{t_0}^{t_0} dt (\psi_t \psi_t^* - c^2 \psi_x \psi_x^* \quad (138)$$

$$- c^2 \psi_y \psi_y^* - m \psi \psi^*) \quad (139)$$

The first functional is discussed on page 43 in the lecture notes on variational calculus. There we proved that the Euler-Lagrange equation for this functional is the quantum mechanical Schrödinger equation. The second functional we discussed in problem 10. There we proved that the corresponding Euler-Lagrange equation is the complex Klein-Gordon equation. Show that the Lagrangian for the functionals (136) and (138) are invariant under an infinitesimal rotation of the complex phase of ψ .

$$\begin{aligned} \psi(t, x, y) &\mapsto \psi(t, x, y) + i\varepsilon \psi(t, x, y) \\ \psi^*(t, x, y) &\mapsto \psi^*(t, x, y) - i\varepsilon \psi^*(t, x, y) \end{aligned} \quad (140)$$

- a) Find the Noether current and its conservation law corresponding to the infinitesimal phase variation (140) for the Schrödinger functional (136).

The Schrödinger equation was derived by Erwin Schrödinger in 1925. Initially it was not at all clear what the physical interpretation of the wave function ψ should be. Schrödinger himself favored initially an interpretation in terms of charge density, but could not make it work and this interpretation was abandoned. The interpretation that lives on to this day was given by Max Born in 1926. In this interpretation $\psi \psi^*(t, x, y)$ is the probability for finding the electron at a point (x, y) at time t . Schrödinger never accepted this interpretation, neither did Einstein.

In what way does the conservation law corresponding to the invariance of the Lagrangian under the infinitesimal phase variation (140) support the interpretation introduced by Max Born?

- b) Find the Noether current and its conservation law corresponding to the infinitesimal phase variation (140) for the complex Klein-Gordon functional (138).

The Klein-Gordon equation was in fact first derived by Erwin Schrödinger in 1925 as a quantum mechanical equation for the electron. In his mind the Klein-Gordon equation was much more likely to be the right equation for the electron than what we today call the Schrödinger equation. It is for example invariant under Lorentz transformations and thus respect the fundamental rules of Einsteins special theory of relativity. Today's Schrödinger equation is an approximation to the Klein-Gordon equation that is valid only for electrons moving slowly compared to the speed of light, and it is not Lorentz invariant and thus does not respect Einsteins special theory of relativity. However

Schrödinger was forced to abandon his fully relativistic Klein-Gordon equation for the electron because it turned out to be inconsistent with known atomic spectral data.

Argue, using the the conservation law corresponding to the invariance of the Lagrangian under the infinitesimal phase variation (140), that it is not possible to generalize Max Born's interpretation to the Klein-Gordon equation by constructing a probability density from ψ . This is another reason why it was abandoned at the time.

It turned out that abandoning the Klein-Gordon equation was premature. It has since then been reintroduced as a quantum equation, but for charged, spin-less particles. In this context the interpretation is nowhere near the original one introduced by Max Born. Today the wave function in the Klein-Gordon equation is interpreted as a field of operators that create particles from the void and return them to the void by annihilation.

One might be surprised at how much insight there is to be gained by using the simple fact that the global phase of the quantum mechanical wave function is arbitrary. Thus nothing change if we make the substitution

$$\psi \mapsto \psi e^{i\alpha}$$

The conserved Noether currents corresponding to invariance with respect to phase, that you have found in problem 12, is however not close to the end of the story. When this phase invariance is coupled to the Gauge Principle, which is closely related to Noether's theorem, the existence of the electromagnetic field and the form of it's interaction with electrons is determined. And there is more: The quantum mechanical wave equations that describe weakly interacting particles like neutrinos, and strongly interacting ones like quarks, also has an invariance with respect to rotation of the global phase. However, for these equations the arbitrary phase is multidimensional and the arbitrary rotation of phase involve 2×2 matrices for the weakly interacting case, and 3×3 matrices for the strongly interacting case. When the Gauge principle is applied to these two cases, the existence of the weak interaction field and the strong interaction field, and their interaction with their respective particles, like neutrinos and quarks, are determined, just like in the case of electromagnetics. The field equations for weak interactions and strong interactions are generalizations of the Maxwell equations involving more than one vector potential. Also these generalized Maxwell-like equations are nonlinear, not linear like the electromagnetic Maxwell equation.

6 Appendix

6.1 The maximum entropy principle for classical systems

Let $x_1, \dots, x_n$ be random variables with an associated probability distribution $\rho(x_1, \dots, x_n)$. Let $f_1(x_1, \dots, x_n), \dots, f_p(x_1, \dots, x_n)$ be functions defined on the space of random variables $\Omega = \{(x_1, x_2, \dots, x_n)\}$ where the variables x_n can run over a finite set, an infinite discrete set, for example a set indexed by a finite set

of integers, or the variables can run over the real numbers. We will usually think about the real number case and will therefore write integrals instead of sums. The functions f_j are our *observables*. Their *expectation* values are as usual defined by

$$\langle f_j \rangle = \int_{\mathbf{R}^n} dV f_j(x_1, \dots, x_n) \rho(x_1, \dots, x_n). \quad (1)$$

The expectation value of a given observable of course depends on which probability distribution, ρ , we use. The challenge in statistics is to figure out which probability distribution one should use in any given situation. Let us say that we for some reason, (expert knowledge, guesswork, hearsay, ...) believe that a probability distribution ρ_0 accurately represents what we currently know about a given system. The probability distribution ρ_0 is called the *prior distribution*, or just the *prior*.

Let us next assume that we measure the mean values of the observables $f_1, \dots, f_p$ and find the values $c_1, \dots, c_p$. If

$$\langle f_j \rangle_0 = \int_{\mathbf{R}^n} dV f_j(x_1, \dots, x_n) \rho_0(x_1, \dots, x_n) = c_j \quad (2)$$

for all j , we are satisfied with our choice of prior. It predicts exactly the mean values that are observed.

But we might not be so lucky. Perhaps

$$\langle f_j \rangle_0 \neq c_j, \quad (3)$$

for at least one j . Our selected ρ_0 is then not the correct one, it predicts expectation values that are not observed. The challenge is to modify ρ_0 into a new distribution ρ that is consistent with *all* the observed mean values.

For this purpose we define a functional $S(\rho)$ by

$$S(\rho) = - \int_{\mathbf{R}^n} dV \rho \ln \left(\frac{\rho}{\rho_0} \right). \quad (4)$$

S is by definition the *relative entropy* of the probability distribution ρ with respect to ρ_0 . We will see later that our use of the word entropy here is consistent with its usage in thermodynamics.

The *maximum entropy* principle states that one should choose the probability distribution that maximizes the functional

$$S(\rho) = - \int_{\mathbf{R}^n} dV \rho \ln \left(\frac{\rho}{\rho_0} \right), \quad (5)$$

subject to the constraints

$$\langle f_j \rangle = \int_{\mathbf{R}^n} dV f_j \rho = c_j, \quad j = 1, 2, \dots, p. \quad (6)$$

6.1.1 The general thermodynamical formalism

In this section we will solve the maximum principle stated in the previous section using the calculus of variations. The problem will initially be solved in the general setting described in the previous section, but we will eventually specialize to the case of statistical mechanics.

In order to proceed we must first recognize that in addition to the p constraints (6), we have one more constraint that simply expresses the fact that ρ is a probability distribution.

$$\langle 1 \rangle = \int_{\mathbf{R}^n} dV \rho(x_1, \dots, x_n) = 1, \quad (7)$$

and we thus have $p+1$ constraints and therefore introduce an extended functional

$$T(\rho) = S(\rho) - \lambda_0 \langle 1 \rangle - \sum_{j=1}^p \lambda_j \langle f_j \rangle, \quad (8)$$

Note that we could have written

$$T(\rho) = S(\rho) - \lambda_0 (\langle 1 \rangle - 1) - \sum_{j=1}^p \lambda_j (\langle f_j \rangle - c_j), \quad (9)$$

in order to make the values of the constraints explicit. However, all constant terms vanish when we take variational derivative, so we might as well drop the constant terms. Also note that our choice of minus sign in front of the Lagrange multiplier terms in (8) and (9) is a convention inspired by the application of this formalism to the case of statistical mechanics.

The integral density corresponding to the extended functional $T(\rho)$ is

$$\mathcal{L} = -\rho \ln\left(\frac{\rho}{\rho_0}\right) - \lambda_0 \rho - \sum_{j=1}^p \lambda_j f_j \rho. \quad (10)$$

Observe that $\mathcal{L}$ does not depend on any derivatives of ρ . The Euler-Lagrange equation for T is therefore simply

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial \rho} &= 0, \\ \Downarrow \\ -\ln\left(\frac{\rho}{\rho_0}\right) - 1 - \lambda_0 - \sum_{j=1}^p \lambda_j f_j &= 0, \end{aligned} \quad (11)$$

whose solution is

$$\rho = \frac{\rho_0}{Z} \exp\left\{-\sum_j \lambda_j f_j\right\},$$

where we have defined $Z = \exp\{(1 + \lambda_0)\}$. In order for the constraint $\langle 1 \rangle = 1$ to be satisfied, we must have

$$\begin{aligned}
\langle 1 \rangle &= 1, \\
&\Downarrow \\
\int_{\mathbf{R}^n} dV \frac{\rho_0}{Z} \exp\left\{-\sum_j \lambda_j f_j\right\} &= 1, \\
&\Downarrow \\
Z = Z(\lambda_1, \dots, \lambda_p) &= \int_{\mathbf{R}^n} dV \rho_0 \exp\left\{-\sum_j \lambda_j f_j\right\}, \quad (12)
\end{aligned}$$

and the stationary distribution is

$$\rho(x_1, \dots, x_n) = \frac{\rho_0(x_1, \dots, x_n)}{Z(\lambda_1, \dots, \lambda_n)} \exp\left\{-\sum_{j=1}^p \lambda_j f_j(x_1, \dots, x_n)\right\}. \quad (13)$$

ρ is called the *maximum entropy distribution* and Z is the *partition function*. Note that we have not proved that the distribution (13) in fact gives a maximum value for S , but this can be done[4].

The Lagrange multipliers $\lambda_1, \dots, \lambda_p$ are chosen so that all the constraints are satisfied

$$\langle f_j \rangle = \int_{\mathbf{R}^n} dV f_j(x_1, \dots, x_n) \rho(x_1, \dots, x_n) = c_j \quad j = 1, \dots, p. \quad (14)$$

The system of equations (14) consists of p equations for the p unknown quantities λ_j .

As it turns out, we almost never need to know the distribution ρ from (13), it is enough to know the partition function. Observe that

$$\begin{aligned}
\langle f_j \rangle &= \int_{\mathbf{R}^n} dV f_j \rho \\
&= \frac{1}{Z} \int_{\mathbf{R}^n} dV f_j \rho_0 \exp\left(-\sum_{i=1}^p \lambda_i f_i\right) \\
&= -\frac{1}{Z} \int_{\mathbf{R}^n} dV \partial_{\lambda_j} \{\rho_0 \exp\left(-\sum_{i=1}^p \lambda_i f_i\right)\} \\
&= -\frac{1}{Z} \partial_{\lambda_j} \int_{\mathbf{R}^n} dV \rho_0 \exp\left(-\sum_{i=1}^p \lambda_i f_i\right) \\
&= -\frac{1}{Z} \partial_{\lambda_j} Z = -\partial_{\lambda_j} \ln Z \\
&\Downarrow \\
\langle f_j \rangle &= -\partial_{\lambda_j} \ln Z, \quad (15)
\end{aligned}$$

and thus we can find the mean of all the quantities f_j by taking partial derivatives of the natural logarithm of the partition function with respect to the Lagrangian multipliers. Moreover, we also have

$$\begin{aligned}
\partial_{\lambda_j \lambda_k} \ln Z &= \partial_{\lambda_j} \left(\frac{1}{Z} \partial_{\lambda_k} Z \right) \\
&= -\frac{1}{Z^2} \partial_{\lambda_j} Z \partial_{\lambda_k} Z + \frac{1}{Z} \partial_{\lambda_j \lambda_k} Z \\
&= -\partial_{\lambda_j} \ln Z \partial_{\lambda_k} \ln Z + \frac{1}{Z} \int_{\mathbf{R}^n} dV f_j f_k \rho_0 \exp \left(-\sum_i \lambda_i f_i \right) \\
&= -\partial_{\lambda_j} \ln Z \partial_{\lambda_k} \ln Z + \langle f_j f_k \rangle.
\end{aligned} \tag{16}$$

Thus

$$\langle f_j f_k \rangle = \partial_{\lambda_j} \ln Z \partial_{\lambda_k} \ln Z + \partial_{\lambda_j \lambda_k} \ln Z \tag{17}$$

In a similar way *all* correlation coefficients $\langle f_1^{n_1} \dots f_p^{n_p} \rangle$ can be expressed through partial derivatives of the partition function.

Inserting the maximum entropy distribution (13) into the entropy functional (5) gives us the following expression for the maximal value of the entropy

$$S = \ln Z + \sum_j \lambda_j \langle f_j \rangle. \tag{18}$$

From a mathematical point of view we now have two sets of variables $\{\langle f_1 \rangle, \dots, \langle f_p \rangle\}$ and $\{\lambda_1, \dots, \lambda_p\}$. Geometrically we imagine that these two pairs of variables, together with S , defines a space Ω of odd dimension $2p + 1$ with coordinates $\{S, \langle f_1 \rangle, \dots, \langle f_p \rangle, \lambda_1, \dots, \lambda_p\}$. The p identities (14) defines a $p + 1$ dimensional surface Λ in Ω .

Taking the differential of the identity (18) we get

$$dS = \sum_j \frac{\partial \ln Z}{\partial \lambda_j} d\lambda_j + \sum_j \{ \langle f_j \rangle d\lambda_j + \lambda_j d\langle f_j \rangle \}.$$

Restricting this differential to the surface Λ , and thus using the identities (15), gives us the following expression for the differential dS restricted to the surface Λ

$$dS = \sum_j \lambda_j d\langle f_j \rangle. \tag{19}$$

The identity (18) defines the entropy as a function depending on all $2p$ variables in Ω . We therefore have

$$dS = \sum_j \frac{\partial S}{\partial \lambda_j} d\lambda_j + \sum_j \frac{\partial S}{\partial \langle f_j \rangle} d\langle f_j \rangle. \tag{20}$$

Comparing (19) and (20) we conclude that on the surface Λ we must have the identities

$$\begin{aligned}
\frac{\partial S}{\partial \lambda_j} &= 0, \\
\frac{\partial S}{\partial \langle f_j \rangle} &= \lambda_j.
\end{aligned} \tag{21}$$

Thus, on the surface Λ , the entropy depends only on the variables $\{\langle f_1 \rangle, \dots, \langle f_p \rangle\}$ and the derivative with respect to these variables determines the values of the Lagrange multipliers in terms of the data $\{c_1, \dots, c_p\}$ of the problem.

It is frequently the case that in addition to the variables $\{x_1, \dots, x_n\}$, the observables depends on parameters. For notational simplicity, let us assume that there is only one parameter denoted by the symbol α . Thus we have observables $\{f_1(x_1, \dots, x_n; \alpha), \dots, f_p(x_1, \dots, x_n; \alpha)\}$. The presence of the parameter does not change the argument leading up to the maximum entropy distribution (13) and thus we have the formulas

$$\begin{aligned}\rho(x_1, \dots, x_n; \alpha) &= \frac{\rho_0(x_1, \dots, x_n)}{Z(\lambda_1, \dots, \lambda_p; \alpha)} \exp \left\{ - \sum_{j=1}^p \lambda_j f_j(x_1, \dots, x_n; \alpha) \right\}, \\ Z(\lambda_1, \dots, \lambda_p; \alpha) &= \int_{\mathbf{R}^n} dV \rho_0 \exp \left\{ - \sum_j \lambda_j f_j(x_1, \dots, x_n; \alpha) \right\}.\end{aligned}\quad (22)$$

Differentiation of the partition function (22) with respect to the parameter α gives us the expression

$$\begin{aligned}\frac{\partial Z}{\partial \alpha} &= - \int_{\mathbf{R}^n} dV \rho_0 \sum_j \lambda_j \frac{\partial f_j}{\partial \alpha} \exp \left\{ - \sum_j \lambda_j f_j \right\}, \\ \Downarrow \\ \frac{\partial \ln Z}{\partial \alpha} &= - \sum_j \lambda_j \left\langle \frac{\partial f_j}{\partial \alpha} \right\rangle.\end{aligned}\quad (23)$$

If we repeat the calculation leading from (18) to (19) for the case when the observables depends on a parameter α , we now get instead of (19) the following more general expression for the differential of the entropy

$$dS = \sum_j \lambda_j d\langle f_j \rangle - \sum_j \lambda_j \left\langle \frac{\partial f_j}{\partial \alpha} \right\rangle d\alpha, \quad (24)$$

where we have used the identity (23).

Note that this differential identity can be written in the form

$$dS = \sum_j \lambda_j dQ_j, \quad (25)$$

where we have introduced the quantities dQ_j representing *generalized heat* associated with the observables

$$\begin{aligned}dQ_j &= d\langle f_j \rangle - \left\langle \frac{\partial f_j}{\partial \alpha} \right\rangle d\alpha \\ &= d\langle f_j \rangle - \left\langle \frac{\partial f_j}{\partial \alpha} d\alpha \right\rangle \\ &= d\langle f_j \rangle - \langle df_j \rangle.\end{aligned}\quad (26)$$

Formula (26) tells us what heat actually represents. Physical systems on the human scale, these are evidently the ones of most immediate interest to us, consists of an immense number of elementary subsystems. The detailed configurational variables for all these elementary systems defines the *microscopic* degrees of freedom of the human scale system. The state of these microscopic degrees of freedom are unknown to us and our ability manipulate them directly is entirely lacking. The few degrees of freedom of the system whose state we *can* know defines the *macroscopic* degrees of freedom for the system. In our description of thermodynamics these are the observables f_j . These macroscopic degrees of freedom typically depend on quantities that, in the context of a given physical system, we can fully control and which consequently are subject to no statistical fluctuations. These quantities are represented by the parameter(s) α . The volume of the container containing a gas, the temperature in the immediate surroundings of the container and the external pressure applied to a fluid-gas interface separating a system from the environment, are examples of quantities represented by the parameter(s) α .

A change in the mean value of a macroscopic degrees of freedom, $d\langle f_j \rangle$, comes from two sources. The first source is a change in the observable representing the said macroscopic degree of freedom. This change is brought about by changing the the parameters α . This quantity is represented by $\langle df_j \rangle$ in formula (26). When this quantity is subtracted from $d\langle f_j \rangle$, what remains is the second source of change of the mean. This second source is a change in the underlying probability distribution, which represents a change in our information about the microscopic state of the system. When our ignorance about the microscopic state of a system increase, the system grows “hotter”, corresponding to an increase in dQ_j . If our ignorance decrease, the system grows “colder”, corresponding to a decrease in dQ_j .

As is usual in thermodynamics, the formalism is misleading in the sense that dQ_j merely denote an infinitesimal amount of generalized heat and is *not* the differential of some function Q_j . No such function exists. The proper mathematical way to think about the identity (24) is that dS and dQ_j are differential forms where dS is an exact differential forms, meaning it is the differential of a function, and dQ_j are inexact differential forms and thus not the differential of a function. However, the mathematical formalism of differential forms must be introduced in the very large context of differential geometry and we will not digress into this area of mathematics.

The above explanation of the nature of heat, referred to the original application of the thermodynamical formalism, where the systems has an immense number of microscopic degrees of freedom, which are in principle knowable and controllable, but as practical matter, not. We however know that the thermodynamical formalism can be applied to any situation where systems has more degrees of freedom than the ones we chose to observe. This might be because the underlying degrees of freedom are unknown but it could also be the case that they are known but that we for various reasons choose to ignore them. In both cases the argument above stands, and the existence of the unknown or ignored degrees of freedom manifest as heat in the theory.

The quantities we can control and which are represented by the parameter(s) α need not necessarily be numerical parameters like temperature, pressure, volume etc. For example, for the situation where the underlying space is the state space a mechanical system consisting of a finite number of particles, a

system of classical fields or even the Fock state space for a quantum mechanical many particle system, the microscopic degrees of freedom are in most cases of interest confined to a bounded spatial domain D . A domain which is defined by its bounding surface Γ . For this case, all observables for the system will depend on the bounding surface Γ , $f_j = f_j(x_1, \dots, x_n; \Gamma)$, and thus for this case the parameter α is a surface and the macroscopic degrees of freedom f_j are functionals.

In order to derive analogs to the important thermodynamical formulas (23) and (24) for this case, we consider a small deformation, $\delta\Gamma$, of the bounding surface Γ . Thus $\Gamma \rightarrow \Gamma + \delta\Gamma$. This deformation leads to variations

$$\begin{aligned}\delta_\Gamma f_j(x_1, \dots, x_n; \Gamma) &= f_j(x_1, \dots, x_n; \Gamma + \delta\Gamma) - f_j(x_1, \dots, x_n; \Gamma), \\ \delta_\Gamma Z(\lambda_1, \dots, \lambda_p; \Gamma) &= Z(\lambda_1, \dots, \lambda_p; \Gamma + \delta\Gamma) - Z(\lambda_1, \dots, \lambda_p; \Gamma).\end{aligned}\quad (27)$$

Arguing exactly like we did for the simple case of a single parameter we now find the important identity

$$\delta_\Gamma \ln Z = - \sum_j \lambda_j \langle \delta_\Gamma f_j \rangle. \quad (28)$$

This identity will, for the special cases mentioned above, lead to the definition of the thermodynamic pressure and related quantities. Corresponding to the *differential* identity for the entropy (24) we now get the following more general *variational* identity

$$\delta S = \sum_j \lambda_j d \langle f_j \rangle - \sum_j \lambda_j \langle \delta_\Gamma f_j \rangle. \quad (29)$$

6.1.2 The thermodynamic formalism in statistical physics

Let us now consider the special case when our underlying space is the classical state space for a mechanical system with n degrees of freedom. This could for example consist of n mass points. We will assume that the system is confined to a bounded domain D in $\mathbf{R}^3$ defined by a bounding surface Γ . The state space is thus a subset of the euclidean space $\mathbf{R}^{6n}$ with coordinates $(\mathbf{q}, \mathbf{p}) = (\mathbf{q}_1, \dots, \mathbf{q}_n, \mathbf{p}_1, \dots, \mathbf{p}_n)$, since we need 3 position coordinates and 3 momentum coordinates for each particle in order to uniquely specify the state of the system. Let $\mathcal{H} = \mathcal{H}(\mathbf{q}, \mathbf{p})$ be the Hamiltonian for the system of mass points. Recall that the value of the Hamiltonian on any given state, $(\mathbf{q}, \mathbf{p})$, is the energy, E , of that state.

When n is large it is very hard, and also mostly useless, to try to track the exact state $(\mathbf{q}(t), \mathbf{p}(t))$ of a system of mass points.

For such a large system it is more useful to consider a probability distribution $\rho(\mathbf{q}, \mathbf{p})$ on the state-space. This is the point of view introduced by Gibbs. We will first consider the simplest, and by far the most common situation, where the Hamiltonian, $\mathcal{H} = \mathcal{H}(\mathbf{q}, \mathbf{p})$ is the only observable. The maximum entropy distribution for this case is

$$\rho(\mathbf{q}, \mathbf{p}) = \frac{\rho_0(\mathbf{q}, \mathbf{p})}{Z} \exp\left(-\frac{\mathcal{H}(\mathbf{q}, \mathbf{p})}{kT}\right), \quad (30)$$

where the partition function is given by

$$Z = Z(T) = \int_{\mathcal{R}^{6n}} d\mathbf{q}d\mathbf{p} \rho_0(\mathbf{q}, \mathbf{p}) \exp\left(-\frac{\mathcal{H}(\mathbf{q}, \mathbf{p})}{kT}\right), \quad (31)$$

and where we have redefined the single Lagrange multiplier using

$$\lambda = \frac{1}{k T}. \quad (32)$$

In this formula, k is the Boltzmann constant and T is a new parameter which by definition is the thermodynamic temperature. The parameter T is determined by

$$\begin{aligned} E &= \langle \mathcal{H} \rangle, \\ &\Updownarrow \\ E &= k T^2 \partial_T \ln Z, \end{aligned} \quad (33)$$

where we have used the chain rule

$$\partial_\lambda = -k T^2 \partial_T, \quad (34)$$

in the general formula (15).

Formula (33) is in statistical mechanics and thermodynamics called the *equation of state*, and all thermodynamic statements that can be made about the system flows from this formula. The formula for the equation of state may look innocent, in order to find it you merely need to take the derivative of the partition function, and partition function also looks innocent, after all it is just a function of one variable, the kind of function we study in first year calculus. However, in order to actually find an expression for this single variable function one needs to do the integral in formula (31), and this is a multiple integral involving something like 10^{27} integration variables in typical situations! Clearly, an exact formula for the partition function can rarely be found. Approximate expressions where the large number of particles are used to ones advantage can more frequently be found, but pushing through calculations like these are as a rule extremely technical. More than one Nobel price has been handed out for developing feasible schemes for calculating the partition function. Given the level of complexity involved in calculating the partition function from the defining formula (31), and the fact that the partition function simply is a function of one or a few variables, it should come as no surprise that the most common approach to finding the equation of state is to fit parametrized functions to experimental data.

The maximum entropy distribution (30) is recognized to essentially be the *Gibb's Canonical ensemble* from statistical physics.

The Gibb's ensemble is the foundation of statistical physics. All results in statistical physics flows from formula (30). Statistical physics is also the foundation of thermodynamics so all conclusions from that subject also flow from the Gibb's ensemble (30). In the thermodynamics context, (33) is, as we have already remarked, nothing but the *equation of state*.

An interesting insight here is that the temperature of a thermodynamic system is in fact a Lagrange multiplier!! This is a profound insight that to this day has not been fully understood or explored.

From this example, it appears useful to think of any application of the maximal entropy principle as an extension of the methods of statistical mechanics to systems that has absolutely nothing to do with the motion of mass points.

This wide general applicability of the methods of statistical physics has lead to deep questions and insights into the nature and significance of the assumption of equilibrium that appears to underline the application of the Gibb's ensemble in statistical physics.

There is also the intriguing fact that the very same functional (5) used in the maximum entropy principle, is also the foundation of information theory which was discovered by Shannon in 1948. This connection between information theory and statistical mechanics (and thermodynamics) has lead to deep insights into the role of information in our fundamental physical theories.

As already discussed in the introduction, the general nature and wide applicability of the maximum entropy principle has been described well by E.T. Jaynes in many papers and the (unfinished) monumental book "Probability theory: The Logic of Science".

As if all this is not impressive enough for one single principle, it is also a very intriguing fact that when one looks deep into the heart of fundamental physics, in the form of quantum field theory, one again finds an appropriately generalized form of the partition function (12). The whole computational engine in the theory of quantum fields revolve around a generalized Gibb's ensemble!

What on earth is going on...

6.1.3 The problem of prior

Note that formula (30) does not uniquely define Gibbs ensemble because of the presence of the prior distribution ρ_0 . The actual Gibbs ensemble corresponds to the choice $\rho_0 = 1$. When using the information theoretical approach to statistical mechanics and thermodynamics, like we do here, one should be very wary when it comes to the choice of the prior distribution. It is simply the most contentious issue in the whole theory. One should ask pointed questions of justification for any proposed choice. What kind of information about the system is it based on, and is it the correct embodiment of said information?

In fact, if one study expositions of statistical mechanics and thermodynamics, which are based on the traditional objective dynamical approach to the subject, one finds that the choice of what from the information theoretical point of view is the prior distribution, is much discussed. The reasons for choosing $\rho_0 = 1$ that have appeared through these discussions are, in our humble opinion, not very convincing.

The problem of determining the prior distribution has been at the center of probability theory and statistics from the very start. The general rules of probability theory tells us how to compute probabilities for derived events from probabilities of primary events. The problem of prior is concerned with the problem of assigning probabilities to primary events. The assignment is supposed to reflect an observers state of knowledge about the primary events. The assignment should be the same for different observers with the same state of knowledge but can be different for observers with different states of knowledge [7]. In this sense probability assignments are subjective [8],[1],[2]. The problem of the prior is how to turn states of knowledge into probability assignments. The first solution to this problem was used by the very founders of probability theory

(Bernoulli and Laplace). If the observers only knowledge of the primary events are their number, then a uniform probability assignment should be used. This idea was later named the principle of indifference by J. M. Keynes. Generalizing this idea to countably infinite or even continuous spaces of primary events has turned out to be very problematic. Laplace himself used such a generalization in his work on probability theory. His probability distribution was uniform and not normalizable since it was defined on the whole real line. Using a uniform distribution for representing indifference about a random variable on a finite interval on the real line would seem to be more reasonable, at least it is normalizable. However even in this case serious problems arise as the well known Bertrand's paradox shows. Problems and paradoxes arising from the various generalizations of the principle of indifference to continuous random variables played no small part in the creation and for a long time complete dominance of the frequency interpretation[3] of probability theory.

The principle of maximum entropy appears first in the writings of W. Gibbs [5] on thermodynamics and statistical physics and later in the fundamental work on information theory by Shannon [10]. However it was E. T. Jaynes [6] who realized the real importance and general nature of the principle of maximum entropy. In his hands it turned into a general method for turning prior knowledge in the form of mean values for observables defined on finite state spaces, into prior probability assignments.

Let us consider this simplest case in more detail. Let $\Omega = \{x_1, x_2, \dots, x_n\}$ be a finite space of primary events. The algebra of possible events is the set of all subsets of Ω . A probability assignment on the set of primary events is a finite set of numbers $p = \{p_i\}$ such that $0 \leq p_i \leq 1$ and $\sum_{i=1}^n p_i = 1$. Let $f_1, \dots, f_k$ be real valued functions on Ω . The principle of maximum entropy states that if the means of the functions $f_1, \dots, f_k$ are known, $\langle f_i \rangle = c_i$, one should, among all probability assignments that satisfy the constraints, pick the one that maximizes the entropy $S = -\sum_{i=1}^n p_i \ln p_i$. The solution to this constrained maximization problem is, as we have seen, the maximum entropy distribution

$$p = \frac{1}{Z(\lambda_1, \dots, \lambda_k)} \exp \left(- \sum_{j=1}^k \lambda_j f_j \right), \quad (35)$$

where Z is the partition function and is given by

$$Z(\lambda_1, \dots, \lambda_k) = \sum_{i=1}^n \exp \left(- \sum_{j=1}^k \lambda_j f_j(i) \right). \quad (36)$$

Observe that for the particular situation where there are no constraints, the principle gives $Z = n$ and the maximum entropy distribution is uniform

$$p_i = \frac{1}{n}. \quad (37)$$

Thus, for an observer that only know that there are n possible primary events, the maximum entropy distribution is exactly the one suggested by the principle of indifferent! The conclusion appeared to be that not only could the maximum entropy distribution tell us how to choose the best distribution in the presence of observed means of a finite number of observables, it could also tell us which

distribution to choose when our ignorance is so profound that the only thing we know about a situation is the number of possible primary events. This distribution is of course exactly what we have called the prior distribution. For a time it looked as if the problem of prior was essentially solved. However continuous valued random variables again turned out to be the Achilles heel. For finite spaces of events the principle will give a unique probability assignment, but when generalizing it to continuous random variables by taking a continuum limit of the finite discrete expression for the entropy, an unknown probability density appears. The density appears because the continuum limit is not unique. Different limiting expressions are found depending on how one approach the continuum through a countable set of discrete spaces. The unknown probability distributions that appears essentially depends on how the discrete points bunch up in the limit. The meaning of this probability density became clear when it was realized that it is the maximum entropy distribution corresponding to no constraints. Thus it was understood that in order to apply the principle of maximum entropy one must start with a prior distribution. The principle of maximum entropy could not determine the prior, it could only tell us how to modify an already existing prior in order to satisfy constraints in the form of mean values. It seemed as if one were back to square one.

There *does* however exist a systematic way to turn prior information on means of observables into prior distributions, and it *does* involve the maximum entropy principle, but not in the direct way just described. In fact, after a certain reformulation it will become evident that the problem of selecting a prior is not merely a side issue that has to be resolved in order to proceed with the real work of applying the maximum entropy principle, the problem of prior is the *only* issue as far as the maximum entropy principle is concerned.

In order to describe this reformulation of the principle of maximum entropy, we will return to the special case of statistical mechanics. In the previous section we discussed the problem of specifying the prior in the context of statistical mechanics and expressed our doubt as to the justifications for making the standard choice $\rho_0(\mathbf{q}, \mathbf{p}) = 1$. Even if we are doubtful about the justification for this particular choice, it is clear that when we apply the maximum entropy principle in statistical mechanics there is a physical context that certainly makes some choices of the prior more reasonable than others. By picking the Hamiltonian function as our observable we must also acknowledge that the system evolve according to the corresponding Hamiltonian equations. It is always the case that the the Hamiltonian function, $\mathcal{H}(\mathbf{q}, \mathbf{p})$, which represents the energy, is a constant of the motion. Depending on the symmetries of the interaction, Hamiltonian systems of equations may also have other conserved quantities. The generic situation is however that the energy is the only conserved quantity. We will assume that this is the case and let the corresponding Hamiltonian function $\mathcal{H}$ be our only observable. The maximum entropy distribution is now given by expression (104) where ρ_0 is the prior distribution. It is in the current context reasonable to impose the condition that the prior is a stationary solution to the corresponding Liouville equation. But this means that the prior distribution, $\rho_0 = \rho_0(\mathbf{q}, \mathbf{p})$, is a conserved quantity for the Hamiltonian system and since the Hamiltonian is the only independent conserved quantity for our generic Hamiltonian system we must have

$$\rho_0(\mathbf{q}, \mathbf{p}) = f_0(\mathcal{H}(\mathbf{q}, \mathbf{p})),$$

where f_0 is an arbitrary function defined on the positive real line. Using this fact we have from (31)

$$\begin{aligned}
Z(T) &= \int_{\mathcal{R}^{6n}} d\mathbf{q} d\mathbf{p} \rho_0(\mathbf{q}, \mathbf{p}) \exp\left(-\frac{\mathcal{H}(\mathbf{q}, \mathbf{p})}{kT}\right) \\
&= \int_{\mathcal{R}^{6n}} d\mathbf{q} d\mathbf{p} f_0(\mathcal{H}(\mathbf{q}, \mathbf{p})) \exp\left(-\frac{\mathcal{H}(\mathbf{q}, \mathbf{p})}{kT}\right) \\
&= \int_0^\infty dE \exp\left(-\frac{E}{kT}\right) f_0(E) \int_{\mathcal{H}=E} d\mathbf{q} d\mathbf{p} \\
&= \int_0^\infty dE \exp\left(-\frac{E}{kT}\right) \rho_0(E),
\end{aligned} \tag{38}$$

where we have defined

$$\rho_0(E) = f_0(E) \int_{\mathcal{H}=E} d\mathbf{q} d\mathbf{p}. \tag{39}$$

The constraints on the *microscopic* prior distribution $\rho_0(\mathbf{q}, \mathbf{p})$ has reduced our original maximum entropy principle on the extremely high dimensional space $\mathbb{R}^{6n}$ with the Hamiltonian as our observable, to a maximum entropy problem on the real line where the coordinate on the line, E , is the observable and the *macroscopic* prior is given by (39). The maximum entropy distribution for this case is

$$\rho(E) = \frac{\rho_0(E)}{Z(T)} \exp\left(-\frac{E}{kT}\right), \tag{40}$$

where the partition function is given by

$$Z(T) = \int_0^\infty dE \rho_0(E) \exp\left(-\frac{E}{kT}\right). \tag{41}$$

This simple situation where we apply the entropy principle to a low dimensional state space $\mathbb{R}^p$ and where the observables are the coordinate functions, $x_1, \dots, x_p$ on the space is not special at all, in fact this is the most common situation when we apply the maximum entropy principle and other applications can almost always be reduced to this situation using an approach similar to the reduction from $\mathbb{R}^{6n}$ to $\mathbb{R}$ described for the case of statistical mechanics.

In most applications of probability theory in statistics there is no underlying high dimensional space of primary events, Ω , like in statistical mechanics and other areas of physics, and the random variables are not some functions, like the Hamiltonian, defined on this space.

Thus in the typical case one can assume that $\Omega = \mathbb{R}^p$, where p is a fairly small number, and that the random variables are just the coordinate function on $\mathbb{R}^p$. The prior probability distribution is then a function, $\rho_0 = \rho_0(x_1, \dots, x_p)$, on $\mathbb{R}^p$, and the partition function is given by the formula

$$Z(\lambda_1, \dots, \lambda_p) = \int_{\mathbb{R}^p} dx_1 dx_2 \dots dx_p \rho_0(x_1, \dots, x_p) \exp\left(-\sum_{j=1}^p \lambda_j x_j\right). \tag{42}$$

The partition function is thus nothing but the multi dimensional Laplace transform of the prior distribution. This relation can be inverted, using analytical

continuation and the multidimensional Fourier transform on the imaginary λ_j axes, and thereby expressing the prior in terms of the partition function

$$\rho_0(x_1, \dots, x_p) = \frac{1}{(2\pi)^p} \int_{\mathbb{R}^p} d\lambda_1 \dots d\lambda_p Z(i\lambda_1, \dots, i\lambda_p) \exp\left(i \sum_{j=1}^p \lambda_j x_j\right). \quad (43)$$

The whole content of the maximum entropy principle is contained in the integral transforms (42) and (43) connecting the partition function and the prior distribution. This is the promised reformulation of the maximum entropy principle, and we understand now that the prior distribution is not merely a bit player in this drama, it is the *only* player. In the next section we will show how this reformulation of the maximum entropy principle gives us a method for solving the problem of prior, which we now see is the only remaining fundamental problem in the statistical modeling of natural or artificial systems.

6.1.4 Solving the problem of prior using stochastic relations

In probability theory and statistics, random variables are often grouped into statistical quantities. These are certain algebraic combinations of means of functions of the random variables. A large set of such statistical quantities are in use, some simple examples are

$$\langle x \rangle \quad \text{The mean of } x. \quad (44)$$

$$\langle x^2 \rangle - \langle x \rangle^2 \quad \text{The variance of } x. \quad (45)$$

$$\langle x^3 \rangle - 3 \langle x \rangle \langle x^2 \rangle + 2 \langle x \rangle^3 \quad \text{The third cumulant.} \quad (46)$$

$$\langle xy \rangle - \langle x \rangle \langle y \rangle \quad \text{The cross variance of } x \text{ and } y. \quad (47)$$

All such quantities can systematically be expressed as functions of the form $F(q_1, \dots, q_k)$ where the variables q_j are means of monomials in the random variables. We will define *stochastic relations* to be systems of equations for the quantities q_j .

$$F_i(q_1, \dots, q_k) = 0 \quad i = 1, \dots, s. \quad (48)$$

Such relations are common in probability and statistics. Examples are zero mean, fixed variance, uncorrelated variables and identities expressing higher order cumulants in terms of lower ones. Identities such as the last ones in the previous list are the fundamental tools used to construct theories of turbulence in fluid, gases and elsewhere. They are also, in their quantum incarnations, the key tools used to find viable simplified models in solid state physics and material science.

In the previous section we have seen that the maximum entropy principle defines a Laplace transform that map the prior distribution to a partition function. As a direct consequence of this transformation, we can express means of monomials in the random variables in terms of partial derivatives of the partition function. For example we have

$$\begin{aligned} \langle f_i \rangle &= -\frac{1}{Z} \partial_{\lambda_i} Z, \\ \text{var}(f_i) &= \langle f_i^2 \rangle - \langle f_i \rangle^2 = -\frac{1}{Z^2} \partial_{\lambda_i}^2 Z + \frac{1}{Z} \partial_{\lambda_i}^2 Z. \end{aligned}$$

This means that the maximum entropy principle turns stochastic relations into systems of partial differential equations for the partition function and therefore imposes constraints on the prior distribution.

The problem is now how to describe the space of solutions of these systems of partial differential equations. In general, not all solutions to the equations can correspond to prior probability distributions. From the definition of the partition function it is for example clear that

$$\begin{aligned} Z(0) &= 1, \\ Z(\lambda_1, \dots, \lambda_p) &\geq 0, \end{aligned}$$

must hold for any acceptable solution.

Finding necessary and sufficient conditions for functions to be the Laplace transform of a probability distribution, and thus be acceptable solutions of the systems of differential equations corresponding to stochastic relations, is not a simple matter, but some results are known [9]. We will not discuss this problem but rather try to explicitly construct the solution space or to say something useful about the structure of the solution space using methods from the formal theory of differential equations. Typically, the solution space is not a linear space and even when it is, the dimension could easily be infinite. However, depending on the number and types of stochastic relations the solution space can end up being parametrized by a finite set of parameters or even be a single point. In this last situation the stochastic relations determine the prior uniquely. Note that in ordinary (parametric) statistics finite parameter families of probability distributions (Gaussian, Poisson, Bernoulli, t-distribution, etc) are assumed to apply in given situations. From the point of view discussed in these notes, this means that in ordinary statistics, stochastic relations constrain the solution space enough for it to be parameterized in terms of a finite number of parameters. Nonparametric statistics correspond to the situation when the solution space is so weakly constrained that it can not be parameterized in terms of a finite number of parameters. Methods from the theory of partial differential equations can in some cases parameterize such weakly constrained solution spaces, not in terms of real numbers, but in terms of arbitrary functions. However for such weakly constrained solution spaces there is another powerful tool available. This is the formal theory of partial differential equations. The main object of study in this theory is the infinitely prolonged hierarchy of a given systems of differential equations. Thus one studies the infinite set of all differential consequences of a given system of equations. Each such differential consequence can be converted back into a stochastic relation by using the relation between mean of monomials and partial derivatives in reverse. One therefore gets the corresponding hierarchy of stochastic relations that are consequences of the original relations induced by the maximum entropy principle and implemented through the Laplace transform.

In the remaining part of this subsection we will discuss several examples that illustrate the method that has been outlined.

Stochastic relations for one random variable Essentially all families of distribution in use in parametric statistics can be derived from simple stochastic relations involving the mean, variance and skewness. In this section we show some examples that support this statement.

Delta distribution Let us consider the stochastic relation corresponding to a fixed mean. It is

$$\langle x \rangle - q = 0. \quad (49)$$

The Laplace transform convert this into the ordinary differential equation

$$Z_\lambda = -qZ. \quad (50)$$

For this simple stochastic relation our system of partial differential equations is a single linear ordinary differential equation. The solution space is linear and parameterized by a single parameter

$$Z(\lambda) = ae^{-q\lambda}. \quad (51)$$

The condition $Z(0) = 1$ fixes the parameter a to be one and we have a unique solution. It is a simple matter to apply the inverse transform (43) to find the corresponding prior distribution

$$\begin{aligned} \rho_0(x) &= \frac{1}{2\pi} \int_{-\infty}^{\infty} d\lambda Z(i\lambda) \exp(i\lambda x) \\ &= \frac{1}{2\pi} \int_{-\infty}^{\infty} d\lambda \exp(-iq\lambda) \exp(i\lambda x) \\ &= \frac{1}{2\pi} \int_{-\infty}^{\infty} d\lambda \exp(i(x-q)\lambda) \\ &= \delta(x-q). \end{aligned} \quad (52)$$

Normal distribution The stochastic relation corresponding to constant variance is

$$\text{var}(x) = q, \quad (53)$$

and the corresponding differential equation is

$$ZZ_{\lambda\lambda} - Z_\lambda^2 - qZ^2 = 0. \quad (54)$$

This is a second order nonlinear ordinary differential equation. The general solution of the nonlinear equation that satisfies the requirement $Z(0) = 1$ is

$$Z(\lambda) = e^{-a\lambda + \frac{1}{2}q\lambda^2}, \quad a \in \mathbb{R}. \quad (55)$$

Using this partition function we can predict the mean of the random variable x to be

$$\begin{aligned} \langle x \rangle &= -\frac{1}{Z(\lambda)} \frac{\partial Z(\lambda)}{\partial \lambda} \\ &= a \end{aligned} \quad (56)$$

and the corresponding prior distribution is found, using (43), to be

$$\begin{aligned}
\rho_0(x) &= \frac{1}{2\pi} \int_{\mathbb{R}} d\lambda Z(i\lambda) \exp(i\lambda x) \\
&= \frac{1}{2\pi} \int_{-\infty}^{\infty} d\lambda \exp\left(-ia\lambda - \frac{1}{2}q\lambda^2\right) \exp(i\lambda x) \\
&= \frac{1}{\sqrt{2\pi q}} e^{-\frac{(x-a)^2}{2q}}.
\end{aligned} \tag{57}$$

which is the normal distribution.

Poisson distribution Let us consider the stochastic relation

$$\text{var}(x) = \langle x \rangle. \tag{58}$$

The corresponding differential equation is

$$ZZ_{\lambda\lambda} - Z_{\lambda}^2 + ZZ_{\lambda} = 0. \tag{59}$$

This equation and most equations derived from stochastic relations simplify considerably if we introduce a new function φ through $Z = e^{\varphi}$. The equation for φ is

$$\varphi_{\lambda\lambda} = -\varphi_{\lambda}. \tag{60}$$

This equation is easy to solve and the corresponding family of partition functions satisfying, as always, the constraint $Z(0) = 1$ is

$$Z(\lambda) = e^{a(e^{-\lambda}-1)}. \tag{61}$$

The corresponding prior distribution is found using (43)

$$\begin{aligned}
\rho_0(x) &= \frac{1}{2\pi} \int_{\mathbb{R}} d\lambda Z(i\lambda) e^{i\lambda x} \\
&= \frac{1}{2\pi} \int_{\mathbb{R}} d\lambda e^{a(e^{-i\lambda}-1)} e^{i\lambda x} \\
&= e^{-a} \sum_{k=0}^{\infty} \frac{a^k}{k!} \frac{1}{2\pi} \int_{\mathbb{R}} d\lambda e^{i\lambda(x-k)} \\
&= e^{-a} \sum_{k=0}^{\infty} \frac{a^k}{k!} \delta(x-k).
\end{aligned}$$

This is a distribution supported on $\Omega = \{0, 1, 2, \dots\}$ and which on this discrete support has the values

$$\rho_0(k) = \frac{e^{-a} a^k}{k!}. \tag{62}$$

This is the Poisson distribution.

Gamma distribution Let us consider a stochastic relation

$$var(x) = \frac{1}{k} \langle x \rangle^2 \quad k > 0. \quad (63)$$

Expressed in terms of φ the corresponding differential equation is

$$\varphi_{\lambda\lambda} = \frac{1}{k} \varphi_{\lambda}^2. \quad (64)$$

The general solution of this equation gives the following family of partition functions

$$Z(\lambda) = (1 + a\lambda)^{-k} \quad a > 0. \quad (65)$$

The corresponding prior distribution is supported on $\Omega = (0, \infty)$ and is given by

$$\rho_0(x) = x^{k-1} \frac{e^{-\frac{x}{a}}}{a^k \Gamma(k)}. \quad (66)$$

This is the Gamma distribution

Bernoulli and Binomial distribution Let the variance be the following quadratic function of the mean

$$var(x) = \langle x \rangle (1 - \langle x \rangle). \quad (67)$$

The corresponding differential equation for φ is

$$\varphi_{\lambda\lambda} = -\varphi_{\lambda}(1 + \varphi_{\lambda}). \quad (68)$$

The solution of the equation gives the following family of partition functions

$$Z(\lambda) = p + qe^{-\lambda} \quad p + q = 1. \quad (69)$$

The corresponding prior distribution is found using (43)

$$\begin{aligned} \rho_0(x) &= \frac{1}{2\pi} \int_{\mathbb{R}} d\lambda Z(i\lambda) e^{i\lambda x} \\ &= \frac{1}{2\pi} \int_{\mathbb{R}} d\lambda (p + qe^{-i\lambda}) e^{i\lambda x} \\ &= \frac{1}{2\pi} \int_{\mathbb{R}} d\lambda p e^{i\lambda x} + \frac{1}{2\pi} \int_{\mathbb{R}} d\lambda q e^{i(x-1)\lambda} \\ &= p\delta(x) + q\delta(x-1). \end{aligned}$$

The distribution is supported on $\Omega = \{0, 1\}$ and is clearly the Bernoulli distribution. If we generalize the stochastic relation to

$$var(x) = \langle x \rangle \left(1 - \frac{1}{n} \langle x \rangle\right). \quad (70)$$

where n is a natural number we get the differential equation

$$\varphi_{\lambda\lambda} = -\varphi_{\lambda} \left(1 + \frac{1}{n} \varphi_{\lambda}\right). \quad (71)$$

The partition function is found to be

$$Z(\lambda) = (p + qe^{-\lambda})^n \quad p + q = 1. \quad (72)$$

The corresponding prior distribution is now

$$\begin{aligned} \rho_0(x) &= \frac{1}{2\pi} \int_{\mathbb{R}} d\lambda Z(i\lambda) e^{i\lambda x} \\ &= \frac{1}{2\pi} \int_{\mathbb{R}} d\lambda (p + qe^{-i\lambda})^n e^{i\lambda x} \\ &= \frac{1}{2\pi} \int_{\mathbb{R}} d\lambda \sum_{k=0}^n \binom{n}{k} p^{n-k} q^k e^{i\lambda(x-k)} \\ &= \sum_{k=0}^n \binom{n}{k} p^{n-k} q^k \delta(x-k). \end{aligned}$$

The prior distribution is supported on $\Omega = \{0, 1, \dots, n\}$ and is on this domain given by

$$\rho_0(k) = \binom{n}{k} p^k q^{n-k}, \quad (73)$$

This is clearly the Binomial distribution.

Stochastic relations for more than one random variable When the number of random variables become larger than one, stochastic relations in general leads to systems of nonlinear partial differential equations. Unless the number and type of relations is right, it is impossible to describe the solution space in terms of a finite number of parameters. This lead us into the domain of nonparametric statistics. This is the domain where the methods from the formal theory of differential equations comes into play. It is not possible to give nontrivial applications of the theory here and we will therefore limit ourselves to two simple examples.

The Multinomial distribution Let $x_1, \dots, x_n$ be n random variables and consider the following system of stochastic relations

$$\begin{aligned} \text{var}(x_i) &= \langle x_i \rangle \left(1 - \frac{1}{n} \langle x_i \rangle\right) \quad i = 1, \dots, n, \\ \text{cov}(x_i, x_j) &= -\frac{1}{n} \langle x_i \rangle \langle x_j \rangle \quad i, j = 1, \dots, n, \quad i \neq j. \end{aligned} \quad (74)$$

The corresponding system of partial differential equations is

$$\begin{aligned} \varphi_{\lambda_i \lambda_i} &= -\varphi_{\lambda_i} \left(1 + \frac{1}{n} \varphi_{\lambda_i}\right), \\ \varphi_{\lambda_i \lambda_j} &= -\frac{1}{n} \varphi_{\lambda_i} \varphi_{\lambda_j}. \end{aligned} \quad (75)$$

The second part of the system of equations has general solutions of the form $\varphi = n \ln(\theta)$ where $\theta(\lambda_1, \dots, \lambda_n) = \sum_{i=1}^n \theta_i(\lambda_i)$. Inserted into the first part of the system this form for φ easily gives the partition function in the form

$$Z(\lambda_1, \dots, \lambda_n) = \left(\sum_{j=1}^n p_j e^{-\lambda_j} + p_{n+1} \right)^n \quad (76)$$

$$= \sum_{k_1 + \dots + k_{n+1} = n} \binom{n}{k_1, \dots, k_{n+1}} p_1^{k_1} \dots p_{n+1}^{k_{n+1}} e^{-\sum_{j=1}^n k_j \lambda_j}. \quad (77)$$

Here we have introduced the standard multinomial symbol

$$\binom{n}{k_1, \dots, k_{n+1}} = \frac{n!}{k_1! \dots k_{n+1}!}.$$

In order to ensure the correct normalization for the corresponding prior distribution we must as usual have

$$\begin{aligned} Z(0, \dots, 0) &= 1 \\ \Updownarrow \\ \sum_{j=1}^{n+1} p_j &= 1. \end{aligned}$$

We now find the prior distribution determined by the partition function (77) using the transform (43)

$$\begin{aligned} \rho_0(x_1, \dots, x_n) &= \frac{1}{(2\pi)^n} \int_{-\infty}^{\infty} d\lambda_1 \dots d\lambda_n Z(\lambda_1, \dots, \lambda_n) e^{-i \sum_{j=1}^n k_j \lambda_j} \\ &= \sum_{k_1 + \dots + k_{n+1} = n} \binom{n}{k_1, \dots, k_{n+1}} p_1^{k_1} \dots p_{n+1}^{k_{n+1}} e^{-\sum_{j=1}^n k_j \lambda_j} \delta(x_1 - k_1) \dots \delta(x_n - k_n). \end{aligned}$$

This is the multinomial distribution. The positivity of ρ_0 is ensured if $p_j \geq 0$. Taken together with the normalization condition this implies that

$$\begin{aligned} 0 \leq p_j \leq 1, \quad j = 1, \dots, n+1 \\ \sum_{j=1}^{n+1} p_j = 1. \end{aligned}$$

Note that for the case discussed in this example the assumed stochastic relations gave rise to a system of partial differential equations for the partition function whose solution space could be parametrized by a finite number of parameters. The corresponding family of prior distributions were then also described in terms of a finite number of parameters. We are thus in the domain of parametric statistics. In general the given set of stochastic relations will give rise to a system of partial differential equations for the partition function whose space of solutions can only be parametrized using an infinite number of parameters. For such cases we are in the realm of non-parametric statistics. In the next example we will see an example of this.

Stochastic relations for the mean For a single random variable, stochastic relations involving only the mean gives distributions located on a finite number of points. For more than one random variable such relations gives rise to nonparametric statistics, or solution spaces parameterized by functions. The theory of partial differential equations can be used to give a full description of these solution spaces. As an example of such a relation consider the case of two random variables whose means are constrained to be on a circle of radius r .

$$\langle x \rangle^2 + \langle y \rangle^2 = r^2. \quad (78)$$

The corresponding partial differential equation is in terms of φ

$$\varphi_\lambda^2 + \varphi_\mu^2 = r^2, \quad (79)$$

and is known from optics as the *Eiconal* equation. The following Z is in the solution space

$$Z = e^{r\sqrt{\lambda^2 + \mu^2}}. \quad (80)$$

This partition functions predicts that the following stochastic relation should hold

$$\text{var}(x) = \left(\frac{\langle y \rangle}{\langle x \rangle} \right)^2 \text{var}(y). \quad (81)$$

The partial differential equation has, however, infinitely many solutions. The method of characteristics can be used to describe the complete solution space. In order to derive stochastic relations that holds for all Z in the solution space, these are the ones that can be said to be consequences of the of the circle constrain, we should consider differential prolongations of the original differential equation. The first prolongation is the system

$$\varphi_\lambda^2 + \varphi_\mu^2 = r^2, \quad (82)$$

$$\varphi_\lambda \varphi_{\lambda\lambda} + \varphi_\mu \varphi_{\mu\lambda} = 0, \quad (83)$$

$$\varphi_\lambda \varphi_{\lambda\mu} + \varphi_\mu \varphi_{\mu\mu} = 0, \quad (84)$$

and this system implies that

$$\varphi_{\lambda\lambda} = \left(\frac{\varphi_\mu}{\varphi_\lambda} \right)^2 \varphi_{\mu\mu}. \quad (85)$$

Translated into stochastic relations this is exactly the one we derived for the special solution $\varphi = r\sqrt{\lambda^2 + \mu^2}$ and it thus holds for all solutions. It is of considerable interest to find a finite set of basic stochastic relations that through some constructive procedure give rise to all consequences of some given system of stochastic relations. This is exactly the kind of question addressed in the formal theory of partial differential equations and the tools developed there can now, through the maximum entropy principle, be applied to solve problems in the area of nonparametric statistics.

6.1.5 Thermodynamic pressure and its cousins

We will now investigate an important consequence of the fundamental variational identity (28) for the thermodynamic case when the total energy is the only observable. For this special case the variational identity turns into

$$\langle \delta_\Gamma \mathcal{H} \rangle = -kT \delta_\Gamma \ln Z. \quad (86)$$

The force acting at a boundary point $\mathbf{x}$ can, taking into account the fact that the state of the system is determined by the position and momenta of all the n particles comprising the system, be given by a function

$$\mathbf{F} = \mathbf{F}(\mathbf{p}, \mathbf{q}, \mathbf{x}), \quad \mathbf{x} \in \Gamma, (\mathbf{p}, \mathbf{q}) \in \mathbb{R}^{6n}. \quad (87)$$

A small deformation of the boundary is determined by an infinitesimal deformation vector field $d\mathbf{r}_\Gamma$ defined on the boundary Γ . The change in total energy induced by this deformation is given by

$$\delta_\Gamma \mathcal{H}(\mathbf{p}, \mathbf{q}; \Gamma) = - \int_\Gamma d\mathbf{x} \mathbf{F}(\mathbf{p}, \mathbf{q}, \mathbf{x}) \cdot d\mathbf{r}_\Gamma. \quad (88)$$

The fundamental variational identity (28) now gives

$$\int_\Gamma d\mathbf{x} \langle \mathbf{F}(\mathbf{p}, \mathbf{q}, \mathbf{x}) \rangle \cdot d\mathbf{r}_\Gamma = kT \delta_\Gamma \ln Z. \quad (89)$$

We will in the following only consider the common situation defined by

$$\langle \mathbf{F}(\mathbf{p}, \mathbf{q}, \mathbf{x}) \rangle = p(\mathbf{x}) \mathbf{n}, \quad (90)$$

where $\mathbf{n}$ is the unit normal for the surface Γ . Note that by definition, p is now the pressure for the system. For this case (91) turns into the identity

$$\int_\Gamma d\mathbf{x} p(\mathbf{x}) \mathbf{n} \cdot d\mathbf{r}_\Gamma = kT \delta_\Gamma \ln Z. \quad (91)$$

Let us first consider the case of a smooth surface, and for this kind of surface, let us consider an infinitesimal variation of the surface that is a pure expansion or contraction. This means that $d\mathbf{r}_\Gamma = \mathbf{n}ds$. For this kind of variation the fundamental variational identity (91) takes the form

$$ds \int_\Gamma d\mathbf{x} p(\mathbf{x}) = kT \delta_\Gamma \ln Z. \quad (92)$$

Using the fact that the volume spanned by the deformation is $d_\Gamma V = A(\Gamma)ds$, where $A(\Gamma)$ is the area of the surface, we have

$$\langle p \rangle_\Gamma = kT \frac{\delta_\Gamma \ln Z}{d_\Gamma V}, \quad (93)$$

where $\langle p \rangle_\Gamma$ is the average of the pressure over the surface of the cavity.

For some important cases the partition function depends on the surface only through the volume. For this situation we have

$$\delta_\Gamma \ln Z = \frac{\partial \ln Z}{\partial V} d_\Gamma V,$$

so that

$$\langle p \rangle_\Gamma = kT \frac{\partial \ln Z}{\partial V}. \quad (94)$$

This is the standard formula for the thermodynamic pressure that one finds in any textbook. It is very frequently true that, independently of the shape, the partition function for large cavities depends only on the volume of the cavity. This may however not be the case for smaller cavities and for such cases we must retreat to the more general identity (93). It is easy to verify that the identity (94) holds for any surface smooth or not. If the deformation is a pure expansion or contraction of a part of the surface defined by $\Gamma_0 \subset \Gamma$, we also get the identity (94), but now with $\Gamma \rightarrow \Gamma_0$.

The general variational identity for the entropy (29) takes for the particular case discussed in this section the form

$$\begin{aligned} \delta S &= \frac{1}{kT} d\langle \mathcal{H} \rangle - \frac{1}{kT} \langle \delta_\Gamma \mathcal{H} \rangle \\ &= \frac{1}{kT} d\langle \mathcal{H} \rangle + \delta_\Gamma \ln Z \\ &= \frac{1}{kT} d\langle \mathcal{H} \rangle + \frac{1}{kT} \langle p \rangle_\Gamma d_\Gamma V, \end{aligned}$$

which can be rewritten as

$$kT dS = dE + \langle p \rangle_\Gamma d_\Gamma V, \quad (95)$$

where we have used (93) and where now $E = \langle \mathcal{H} \rangle$ is the energy of the system. We recognize (95) as one of the fundamental formulas from conventional thermodynamics.

In this section we have done the derivation of the formulas for the Thermodynamical pressure for the case of a classical system. However, the derivation of the pressure formula for the case of quantum systems leads the exact same formulas. If there are more observables in addition to the energy, for example total momentum and/or total angular momentum, the pressure formulas must be generalized. The derivation of the generalizations follow the pattern laid down in this section.

6.2 The maximum entropy principle for quantum systems

We have in the first section of these notes introduced the thermodynamical formalism in the context of classical physics and classical observables. It involved a state space that was finite or at least finite dimensional, and the challenge was to determine which probability distribution on the state space is the best to use, given the means of a finite number of observables of the system. The solution to this problem was to choose the probability distribution that maximized the entropy functional (4) under the constraints determined by the given means. At the face of it, for a quantum system, the situation appears to be very different. For this case the state space is an infinite dimensional Hilbert space and the full information that an observer has is encoded in the *density operator* for the system. This is a self-adjoint positive operator $\hat{\rho}$, on the Hilbert space with trace equal to one.

$$\text{Tr}(\hat{\rho}) = 1.$$

The expectation value of any quantum observable, $\hat{A}$ is by definition

$$\langle \hat{A} \rangle = \text{Tr}(\hat{\rho}\hat{A}). \quad (96)$$

The question one poses is which density operator should be used if we only know the expectation value of a finite number of quantum observables $\hat{A}_i$.

$$\langle \hat{A}_i \rangle = a_i, \quad i = 1, 2, \dots, p. \quad (97)$$

Even though there are real differences between the classical and the quantum case, much is also the same.

The analog of the Gibbs entropy measure (5) is the Von Neumann entropy measure for density operators given by

$$S(\hat{\rho}) = -\text{Tr}(\hat{\rho} \ln \hat{\rho}). \quad (98)$$

The solution to the question posed on the previous page proposed by the Maximum entropy method is to choose the density operator that maximize the Von Neumann entropy while satisfying the constraints (97). It is a simple exercise to show that the unique solution to this maximization problem is the following density operator

$$\hat{\rho} = \frac{1}{Z(\lambda_1, \dots, \lambda_p)} \exp\left(-\sum_i \lambda_i \hat{A}_i\right). \quad (99)$$

This operator is the *maximum entropy density operator*. The function Z is the partition function and is given by

$$Z(\lambda_1, \dots, \lambda_p) = \text{Tr} \left\{ \exp\left(-\sum_i \lambda_i \hat{A}_i\right) \right\}. \quad (100)$$

Arguing like in the classical case we find that much of the thermodynamic formalism is the same as before. Specifically we have

$$S = \ln Z + \sum_{i=1}^p \lambda_i \langle \hat{A}_i \rangle,$$

$$\begin{aligned} \langle \hat{A}_i \rangle &= -\frac{\partial \ln Z}{\partial \lambda_i}, \\ \lambda_i &= \frac{\partial S}{\partial \langle \hat{A}_i \rangle}, \end{aligned} \quad (101)$$

$$\sum_i \lambda_i \left\langle \frac{\partial \hat{A}_i}{\partial \alpha} \right\rangle = -\frac{\partial \ln Z}{\partial \alpha}, \quad (102)$$

where in the last identity we assume that all the observables depend on some parameter α . Like for the classical case, it is common that the system of interest is restricted to some spatial domain D . In this case we get a quantum analog to the classical formula (28)

$$\delta_\Gamma \ln Z = -\sum_j \lambda_j \langle \delta_\Gamma \hat{A}_j \rangle. \quad (103)$$

As we can see, much of the thermodynamic formalism is the same for the classical and the quantum case. However some things are different, or they at least appear to be different. In the classical case we can find correlations between different observables by computing mixed partial derivatives of the partition function as shown in (16) and (17). In the quantum case this is problematic unless the operators commute. For the case of two observables $\hat{A}$ and $\hat{B}$ we have for example

$$\frac{\partial^2 Z}{\partial_\mu \partial_\lambda} = \frac{\partial^2 Z}{\partial_\lambda \partial_\mu} = 2 \langle \hat{A} \hat{B} + \hat{B} \hat{A} \rangle - \langle \hat{A} \rangle \langle \hat{B} \rangle.$$

In a sense this should not come as a surprise. The reason for this is that $\hat{A} \hat{B}$ is not in general Hermitian even if both $\hat{A}$ and $\hat{B}$ are. Thus $\langle \hat{A} \hat{B} \rangle$ is not something that you could ever measure, so it does not make sense to try to predict it. However, $\hat{A} \hat{B} + \hat{B} \hat{A}$ is a Hermitian operator and thus $\langle \hat{A} \hat{B} + \hat{B} \hat{A} \rangle$ is something one would want to predict. And, this is exactly what you would be able to predict using the thermodynamical formalism.

There is another important way in which the classical and quantum cases are different; namely the question of how to include prior information about the system into the thermodynamical formalism. In the classical case this was taken care of by using the entropy measure (5) that included the prior distribution ρ_0 . We have seen how maximization the entropy in the context of statistical mechanics leads to the distribution

$$\rho(\mathbf{x}_1, \dots, \mathbf{x}_n, \mathbf{p}_1, \dots, \mathbf{p}_n) = \frac{\rho_0}{Z} \exp\left(-\frac{\mathcal{H}}{kT}\right), \quad (104)$$

which we recognized to be the Gibbs canonical ensemble. This is however not entirely correct, the canonical ensemble corresponds to the case when we have a uniform prior. We could have gotten this distribution directly from a maximization of the functional

$$S(\rho) = - \int_{\mathbf{R}^n} dV \rho \ln(\rho). \quad (105)$$

This is in fact the functional used by Gibbs in his foundational studies of thermodynamics. It is this functional that is called the Gibbs entropy measure. The functional we introduced in (5) measure by definition the relative entropy of ρ with respect to ρ_0 . It is also denoted by other names in the research literature.

The Von Neumann entropy introduced in (98) is the direct analog of the Gibbs entropy measure (105). However, in contrast to the classical case, there does not exists a measure of information for quantum systems that is universally recognized to be the best measure to use in all cases where there is prior information to take into account. Many different kind of measures has been studied by physicists and mathematicians over the years. Today these questions are intensely pursued in the topical field of quantum information theory.

Note that the classical relative entropy measure can be written in the form

$$S(\rho | \rho_0) = - \int_{\mathbf{R}^n} dV \rho (\ln(\rho) - \ln(\rho_0)). \quad (106)$$

One approach to a quantum version of relative entropy is to directly generalize (106) to the quantum case. This gives us for any pair of density operators $\hat{\rho}$ and $\hat{\rho}_0$ the relative quantum entropy in the form

$$S_Q(\hat{\rho}|\hat{\rho}_0) = -Tr(\hat{\rho} (\ln \hat{\rho} - \ln \hat{\rho}_0)). \quad (107)$$

One could now guess that the corresponding maximal entropy distribution, when the only observable is the total energy, will take the form

$$\hat{\rho} = \frac{\hat{\rho}_0}{Z} \exp\left(-\frac{\hat{\mathcal{H}}}{kT}\right), \quad (108)$$

However, a formula like this can not possibly be correct because the right hand side of (108) is not a Hermitian operator unless the prior $\hat{\rho}_0$ commutes with the total energy operator $\hat{\mathcal{H}}$.

Here we will not pursue these matters, but we should keep in mind that there is a real issue here concerning the general validity of the thermodynamical formalism for quantum systems in cases when there is prior information present.

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